

1.2.2. Percentage of Programmes in which Choice Based Credit System (CBCS)/elective course system has been implemented

The Institution follows Choice Based Credit System (CBCS) / elective course system in all the programmes offered under Engineering, Architecture, Arts & Science, Pharmacy and law. The curriculum of the programmes offered in the institution following CBCS / elective course system are enlisted as Annexures.


14/02/2020
Dean, Academic Affairs
DEAN (ACADEMIC AFFAIRS)
B.S. Abdur Rahman Crescent Institute
of Science & Technology
Vandalur, Chennai - 600 048, India



Registrar

CURRICULUM FOR UG & PG

PROGRAMMES OFFERED DURING THE LAST FIVE YEARS

TABLE OF CONTENTS

S.No.	Name of the Programme
UG Programmes	
1	B.Tech Civil Engineering
2	B.Tech Mechanical Engineering
3	B.Tech Electrical and Electronics Engineering
4	B.Tech Electronics and Communications Engineering
5	B.Tech Computer Science Engineering
6	B.Tech Electronics and Instrumentation Engineering
7	B.Tech Polymer Engineering
8	B.Tech Information Technology
9	B.Tech Aeronautical Engineering
10	B.Tech Automobile Engineering
11	B.Tech Biotechnology
12	B.A. Islamic Studies
13	B.Com General
14	B.Com Accounts and Finance
15	BBA LLB (Hons.)
16	B.Com LLB (Hons.)
17	BBA Financial Services
18	B.Sc. Biotechnology
19	B. Pharm
20	BCA
21	B. Arch
22	B.Sc. Computer Science
23	B. Des
P.G. Programmes	
1.	MBA
2.	MCA
3.	M.Tech CAD-CAM
4.	M. Tech Computer Science Engineering
5.	M.Tech Power Systems Engineering
6.	M.Tech Communication Systems

7.	M.Tech VLSI and Embedded System
8.	M. Tech Construction Engineering & Project Management
9.	M.Tech Power Electronics Drive
10.	M.Tech Manufacturing
11.	M. Tech Electronics and Instrumentation Engineering
12.	M. Tech Information Technology
13.	M.Tech Structural Engineering
14.	M.Sc. Actuarial Science
15.	M.Sc. Chemistry
16.	M.Sc. Physics
17.	M.Sc. Biotechnology
18.	M.Sc. Microbiology
19.	M.A. Islamic Studies
20.	M.Tech. Biotechnology
21.	M.Sc Biochemistry and Molecular Biology
22.	M.Arch



UG PROGRAMME

CURRICULUM

B.TECH PROGRAMMES & ARTS AND SCIENCE PROGRAMME

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE AND TECHNOLOGY**B.TECH. CIVIL ENGINEERING****SEMESTER I**

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC 1181	Differential Calculus and Geometry	3	1	0	4	
2.	HS	ENC 1181/ ISC 1181/ LNC 1181/ LNC 1182 / LNC 1183	English / Arabic / Mandarin / German / Japanese	3	0	0	3	
3.	BS	PHC 1181	Physics	3	0	2	4	
4.	BS	CHC 1181	Chemistry	3	0	2	4	
5.	ESF	GEC 1101	Engineering Graphics	2	0	2	3	
6.	ESF	GEC 1102	Engineering Design	2	0	0	2	
7.	ESF	GEC 1103	Basic Engineering Practices Laboratory	0	0	2	1	
8.	ESF	GEC 1104	Computer Programming I	1	0	2	2	23

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC 1281	Advanced Calculus	3	1	0	4	
2.	BS	-	Physics Elective	2	0	2	3	
3.	BS	-	Chemistry Elective	2	0	2	3	
4.	ESF	GEC 1211	Basic Engineering Mechanics	3	1	0	4	
5.	BS	GEC 1212	Environmental Studies	2	0	0	2	
6.	ESF	GEC 1213	Computer Programming II	1	0	2	2	
7.	EC	CEC 1211	Engineering Geology	2	0	0	2	
8.	EC	CEC 1212	Building Construction and Practices	3	0	0	3	
9.	EC	CEC 1213	Civil Engineering Drawing	0	0	2	1	24

SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	MAC2181	Partial Differential Equations and Transforms	3	1	0	4
2.	HS	-	Humanities Elective I	2	0	0	2
3.	HS	ENC 2181	Oral Communication	0	0	2	1
4.	EC	CEC2101	Mechanics of Solids	3	0	2	4
5.	EC	CEC2102	Surveying	3	0	0	3
6.	EC	CEC2103	Mechanics of Fluids	3	0	0	3
7.	EC	CEC2104	Concrete Technology	3	0	2	4
8.	EC	CEC2105	Surveying Lab	0	0	2	1
							22

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	-	Mathematics Elective I	3	1	0	4
2.	HS	-	Humanities Elective II	2	0	0	2
3.	HS	ENC2282	Written Communication	0	0	2	1
4.	EC	CEC2211	Hydraulic and Hydraulic Machinery	3	0	2	4
5.	EC	CEC2212	Water and Wastewater Engineering	3	0	0	3
6.	EC	CEC2213	Strength of Materials	3	0	0	3
7.	EC	CEC2214	Transportation Engineering I	3	0	0	3
8.	EC	CEC2215	Environmental Engineering Lab	0	0	2	1
9.	PE		Programme Elective				3*
							24

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181 MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	GE	-	General Elective I	3	0	0	3
3.	HS	ENC3181	Communication and soft skill I – Career Choice	0	0	2	1
4.	EC	CEC3101	Structural Analysis I	3	0	0	3
5.	EC	CEC3102	Design of Reinforced Concrete Structures	3	0	0	3
6.	EC	CEC3103	Transportation Engineering II	3	0	0	3
7.	EC	CEC3104	Geotechnical Engineering I	3	0	2	4
8.	PE	-	Programme Elective III				4** 24

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181 MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	BS	-	Mathematics Elective II	2	0	0	2
3.	HS	ENC3281	Communication and soft skill II – Confidence Building	0	0	2	1
4.	EC	CEC3211	Structural Analysis – II	3	0	0	3
5.	EC	CEC3212	Geotechnical Engineering II	3	0	0	3
6.	EC	CEC3213	Design of Steel Structures	3	0	0	3
7.	EC	CEC3214	Computer Modeling and Structural Design Laboratory	0	0	2	1
8.	EC	CEC3215	Survey and Geotechnical Investigation Camp	0	0	2	1

9.	PE	-	Programme Elective	6#	23
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SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	GE	-	General Elective II	3	0	0	3
2.	EC	CEC4101	Estimation and Costing	2	0	2	3
3.	EC	CEC4102	Design of Hydraulic Structures	2	0	0	2
4.	EC	CEC4103	Prestressed Concrete	3	0	0	3
5.	EC	CEC4104	Construction Management	3	0	0	3
6.	PE	-	Programme Elective	3	0	0	8##
7.	EC	CEC4105	Internship				1*** 23

SEMESTER VIII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	EC	CEC4211	Project Work	0	0	24	12 12
Total credits – 175							

* Student has to take courses for a minimum of 3 credits from the list of electives for the corresponding semester.

** Student has to take courses for a minimum of 4 credits from the list of electives for the corresponding semester.

Student has to take courses for a minimum of 6 credits from the list of electives for the corresponding semester.

Student has to take courses for a minimum of 8 credits from the list of electives for the corresponding semester.

*** Industrial training will be undertaken during Third year summer vacation for 15 days. The credit will be awarded in the 7th Semester.

**PROGRAMME ELECTIVES
IV SEMESTER**

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	CECX01	Modern Construction Materials	2	0	0	2
2.	PE	CECX02	Construction Equipment and Techniques	2	0	0	2
3.	PE	CECX03	Principles of Architecture	2	0	0	2
4.	PE	CECX04	Safety in Construction	1	0	0	1
5.	PE	CECX05	Real Estate Laws	1	0	0	1
6.	PE	CECX06	Vaastu Satra and Architectural Design	1	0	0	1

V SEMESTER

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	CECX10	Hydrology	3	0	0	3
2.	PE	CECX11	Air Pollution and Control	3	0	0	3
3.	PE	CECX12	Traffic Engineering	3	0	0	3
4.	PE	CECX13	Solid Waste Management	3	0	0	3
5.	PE	CECX14	Rock Mechanics	2	0	0	2
6.	PE	CECX15	Form Work	2	0	0	2
7.	PE	CECX16	Industrial Waste Water Treatment	2	0	0	2
8.	PE	CECX17	Hazardous Waste Management	2	0	0	2
9.	PE	CECX18	Environmental Geotechnology	2	0	0	2
10.	PE	CECX19	Durability properties of Concrete	2	0	0	2
11.	PE	CECX20	Industrial health & Safety	2	0	0	2
12.	PE	CECX21	Urban Design	2	0	0	2

13.	PE	CECX22	Water proofing of RCC structures	1	0	0	1
14.	PE	CECX23	Construction chemicals for repair works	1	0	0	1
15.	PE	CECX24	Geo Synthetics	1	0	0	1

VI SEMESTER

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	CECX30	Dynamics of Structures	3	0	0	3
2.	PE	CECX31	Advanced Concrete Design	3	0	0	3
3.	PE	CECX32	Repair and Rehabilitation of Structures	3	0	0	3
4.	PE	CECX33	Ground Improvement Techniques	3	0	0	3
5.	PE	CECX34	Pavement Design	3	0	0	3
6.	PE	CECX35	Housing, Planning and Management	3	0	0	3
7.	PE	CECX36	Building Services	3	0	0	3
8.	PE	CECX37	Engineering Ethics	3	0	0	3
9.	PE	CECX38	Environmental Risk Assessment	3	0	0	3
10.	PE	CECX39	Remote Sensing	3	0	0	3
11.	PE	CECX40	Modern Surveying	3	0	0	3

VII SEMESTER

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	CECX45	Bridge Engineering	3	0	0	3
2.	PE	CECX46	Earthquake Engineering	3	0	0	3
3.	PE	CECX47	Industrial Structures	3	0	0	3
4.	PE	CECX48	Prefabricated Structures	3	0	0	3
5.	PE	CECX49	Tall Structures	3	0	0	3
6.	PE	CECX50	Design of Machine Foundation	3	0	0	3

7.	PE	CECX51	Ground Water Engineering	3	0	0	3
8.	PE	CECX52	Environmental Impact Assessment	3	0	0	3
9.	PE	CECX53	Coastal Engineering	3	0	0	3
10.	PE	CECX54	GIS & its Applications	3	0	0	3
11.	PE	CECX55	Geodesy	2	0	0	2
12.	PE	CECX56	Building Information Modeling	2	0	0	2
13.	PE	CECX57	Design of cold formed Steel Structures	2	0	0	2
14.	PE	CECX58	Corrosion Control of Steel in Concrete	2	0	0	2
15.	PE	CECX59	Flood Management	2	0	0	2
16.	PE	CECX60	Urban & Indoor Air Quality Management	2	0	0	2
17.	PE	CECX61	Contract laws and Regulations	2	0	0	2
18.	PE	CECX62	Smart Cities	2	0	0	2
19.	PE	CECX63	Environmental Monitoring	2	0	0	2

Physics Elective Courses
(To be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
1.	PHCX 01	Fundamentals of Engineering Materials	2	0	2	3
2.	PHCX 02	Heat and Thermodynamics	2	0	2	3
3.	PHCX 03	Introduction to Nanoscience and Technology	2	0	2	3
4.	PHCX 04	Lasers and their applications	2	0	2	3
5.	PHCX 05	Materials Science	2	0	2	3
6.	PHCX 06	Non-Destructive Testing	2	0	2	3
7.	PHCX 07	Properties of Matter and Acoustics	2	0	2	3
8.	PHCX 08	Properties of Matter and Nondestructive Testing	2	0	2	3
9.	PHCX 09	Semiconductor Physics and Optoelectronics	2	0	2	3

Chemistry Elective Courses
(To be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
1.	CHCX01	Analytical Instrumentation	2	0	2	3
2.	CHCX02	Corrosion and its Control	2	0	2	3
3.	CHCX03	Electrical Materials and Batteries	2	0	2	3
4.	CHCX04	Engineering Materials	2	0	2	3
5.	CHCX05	Fuels and Combustion	2	0	2	3
6.	CHCX06	Fundamentals of Physical Chemistry	2	0	2	3
7.	CHCX07	Green Technology	2	0	2	3
8.	CHCX08	Organic Chemistry of Biomolecules	2	0	2	3
9.	CHCX09	Polymer Science and Technology	2	0	2	3

Maths Elective Courses
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
1.	MACX 01	Discrete Mathematics And Graph Theory	3	1	0	4
2.	MACX 02	Probability And Statistics	3	1	0	4
3.	MACX 03	Random Processes	3	1	0	4
4.	MACX 04	Applied Numerical Methods	3	1	0	4

Maths Elective Courses
(To be offered in VI Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
1.	MACX 05	Mathematical Programming	2	0	0	2
2.	MACX 06	Statistical Methods for Data Analysis	2	0	0	2
3.	MACX 07	Numerical Methods for Integral and Differential Equations	2	0	0	2
4.	MACX 08	Mathematical Modelling	2	0	0	2
5.	MACX 09	Graph Theory	2	0	0	2

Humanities Elective I
(To be offered in III Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
1.	SSCX01	Fundamentals of Economics	2	0	0	2
2.	SSCX02	Principles of Sociology	2	0	0	2
3.	SSCX03	Sociology of Indian Society	2	0	0	2

Humanities Elective II
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
1.	SSCX04	Economics of Sustainable Development	2	0	0	2
2.	SSCX05	Industrial Sociology	2	0	0	2
3.	SSCX06	Law for Engineers	2	0	0	2

General Elective
Group I Courses
(To be offered in V semester)

Sl. No.	Course Code	Course Title	Offering Department
1.	GECX101	Disaster Management	Civil
2.	GECX102	Total Quality Management	Mechanical
3.	GECX103	Energy Studies	Mechanical
4.	GECX104	Robotics	Mechanical
5.	GECX105	Transport Management	Automobile
6.	GECX106	Control Systems	EEE
7.	GECX107	Introduction to VLSI Design	ECE
8.	GECX108	Plant Engineering	EIE
9.	GECX109	Network Security	CSE
10.	GECX110	Knowledge management	CSE
11.	GECX111	Cyber security	IT
12.	GECX112	Genetic Engineering	LS
13.	GECX113	Fundamentals of Project Management	CBS
14.	GECX114	Operations Research	Mathematics
15.	GECX115	Nano Technology	Physics / Chemistry
16.	GECX116	Vehicle Maintenance	Automobile
17.	GECX117	Fundamentals of Digital Image Processing	ECE

Group II Courses
(To be offered in VII semester)

Sl. No.	Course Code	Course Title	Offering Department
1.	GECX201	Green Design and Sustainability	Civil
2.	GECX202	Appropriate Technology	Civil / Mechanical
3.	GECX203	Engineering System Modelling and Simulation	Mechanical
4.	GECX204	Value Analysis and Engineering	Mechanical
5.	GECX205	Industrial Safety	Mechanical
6.	GECX206	Advanced Optimization Techniques	Mechanical
7.	GECX207	Mat Lab Simulation	EEE
8.	GECX208	Embedded Systems and its Applications	ECE
9.	GECX209	Usability Engineering	CSE
10.	GECX210	Supply Chain Management	CBS
11.	GECX211	System Analysis and Design	CA
12.	GECX212	Advanced Materials	Physics & Chemistry
13.	GECX213	National Service Scheme	School of Humanities
14.	GECX214	Automotive Pollution and Control	Automobile
15.	GECX215	Motor Vehicle Act, Insurance and Policy	Automobile
16.	GECX216	Principles of Communication Systems	ECE
17.	GECX217	Lean Management	Civil
18.	GECX218	Spatial Data Modeling & Analysis	Civil

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE AND TECHNOLOGY
B.TECH. MECHANICAL ENGINEERING

SEMESTER I

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	-
1.	BS	MAC 1181	Differential Calculus and Geometry	3	1	0	4	
2.	HS	ENC 1181/ ISC 1181/ LNC 1181/ LNC 1182 / LNC 1183	English / Arabic / Mandarin / German / Japanese	3	0	0	3	
3.	BS	PHC 1181	Physics	3	0	2	4	
4.	BS	CHC 1181	Chemistry	3	0	2	4	
5.	ESF	GEC 1101	Engineering Graphics	2	0	2	3	
6.	ESF	GEC 1102	Engineering Design	2	0	0	2	
7.	ESF	GEC 1103	Basic Engineering Practices Laboratory	0	0	2	1	
8.	ESF	GEC 1104	Computer Programming I	1	0	2	2	23

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	MAC 1281	Advanced Calculus	3	1	0	4
2.	BS	-	Physics Elective	2	0	2	3
3.	BS	-	Chemistry Elective	2	0	2	3
4.	ESF	GEC1211	Basic Engineering Mechanics	3	1	0	4
5.	BS	GEC 1212	Environmental Studies	2	0	0	2
6.	ESF	GEC 1213	Computer Programming II	1	0	2	2
7.	EC	MEC1211	Material Science	3	0	0	3
8.	EC	MEC1212	Design Appreciation Lab	0	0	3	1

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
9.	ESF	EEC 1283	Basic Electrical Engineering	2	0	0	2	
10.	ESF	EEC 1284	Electrical Engineering Laboratory	0	0	2	1	25

SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC2181	Partial Differential Equations and Transforms	3	1	0	4	
2.	HS	-	Humanities Elective I	2	0	0	2	
3.	HS	ENC 2181	Oral Communication	0	0	2	1	
4.	EC	MEC2101	Manufacturing Processes	3	0	0	3	
5.	EC	MEC2102	Engineering Metallurgy	3	0	0	3	
6.	EC	MEC2103	Engineering Thermodynamics	3	1	0	4	
7.	EC	MEC2104	Solid Mechanics	3	1	0	4	
8.	EC	MEC2105	Manufacturing Processes Lab	0	0	3	1	
9.	EC	MEC2106	Material Testing and Metallurgy Lab	0	0	3	1	
10.	EC	MEC2107	Part Modeling Lab	0	0	3	1	24

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	-	Mathematics Elective I	3	1	0	4	
2.	HS	-	Humanities Elective II	2	0	0	2	
3.	HS	ENC2282	Written Communication	0	0	2	1	
4.	EC	MEC2211	Kinematics of Machinery	3	0	0	3	
5.	EC	MEC2212	Fluid Mechanics and Machinery	3	1	0	4	

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
6.	EC	MEC2213	Metal Cutting and Machine Tools	3	0	0	3	
7.	EC	MEC2214	Machining Lab	0	0	3	1	
8.	EC	MEC2215	Product Modelling Lab	0	0	3	1	
9.	EC	MEC2216	Fluid Mechanics and Machinery Lab	0	0	3	1	20

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	MS	MSC 3181 MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3	
2.	GE	-	General Elective I	3	0	0	3	
3.	HS	ENC3181	Communication and soft skill I Career Choice	0	0	2	1	
4.	EC	MEC3101	Dynamics of Machinery	3	1	0	4	
5.	EC	ECC3181	Electronics for Mechanical Systems	2	0	0	2	
6.	EC	MEC3102	Thermal Engineering (Integrated Lab)	3	0	3	4	
7.	EC	MEC3103	Mechanics Lab	0	0	3	1	
8.	EC	ECC31812	Electronics Lab	0	0	3	1	
9.	PE		Programme Elective ^{##}				6**	25

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181 MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	BS	-	Mathematics Elective II	2	0	0	2
3.	HS	ENC3281	Communication and soft skill II – Confidence Building	0	0	2	1
4.	EC	MEC3211	Machine Design	3	1	0	4
5.	EC	MEC3212	Metrology and Mechanical Measurements (Integrated Lab)	3	0	3	4
6.	EC	MEC3213	Heat and Mass Transfer	3	0	0	3
7.	EC	MEC3214	Heat and Mass Transfer Lab	0	0	3	1
8.	EC	MEC3215	CNC Lab	0	0	3	1
9.	PE		Programme Elective ##				6** 25

SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	GE	-	General Elective II	3	0	0	3
2.	EC	MEC4101	Finite Element Analysis (Integrated Lab)	3	0	3	4
3.	EC	MEC4102	Mechatronics (Integrated Lab)	3	0	3	4
4.	EC	MEC4103	Design of Transmission Systems	3	0	0	3
5.	EC	MEC4104	Simulation Lab	0	0	3	1
6.	EC	MEC4105	Automobile Lab	0	0	3	1
7.	PE		Programme Elective ##				6**

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
8.	EC	MEC4106	Internship				1 23

SEMESTER VIII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	EC	MEC4211	Project Work	0	0	24	12 12

Total credits – 177

- * Industrial training will be undertaken during Third year summer vacation. The credit will be awarded in the 7th Semester.

PROGRAMME ELECTIVES

DESIGN ELECTIVES

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
7.	PE	MECX01	Advanced Strength of Materials	3	0	0	3
8.	PE	MECX02	Composite Materials for Manufacture	3	0	0	3
9.	PE	MECX03	Design of Hydraulics and Pneumatics	3	0	0	3
10.	PE	MECX04	Noise, Vibration and Harshness	3	0	0	3
11.	PE	MECX05	Design of Jigs, Fixtures and Press Tools	3	0	0	3
12.	PE	MECX06	Industrial Problem Solving Techniques	3	0	0	3
13.	PE	MECX07	Geometric Modelling	2	0	0	2
14.	PE	MECX08	Reverse Engineering	2	0	0	2
15.	PE	MECX09	Reliability Engineering	2	0	0	2
16.	PE	MECX10	Micro Electro Mechanical Systems (MEMS)	2	0	0	2
17.	PE	MECX11	Geometric Dimensioning and Tolerancing	1	0	0	1
18.	PE	MECX12	Tool and Die Design	1	0	0	1

THERMAL ELECTIVES

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	MECX21	Refrigeration and Air Conditioning	3	0	0	3
2.	PE	MECX22	Advanced I.C. Engines	3	0	0	3
3.	PE	MECX23	Nuclear Engineering	3	0	0	3
4.	PE	MECX24	Gas Dynamics and Jet Propulsion	3	0	0	3
5.	PE	MECX25	Energy Conversion	3	0	0	3
6.	PE	MECX26	Turbomachinery	3	0	0	3
7.	PE	MECX27	Energy Conservation and Management	3	0	0	3
8.	PE	MECX28	Computational Flow and Heat Transfer	3	0	0	3
9.	PE	MECX29	Renewable Sources of Energy	3	0	0	3
10.	PE	MECX30	Solar Engineering	3	0	0	3
11.	PE	MECX31	Design of Thermal Systems	3	0	0	3
12.	PE	MECX32	Automobile Engineering	3	0	0	3
13.	PE	MECX33	Automotive Pollution and Control	2	0	0	2
14.	PE	MECX34	Fuels and Combustion	2	0	0	2
15.	PE	MECX35	Alternate Fuels	1	0	0	1

MANUFACTURING AND MATERIALS ELECTIVES

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	MECX41	Modern Manufacturing Systems	3	0	0	3
2.	PE	MECX42	Process Planning and Cost Estimation	3	0	0	3
3.	PE	MECX43	Production Planning and Control	3	0	0	3
4.	PE	MECX44	Statistics and Quality Control	3	0	0	3
5.	PE	MECX45	Modern Production Management	3	0	0	3
6.	PE	MECX46	Advanced Optimisation Techniques	3	0	0	3
7.	PE	MECX47	Mechanical Maintenance	2	0	0	2
8.	PE	MECX48	Robotics and Automation	2	0	0	2
9.	PE	MECX49	Advanced Production Process for Automotive Components	2	0	0	2
10.	PE	MECX50	Plant Layout and Material Handling	2	0	0	2
11.	PE	MECX51	Industrial Safety Engineering	2	0	0	1
12.	PE	MECX52	Operations Research & System Analysis	2	0	0	1
13.	PE	MECX53	Nano Materials & Fabrications	2	0	0	2
14.	PE	MECX54	Advanced materials	2	0	0	2
15.	PE	MECX55	Prototyping Techniques	2	0	0	2
16.	PE	MECX57	Industrial Marketing	1	0	0	1
17.	PE	MECX58	Entrepreneurial Development	1	0	0	1
18.	PE	MECX59	Digital Manufacturing	1	0	0	1
19.	PE	MECX60	Additive Manufacturing	1	0	0	1

20.	PE	MECX61	Failure Analysis and techniques	1	0	0	1
21.	PE	MECX62	Advanced System Simulation (1 D Modeling)	1	0	0	1

Physics Elective Courses

(To be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
10.	PHCX 01	Fundamentals of Engineering Materials	2	0	2	3
11.	PHCX 02	Heat and Thermodynamics	2	0	2	3
12.	PHCX 03	Introduction to Nanoscience and Technology	2	0	2	3
13.	PHCX 04	Lasers and their applications	2	0	2	3
14.	PHCX 05	Materials Science	2	0	2	3
15.	PHCX 06	Non-Destructive Testing	2	0	2	3
16.	PHCX 07	Properties of Matter and Acoustics	2	0	2	3
17.	PHCX 08	Properties of Matter and Nondestructive Testing	2	0	2	3
18.	PHCX 09	Semiconductor Physics and Optoelectronics	2	0	2	3

Chemistry Elective Courses

(To be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
10.	CHCX01	Analytical Instrumentation	2	0	2	3
11.	CHCX02	Corrosion and its Control	2	0	2	3
12.	CHCX03	Electrical Materials and Batteries	2	0	2	3
13.	CHCX04	Engineering Materials	2	0	2	3
14.	CHCX05	Fuels and Combustion	2	0	2	3
15.	CHCX06	Fundamentals of Physical Chemistry	2	0	2	3

16.	CHCX07	Green Technology	2	0	2	3
17.	CHCX08	Organic Chemistry of Biomolecules	2	0	2	3
18.	CHCX09	Polymer Science and Technology	2	0	2	3

Maths Elective Courses

(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
5.	MACX 01	Discrete Mathematics And Graph Theory	3	1	0	4
6.	MACX 02	Probability And Statistics	3	1	0	4
7.	MACX 03	Random Processes	3	1	0	4
8.	MACX 04	Applied Numerical Methods	3	1	0	4

Maths Elective Courses

(To be offered in VI Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
6.	MACX 05	Mathematical Programming	2	0	0	2
7.	MACX 06	Statistical Methods for Data Analysis	2	0	0	2
8.	MACX 07	Numerical Methods for Integral and Differential Equations	2	0	0	2
9.	MACX 08	Mathematical Modelling	2	0	0	2
10.	MACX 09	Graph Theory	2	0	0	2

Humanities Elective I
(To be offered in III Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
4.	SSCX01	Fundamentals of Economics	2	0	0	2
5.	SSCX02	Principles of Sociology	2	0	0	2
6.	SSCX03	Sociology of Indian Society	2	0	0	2

Humanities Elective II
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
4.	SSCX04	Economics of Sustainable Development	2	0	0	2
5.	SSCX05	Industrial Sociology	2	0	0	2
6.	SSCX06	Law for Engineers	2	0	0	2

**General Elective
Group I Courses**
(To be offered in V semester)

Sl. No.	Course Code	Course Title	Offering Department
18.	GECX101	Disaster Management	Civil
19.	GECX102	Total Quality Management	Mechanical
20.	GECX103	Energy Studies	Mechanical
21.	GECX104	Robotics	Mechanical
22.	GECX105	Transport Management	Automobile
23.	GECX106	Control Systems	EEE
24.	GECX107	Introduction to VLSI Design	ECE
25.	GECX108	Plant Engineering	EIE
26.	GECX109	Network Security	CSE
27.	GECX110	Knowledge management	CSE
28.	GECX111	Cyber security	IT
29.	GECX112	Genetic Engineering	LS
30.	GECX113	Fundamentals of Project Management	CBS
31.	GECX114	Operations Research	Mathematics
32.	GECX115	Nano Technology	Physics / Chemistry
33.	GECX116	Vehicle Maintenance	Automobile
34.	GECX117	Fundamentals of Digital Image Processing	ECE

Group II Courses
(To be offered in VII semester)

Sl. No.	Course Code	Course Title	Offering Department
19.	GECX201	Green Design and Sustainability	Civil
20.	GECX202	Appropriate Technology	Civil / Mechanical
21.	GECX203	Engineering System Modelling and Simulation	Mechanical
22.	GECX204	Value Analysis and Engineering	Mechanical
23.	GECX205	Industrial Safety	Mechanical
24.	GECX206	Advanced Optimization Techniques	Mechanical
25.	GECX207	Mat Lab Simulation	EEE
26.	GECX208	Embedded Systems and its Applications	ECE
27.	GECX209	Usability Engineering	CSE
28.	GECX210	Supply Chain Management	CBS
29.	GECX211	System Analysis and Design	CA
30.	GECX212	Advanced Materials	Physics & Chemistry
31.	GECX213	National Service Scheme	School of Humanities
32.	GECX214	Automotive Pollution and Control	Automobile
33.	GECX215	Motor Vehicle Act, Insurance and Policy	Automobile
34.	GECX216	Principles of Communication Systems	ECE
35.	GECX217	Lean Management	Civil
36.	GECX218	Spatial Data Modeling & Analysis	Civil

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B.TECH. ELECTRICAL AND ELECTRONICS ENGINEERING

SEMESTER I

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC 1181	Differential Calculus and Geometry	3	1	0	4	
2.	HS	ENC 1181/ ISC 1181/ LNC 1181/ LNC 1182 / LNC 1183	English / Arabic / Mandarin / German / Japanese	3	0	0	3	
3.	BS	PHC 1181	Physics	3	0	2	4	
4.	BS	CHC 1181	Chemistry	3	0	2	4	
5.	ESF	GEC 1101	Engineering Graphics	2	0	2	3	
6.	ESF	GEC 1102	Engineering Design	2	0	0	2	
7.	ESF	GEC 1103	Basic Engineering Practices Laboratory	0	0	2	1	
8.	ESF	GEC 1104	Computer Programming I	1	0	2	2	23

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC 1281	Advanced Calculus	3	1	0	4	
2.	BS	-	Physics Elective	2	0	2	3	
3.	BS	-	Chemistry Elective	2	0	2	3	
4.	ESF	GEC 1211	Basic Engineering Mechanics	3	1	0	4	
5.	BS	GEC 1212	Environmental Studies	2	0	0	2	
6.	ESF	GEC 1213	Computer Programming II	1	0	2	2	
7.	EC	EEC 1211	Electric Circuit Analysis	3	1	0	4	
8.	EC	EEC 1212	Electric Circuits and Simulation Lab	0	0	3	1	23

SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	MAC2181	Partial Differential Equations and Transforms	3	1	0	4
2.	HS	-	Humanities Elective I	2	0	0	2
3.	HS	ENC 2181	Oral Communication	0	0	2	1
4.	EC	EEC2101	Electro Magnetic Theory	3	0	0	3
5.	EC	EEC2102	Signals and Systems	3	0	0	3
6.	EC	EEC2103	Electronic Devices & Circuits	3	0	0	3
7.	EC	EEC2104	Electromechanical Energy Conversion	3	0	2	4
8.	EC	EEC2105	Electronic Devices & Circuits Lab	0	0	3	1
9.	EC	EEC2106	Electromechanical Energy Conversion Lab	0	0	3	1

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	-	Mathematics Elective I	3	1	0	4
2.	HS	-	Humanities Elective II	2	0	0	2
3.	HS	ENC2282	Written Communication	0	0	2	1
4.	EC	EEC2211	Power System - I	3	1	0	4
5.	EC	EEC2212	AC Machines	3	1	0	4
6.	EC	EEC2213	Power Electronics	3	1	0	4
7.	EC	EEC2214	AC Machines Lab	0	0	3	1
8.	EC	EEC2215	Power Electronics Lab	0	0	3	1
9.	PE		Programme Elective				3
							24

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181 MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	GE	-	General Elective I	3	0	0	3
3.	HS	ENC3181	Communication and soft skill I Career Choice	0	0	2	1
4.	EC	EEC3101	Power System - II	3	1	0	4
5.	EC	EEC3102	Measurements & Instrumentation	3	0	0	3
6.	EC	EEC3103	Digital Systems and Integrated Circuits	3	0	0	3
7.	EC	EEC3104	Power System Simulation Lab	0	0	3	1
8.	EC	EEC3105	Measurements & Instrumentation Lab	0	0	3	1
9.	EC	EEC3106	Digital Systems & Integrated Circuits Lab	0	0	3	1
10.	PE	-	Programme Elective				3
							23

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181 MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	BS	-	Mathematics Elective II	2	0	0	2
3.	HS	ENC3281	Communication & soft skill II Confidence Building	0	0	2	1
4.	EC	EEC3211	Embedded Systems	3	1	0	4
5.	EC	EEC3212	Control Systems	3	1	0	4
6.	EC	EEC3213	Embedded Systems Lab	0	0	3	1
7.	EC	EEC3214	Control Systems Lab	0	0	3	1
8.	EC	EEC3215	Self learning	0	2	0	2

9.	PE	-	Programme Elective	6	24
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SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	GE	-	General Elective II	3	0	0	3
2.	EC	EEC4101	Power System Protection & Switchgear	3	0	0	3
3.	EC	EEC4102	Design of Electrical Apparatus	3	0	0	3
4.	EC	EEC4103	PLC & DCS*	3	0	2	5
5.	PE	-	Programme Elective	3	0	0	9
6.	EC	EEC3216	Internship*				1** 24

SEMESTER VIII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	EC		Project Work	0	0	24	12 12

Total credits – 175

* 15 days

** Internship will be undertaken during Third year summer vacation. The credit will be awarded in the 7th Semester.

PROGRAMME ELECTIVES**POWER SYSTEM**

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
19.	PE	EECX01	Power Distribution System	3	0	0	3
20.	PE	EECX02	Power System Planning & Reliability	3	0	0	3
21.	PE	EECX03	Power System Dynamics	3	0	0	3
22.	PE	EECX04	Power System Transients	3	0	0	3
23.	PE	EECX05	Smart Power Grid	3	0	0	3
24.	PE	EECX06	Wind Energy Conversion systems	3	0	0	3
25.	PE	EECX07	Flexible AC Transmission Systems	3	0	0	3
26.	PE	EECX08	Industrial Power System Analysis & Design	3	0	0	3
27.	PE	EECX09	Electric Energy Generation, Utilization and Conservation	3	0	0	3
28.	PE	EECX10	Restructured power systems	3	0	0	3
29.	PE	EECX11	Solar Energy Technology	2	0	0	2
30.	PE	EECX12	Micro-grid Protection	3	0	0	3
31.	PE	EECX13	Electric Vehicles	1	0	0	1
32.	PE	EECX14	Power System Simulation Software	1	0	0	1
33.	PE	EECX15	Wide Area Measurement Systems	2	0	0	2
34.	PE	EECX16	Energy Auditing	2	0	0	2

POWER ELECTRONICS & DRIVES

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	EECX21	Special Electrical Machines	3	0	0	3
2.	PE	EECX22	CAD for Electrical Apparatus	3	0	0	3
3.	PE	EECX23	Software for Circuit Simulation	3	0	0	3
4.	PE	EECX24	Electromagnetic Field Computation and Modelling	3	0	0	3
5.	PE	EECX25	Solid State AC & DC Drives	3	0	0	3
6.	PE	EECX26	Converters, Applications & Design	3	0	0	3
7.	PE	EECX27	Power Electronics Application to Renewable Energy System	3	0	0	3
8.	PE	EECX28	Embedded Control of Electric Drives	3	0	0	3
9.	PE	EECX29	Power Quality	3	0	0	3
10.	PE	EECX33	IoT for Electrical Engineers	3	0	0	3

HIGH VOLTAGE ENGINEERING

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	EECX36	High Voltage Engineering	3	0	0	3
2.	PE	EECX37	Bio-Electrics	3	0	0	3
3.	PE	EECX38	EHV AC & DC Transmission Engineering	3	0	0	3
4.	PE	EECX39	High Voltage DC Transmission	3	0	0	3
5.	PE	EECX40	Electromagnetic Interference and Electromagnetic Compatibility	3	0	0	3

6.	PE	EECX41	Outdoor Insulators	3	0	0	3
7.	PE	EECX44	High Voltage Testing Techniques	3	0	0	3
8.	PE	EECX45	Pulsed Electric Field and Food Preservation	3	0	0	3

COMPUTER SCIENCE & INFORMATION TECHNOLOGY

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	ITCX81	Database Management Systems	3	0	0	3
4.	PE	ITCX82	Computer Networks	3	0	0	3
5.	PE	ITCX83	Java Programming	2	0	2	3
6.	PE	ITCX84	Computer Hardware and Interfacing	2	0	2	3
7.	PE	ITCX85	Cloud Computing	3	0	0	3
8.	PE	ITCX86	Operating Systems	3	0	0	3
9.	PE	ITCX87	Internet of Things	3	0	0	3
10.	PE	ITCX88	Python Programming	3	0	0	3

ELECTRONICS, COMMUNICATION, INSTRUMENTATION AND CONTROL

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	ECCX81	Image Processing	3	0	0	3
2.	PE	ECCX82	VLSI Design	3	0	0	3
3.	PE	ECCX83	Integrated Circuits and System Design	3	0	0	3
4.	PE	ECCX84	Communication System Security	3	0	0	3
5.	PE	ECCX85	Embedded Hardware & Software System Design	3	0	0	3

6.	PE	ECCX86	Digital Signal Processors	3	0	0	3
7.	PE	EICX81	Bio Instrumentation and Signal Analysis	3	0	0	3
8.	PE	EICX82	Sensors for Bio-Medical Application	3	0	0	3
9.	PE	EICX83	Intelligent Control	3	0	0	3
10.	PE	EICX84	Advanced Control System	3	0	0	3
11.	PE	EICX85	Power Plant Instrumentation	3	0	0	3
12.	PE	EICX86	Instrumentation and Control in Petrochemical Industries	3	0	0	3

**Physics Elective Courses
(To be offered in II Semester)**

Sl. No.	Course Code	Course Title	L	T	P	C
19.	PHCX 01	Fundamentals of Engineering Materials	2	0	2	3
20.	PHCX 02	Heat and Thermodynamics	2	0	2	3
21.	PHCX 03	Introduction to Nanoscience and Technology	2	0	2	3
22.	PHCX 04	Lasers and their applications	2	0	2	3
23.	PHCX 05	Materials Science	2	0	2	3
24.	PHCX 06	Non-Destructive Testing	2	0	2	3
25.	PHCX 07	Properties of Matter and Acoustics	2	0	2	3
26.	PHCX 08	Properties of Matter and Nondestructive Testing	2	0	2	3
27.	PHCX 09	Semiconductor Physics and Optoelectronics	2	0	2	3

**Chemistry Elective Courses
(To be offered in II Semester)**

Sl. No.	Course Code	Course Title	L	T	P	C
19.	CHCX01	Analytical Instrumentation	2	0	2	3
20.	CHCX02	Corrosion and its Control	2	0	2	3
21.	CHCX03	Electrical Materials and Batteries	2	0	2	3
22.	CHCX04	Engineering Materials	2	0	2	3
23.	CHCX05	Fuels and Combustion	2	0	2	3
24.	CHCX06	Fundamentals of Physical Chemistry	2	0	2	3
25.	CHCX07	Green Technology	2	0	2	3
26.	CHCX08	Organic Chemistry of Biomolecules	2	0	2	3
27.	CHCX09	Polymer Science and Technology	2	0	2	3

Maths Elective Courses
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
9.	MACX 01	Discrete Mathematics And Graph Theory	3	1	0	4
10.	MACX 02	Probability And Statistics	3	1	0	4
11.	MACX 03	Random Processes	3	1	0	4
12.	MACX 04	Applied Numerical Methods	3	1	0	4

Maths Elective Courses
(To be offered in VI Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
11.	MACX 05	Mathematical Programming	2	0	0	2
12.	MACX 06	Statistical Methods for Data Analysis	2	0	0	2
13.	MACX 07	Numerical Methods for Integral and Differential Equations	2	0	0	2
14.	MACX 08	Mathematical Modelling	2	0	0	2
15.	MACX 09	Graph Theory	2	0	0	2

Humanities Elective I
(To be offered in III Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
7.	SSCX01	Fundamentals of Economics	2	0	0	2
8.	SSCX02	Principles of Sociology	2	0	0	2
9.	SSCX03	Sociology of Indian Society	2	0	0	2

Humanities Elective II
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
7.	SSCX04	Economics of Sustainable Development	2	0	0	2
8.	SSCX05	Industrial Sociology	2	0	0	2
9.	SSCX06	Law for Engineers	2	0	0	2

General Elective
Group I Courses
(To be offered in V semester)

Sl. No.	Course Code	Course Title	Offering Department
35.	GECX101	Disaster Management	Civil
36.	GECX102	Total Quality Management	Mechanical
37.	GECX103	Energy Studies	Mechanical
38.	GECX104	Robotics	Mechanical
39.	GECX105	Transport Management	Automobile
40.	GECX106	Control Systems	EEE
41.	GECX107	Introduction to VLSI Design	ECE
42.	GECX108	Plant Engineering	EIE
43.	GECX109	Network Security	CSE
44.	GECX110	Knowledge management	CSE
45.	GECX111	Cyber security	IT
46.	GECX112	Genetic Engineering	LS
47.	GECX113	Fundamentals of Project Management	CBS
48.	GECX114	Operations Research	Mathematics
49.	GECX115	Nano Technology	Physics / Chemistry
50.	GECX116	Vehicle Maintenance	Automobile
51.	GECX117	Fundamentals of Digital Image Processing	ECE

Group II Courses
(To be offered in VII semester)

Sl. No.	Course Code	Course Title	Offering Department
1.	GECX201	Green Design and Sustainability	Civil
2.	GECX202	Appropriate Technology	Civil / Mechanical
3.	GECX203	Engineering System Modelling and Simulation	Mechanical
4.	GECX204	Value Analysis and Engineering	Mechanical
5.	GECX205	Industrial Safety	Mechanical
6.	GECX206	Advanced Optimization Techniques	Mechanical
7.	GECX207	Mat Lab Simulation	EEE
8.	GECX208	Embedded Systems and its Applications	ECE
9.	GECX209	Usability Engineering	CSE
10.	GECX210	Supply Chain Management	CBS
11.	GECX211	System Analysis and Design	CA
12.	GECX212	Advanced Materials	Physics & Chemistry
13.	GECX213	National Service Scheme	School of Humanities
14.	GECX214	Automotive Pollution and Control	Automobile
15.	GECX215	Motor Vehicle Act, Insurance and Policy	Automobile
16.	GECX216	Principles of Communication Systems	ECE
17.	GECX217	Lean Management	Civil
18.	GECX218	Spatial Data Modeling & Analysis	Civil

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Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC1181	Differential Calculus and Geometry	3	1	0	4	
2.	HS	ENC1181/ ISC 1181/ LNC1181/ LNC1182 / LNC1183	English / Arabic / Mandarin / German / Japanese	3	0	0	3	
3.	BS	PHC1181	Physics	3	0	2	4	
4.	BS	CHC1181	Chemistry	3	0	2	4	
5.	ESF	GEC1101	Engineering Graphics	2	0	2	3	
6.	ESF	GEC1102	Engineering Design	2	0	0	2	
7.	ESF	GEC1103	Basic Engineering Practices Laboratory	0	0	2	1	
8.	ESF	GEC1104	Computer Programming I	1	0	2	2	23

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC1281	Advanced Calculus	3	1	0	4	
2.	BS	-	Physics Elective	2	0	2	3	
3.	BS	-	Chemistry Elective	2	0	2	3	
4.	ESF	GEC1211	Engineering Mechanics	3	1	0	4	
5.	BS	GEC1212	Environmental Science	2	0	0	2	
6.	ESF	GEC1213	Computer Programming II	1	0	2	2	
7.	EC	ECC1201	Circuit and Network Analysis	2	0	2	3	
8.	EC	ECC1202	Electron Devices	3	0	0	3	
9.	EC	ECC1203	Electron Devices Laboratory	0	0	3	1	25

SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	MAC2181	Partial Differential Equations and Transforms	3	1	0	4
2.	HS	-	Humanities Elective I	2	0	0	2
3.	HS	ENC2181	Oral Communication	0	0	2	1
4.	EC	ECC2101	Electronic Circuits	3	1	0	4
5.	EC	ECC2102	Signals and Systems	3	0	2	4
6.	EC	ECC2103	Digital Electronics	3	0	0	3
7.	EC	ECC2104	Electromagnetic Fields and Transmission Lines	3	1	0	4
8.	EC	ECC2105	Electronic Circuits Laboratory	0	0	2	1
9.	EC	ECC2106	Digital Electronics Laboratory	0	0	2	1
							24

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	-	Mathematics Elective I	3	1	0	4
2.	HS	-	Humanities Elective II	2	0	0	2
3.	HS	ENC2282	Written Communication	0	0	2	1
4.	EC	ECC2201	Digital Signal Processing	3	1	0	4
5.	EC	ECC2202	Linear Integrated Circuits	3	0	0	3
6.	EC	ECC2203	Communication Theory and systems	3	0	0	3
7.	EC	ECC2204	Microprocessors and Microcontrollers	3	0	0	3
8.	EC	ECC2205	Microprocessor & Microcontroller Laboratory	0	0	2	1
9.	EC	ECC2206	Digital Signal Processing Laboratory	0	0	2	1
10.	EC	ECC2207	Linear Integrated Circuits Laboratory	0	0	2	1
				23			

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181/ MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	GE	-	General Elective I	3	0	0	3
3.	HS	ENC3181	Communication and soft skill – I Career Choice	0	0	2	1
4.	EC	ECC3101	Digital Communication	3	1	0	4
5.	EC	ECC3102	Computer Networks	3	0	0	3
6.	EC	ECC3103	VLSI Design	3	0	0	3
7.	EC	ECC3104	VLSI Laboratory	0	0	2	1
8.	EC	ECC3105	Computer Networks Laboratory	0	0	2	1
9.	EC	ECC3106	Communication System Laboratory	0	0	2	1
10.	PE	-	Professional Elective I	3	0	0	3*
11.	PE	-	Professional Elective II	3	0	0	3* 26

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181/ MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	BS	-	Mathematics Elective II	2	0	0	2
3.	HS	ENC3281	Communication and soft skill – II Confidence Building	0	0	2	1
4.	EC	ECC3201	RF & Microwave Engineering	3	0	0	3
5.	EC	ECC3202	Embedded Systems	3	0	0	3
6.	EC	ECC3203	RF and Microwave Laboratory	0	0	2	1
7.	EC	ECC3204	Embedded System Design Laboratory	0	0	2	1

8.	PE	-	Professional Elective III	3	0	0	3*	
9.	PE	-	Professional Elective IV	3	0	0	3*	
10.	PE	-	Professional Elective V	3	0	0	3*	23

SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	GE		General Elective II	3	0	0	3	
2.	EC	ECC4101	Introduction to Wireless Communication	3	0	0	3	
3.	EC	ECC4102	Optical Communication	3	0	0	3	
4.	EC	ECC4103	Optical Communication Laboratory	0	0	2	1	
5.	EC	ECC4104	Wireless Communication Laboratory	0	0	2	1	
6.	PE		Professional Elective VI	3	0	0	3*	
7.	PE		Professional Elective VII	3	0	0	3*	
8.	PE		Professional Elective VIII	3	0	0	3*	
9.	EC	ECC4105	Internship / Industry Training / Mini Project				1 [#]	21

SEMESTER VIII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	EC	ECC4201	Project Work	0	0	24	12	12

Total credits – 177

[#] Industrial training will be undertaken during Third year summer vacation for 15 days.

The credit will be awarded in the 7th Semester.

* Student has to take courses for a minimum of 3 credits from the list of electives for the corresponding semester.

**PROFESSIONAL ELECTIVES
RF COMMUNICATION STREAM**

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
35.	PE	ECCX01	Digital System Design	3	0	0	3
36.	PE	ECCX02	Computer Architecture	3	0	0	3
37.	PE	ECCX03	Control Systems	3	0	0	3
38.	PE	ECCX04	Biomedical Electronics	3	0	0	3
39.	PE	ECCX05	Image Processing	2	0	2	3
40.	PE	ECCX07	PCB Design	1	0	2	2
41.	PE	ECCX08	Data structure and its algorithms	1	0	2	2
42.	PE	ECCX09	Java Programming	1	0	0	1

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	ECCX13	Telecommunication Switching Networks	3	0	0	3
2.	PE	ECCX14	Artificial Intelligence	3	0	0	3
3.	PE	ECCX15	Wireless Networks & its Applications	3	0	3	3
4.	PE	ECCX16	Multimedia Communication Systems	3	0	0	3
5.	PE	ECCX17	Principles of Wireless Sensor network	3	0	0	3
6.	PE	ECCX18	Fundamentals of Nano technology	3	0	0	3
7.	PE	ECCX20	Introduction to Network Security	3	0	0	3
8.	PE	ECCX21	Information coding Techniques	3	0	0	3

9.	PE	ECCX22	Python Programming	1	0	0	1
10.	PE	ECCX23	Design using EDA Tools	0	0	2	1
11.	PE	ECCX24	Predictive analytics	1	0	0	1
12.	PE	ECCX25	IC technology	3	0	0	3
13.	PE	ECCX26	Simulation of RF Circuits and Components	0	0	2	1
14.	PE	ECCX27	RF Test and measurement	1	0	0	1

SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	ECCX35	Electromagnetic Interference & Compatibility	3	0	0	3
2.	PE	ECCX36	Introduction to Internet of Things	3	0	0	3
3.	PE	ECCX37	RF System Design	3	0	0	3
4.	PE	ECCX38	Introduction to Satellite Communication	3	0	0	3
5.	PE	ECCX39	Microwave & Millimeter Wave Systems	3	0	0	3
6.	PE	ECCX40	Radar & Navigational Aids	3	0	0	3
7.	PE	ECCX41	Advanced Antenna Design	2	0	2	3
8.	PE	ECCX42	Introduction to Error control coding	3	0	0	3
9.	PE	ECCX43	Satellite Image processing	3	0	0	3
10.	PE	ECCX44	Software defined radio	3	0	0	3
11.	PE	ECCX45	GNU radio realization through python and C++	1	0	2	2
12.	PE	ECCX46	Implementation of GSM 2G, 3G stack	1	0	2	2
13.	PE	ECCX47	IoT for Industrial Applications	1	0	2	2
14.	PE	ECCX48	Television Engineering	1	0	2	2
15.	PE	ECCX49	Mobile Networks	1	0	0	1

16.	PE	ECCX50	Optical networks	1	0	0	1
17.	PE	ECCX51	Adaptive signal processing	1	0	0	1
18.	PE	ECCX52	Study on Network Simulators	1	0	0	1
19.	PE	ECCX53	MIMO OFDM Technology	1	0	0	1

SIGNAL PROCESSING STREAM

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	ECCX01	Digital System Design	3	0	0	3
2.	PE	ECCX02	Computer Architecture	3	0	0	3
3.	PE	ECCX03	Control Systems	3	0	0	3
4.	PE	ECCX04	Biomedical Electronics	3	0	0	3
5.	PE	ECCX05	Image Processing	2	0	2	3
6.	PE	ECCX07	PCB Design	1	0	2	2
7.	PE	ECCX08	Data structure and its algorithms	1	0	2	2
8.	PE	ECCX09	Java Programming	1	0	0	1
9.	PE	ECCX10	MATLAB Programming	1	0	0	1

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	ECCX12	Biomedical Signal Processing	3	0	0	3
2.	PE	ECCX14	Artificial Intelligence	3	0	0	3
3.	PE	ECCX16	Multimedia Communication Systems	3	0	0	3
4.	PE	ECCX17	Principles of Wireless Sensor network	3	0	0	3
5.	PE	ECCX20	Introduction to Network Security	3	0	0	3

6.	PE	ECCX28	DSP Architecture and Programming	3	0	0	3
7.	PE	ECCX21	Information coding Techniques	3	0	0	3
8.	PE	ECCX27	RF Test and measurement	1	0	0	1
9.	PE	ECCX22	Python Programming	1	0	0	1
10.	PE	ECCX23	Design using EDA Tools	0	0	2	1
11.	PE	ECCX24	Predictive analytics	1	0	0	1
12.	PE	ECCX26	Simulation of RF Circuits and Components	0	0	2	1
13.	PE	ECCX29	DSP applications	0	0	2	1

SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	ECCX34	VLSI Signal Processing	3	0	0	3
2.	PE	ECCX36	Introduction to Internet of Things	3	0	0	3
3.	PE	ECCX44	Software defined radio	3	0	0	3
4.	PE	ECCX54	Medical Image Processing	3	0	0	3
5.	PE	ECCX55	Computer Vision and its Applications	3	0	0	3
6.	PE	ECCX56	Signal Detection and Estimation	3	0	0	3
7.	PE	ECCX45	GNU radio realization through python and C++	1	0	2	2
8.	PE	ECCX46	Implementation of GSM (2G), 3G stack	1	0	2	2
9.	PE	ECCX47	IoT for Industrial Applications	1	0	2	2
10.	PE	ECCX48	Television Engineering	1	0	2	2
11.	PE	ECCX57	Video Processing Techniques	2	0	0	2
12.	PE	ECCX49	Mobile Networks	1	0	0	1
13.	PE	ECCX50	Optical networks	1	0	0	1
14.	PE	ECCX51	Adaptive signal processing	1	0	0	1

15.	PE	ECCX52	Study on Network Simulators	1	0	0	1
16.	PE	ECCX53	MIMO OFDM Technology	1	0	0	1

VLSI & EMBEDDED SYSTEM STREAM

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	ECCX01	Digital System Design	3	0	0	3
2.	PE	ECCX02	Computer Architecture	3	0	0	3
3.	PE	ECCX03	Control Systems	3	0	0	3
4.	PE	ECCX04	Biomedical Electronics	3	0	0	3
5.	PE	ECCX05	Image Processing	2	0	2	3
6.	PE	ECCX06	Advanced Microprocessor and Microcontrollers	3	0	0	3
7.	PE	ECCX11	Introduction to ASIC Design	3	0	0	3
8.	PE	ECCX07	PCB Design	1	0	2	2
9.	PE	ECCX08	Data structure and its algorithms	1	0	2	2
10.	PE	ECCX09	Java Programming	1	0	0	1
11.	PE	ECCX10	MATLAB Programming	1	0	0	1

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	ECCX14	Artificial Intelligence	3	0	0	3
2.	PE	ECCX17	Principles of Wireless Sensor Network	3	0	0	3
3.	PE	ECCX18	Fundamentals of Nano technology	3	0	0	3
4.	PE	ECCX19	Embedded Signal Processing	3	0	0	3
5.	PE	ECCX20	Introduction to Network Security	3	0	0	3

6.	PE	ECCX28	DSP Architecture and Programming	3	0	0	3
7.	PE	ECCX30	Introduction to Reconfigurable Computing	3	0	0	3
8.	PE	ECCX31	Programming in Embedded systems	3	0	0	3
9.	PE	ECCX32	Multicore Architecture and Parallel Programming	3	0	0	3
10.	PE	ECCX21	Information coding Techniques	3	0	0	3
11.	PE	ECCX22	Python Programming	1	0	0	1
12.	PE	ECCX23	Design using EDA Tools	0	0	2	1
13.	PE	ECCX25	IC technology	3	0	0	3
14.	PE	ECCX26	Simulation of RF Circuits and Components	0	0	2	1

SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	ECCX33	Robotics	3	0	0	3
2.	PE	ECCX34	VLSI Signal Processing	3	0	0	3
3.	PE	ECCX36	Introduction to Internet of Things	3	0	0	3
4.	PE	ECCX58	Digital VLSI Testing	3	0	0	3
5.	PE	ECCX59	Fin FET Technology	3	0	0	3
6.	PE	ECCX60	Real Time Embedded Systems	3	0	0	3
7.	PE	ECCX61	Introduction to RTOS	3	0	0	3
8.	PE	ECCX62	Mechatronics	3	0	0	3
9.	PE	ECCX45	GNU radio realization through python and C++	1	0	2	2
10.	PE	ECCX46	Implementation of GSM(2G), 3G stack	1	0	2	2
11.	PE	ECCX47	IoT for Industrial Applications	1	0	2	2
12.	PE	ECCX48	Television Engineering	1	0	2	2

13.	PE	ECCX57	Video Processing Techniques	2	0	0	2
14.	PE	ECCX49	Mobile Networks	1	0	0	1
15.	PE	ECCX50	Optical networks	1	0	0	1
16.	PE	ECCX52	Study on Network Simulators	1	0	0	1

Physics Elective Courses
(to be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
28.	PHCX 01	Fundamentals of Engineering Materials	2	0	2	3
29.	PHCX 02	Heat and Thermodynamics	2	0	2	3
30.	PHCX 03	Introduction to Nanoscience and Technology	2	0	2	3
31.	PHCX 04	Lasers and their applications	2	0	2	3
32.	PHCX 05	Materials Science	2	0	2	3
33.	PHCX 06	Non-Destructive Testing	2	0	2	3
34.	PHCX 07	Properties of Matter and Acoustics	2	0	2	3
35.	PHCX 08	Properties of Matter and Nondestructive Testing	2	0	2	3
36.	PHCX 09	Semiconductor Physics and Optoelectronics	2	0	2	3

Chemistry Elective Courses
(to be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
28.	CHCX01	Analytical Instrumentation	2	0	2	3
29.	CHCX02	Corrosion and its Control	2	0	2	3
30.	CHCX03	Electrical Materials and Batteries	2	0	2	3
31.	CHCX04	Engineering Materials	2	0	2	3
32.	CHCX05	Fuels and Combustion	2	0	2	3
33.	CHCX06	Fundamentals of Physical Chemistry	2	0	2	3
34.	CHCX07	Green Technology	2	0	2	3
35.	CHCX08	Organic Chemistry of Biomolecules	2	0	2	3
36.	CHCX09	Polymer Science and Technology	2	0	2	3

Maths Elective Courses
(to be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
13.	MACX 01	Discrete Mathematics And Graph Theory	3	1	0	4
14.	MACX 02	Probability And Statistics	3	1	0	4
15.	MACX 03	Random Processes	3	1	0	4
16.	MACX 04	Applied Numerical Methods	3	1	0	4

Maths Elective Courses
(to be offered in VI Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
16.	MACX 05	Mathematical Programming	2	0	0	2
17.	MACX 06	Statistical Methods for Data Analysis	2	0	0	2
18.	MACX 07	Numerical Methods for Integral and Differential Equations	2	0	0	2
19.	MACX 08	Mathematical Modelling	2	0	0	2
20.	MACX 09	Graph Theory	2	0	0	2

Humanities Elective I
(to be offered in III Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
10.	SSCX01	Fundamentals of Economics	2	0	0	2
11.	SSCX02	Principles of Sociology	2	0	0	2
12.	SSCX03	Sociology of Indian Society	2	0	0	2

Humanities Elective II
(to be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
10.	SSCX04	Economics of Sustainable Development	2	0	0	2
11.	SSCX05	Industrial Sociology	2	0	0	2
12.	SSCX06	Law for Engineers	2	0	0	2

General Elective
Group I Courses
(To be offered in V semester)

Sl. No.	Course Code	Course Title	Offering Department
52.	GECX101	Disaster Management	Civil
53.	GECX102	Total Quality Management	Mechanical
54.	GECX103	Energy Studies	Mechanical
55.	GECX104	Robotics	Mechanical
56.	GECX105	Transport Management	Automobile
57.	GECX106	Control Systems	EEE
58.	GECX107	Introduction to VLSI Design	ECE
59.	GECX108	Plant Engineering	EIE
60.	GECX109	Network Security	CSE
61.	GECX110	Knowledge management	CSE
62.	GECX111	Cyber security	IT
63.	GECX112	Genetic Engineering	LS
64.	GECX113	Fundamentals of Project Management	CBS
65.	GECX114	Operations Research	Mathematics
66.	GECX115	Nano Technology	Physics / Chemistry
67.	GECX116	Vehicle Maintenance	Automobile
68.	GECX117	Fundamentals of Digital Image Processing	ECE

Group II Courses
(To be offered in VII semester)

Sl. No.	Course Code	Course Title	Offering Department
37.	GECX201	Green Design and Sustainability	Civil
38.	GECX202	Appropriate Technology	Civil / Mechanical
39.	GECX203	Engineering System Modelling and Simulation	Mechanical
40.	GECX204	Value Analysis and Engineering	Mechanical
41.	GECX205	Industrial Safety	Mechanical
42.	GECX206	Advanced Optimization Techniques	Mechanical
43.	GECX207	Mat Lab Simulation	EEE
44.	GECX208	Embedded Systems and its Applications	ECE
45.	GECX209	Usability Engineering	CSE
46.	GECX210	Supply Chain Management	CBS
47.	GECX211	System Analysis and Design	CA
48.	GECX212	Advanced Materials	Physics & Chemistry
49.	GECX213	National Service Scheme	School of Humanities
50.	GECX214	Automotive Pollution and Control	Automobile
51.	GECX215	Motor Vehicle Act, Insurance and Policy	Automobile
52.	GECX216	Principles of Communication Systems	ECE
53.	GECX217	Lean Management	Civil
54.	GECX218	Spatial Data Modeling & Analysis	Civil

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE AND TECHNOLOGY**B.TECH. COMPUTER SCIENCE ENGINEERING****SEMESTER I**

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC 1181	Differential Calculus and Geometry	3	1	0	4	
2.	HS	ENC 1181/ ISC 1181/ LNC 1181/ LNC 1182 / LNC 1183	English / Arabic / Mandarin / German / Japanese	3	0	0	3	
3.	BS	PHC 1181	Physics	3	0	2	4	
4.	BS	CHC 1181	Chemistry	3	0	2	4	
5.	ESF	GEC 1101	Engineering Graphics	2	0	2	3	
6.	ESF	GEC 1102	Engineering Design	2	0	0	2	
7.	ESF	GEC 1103	Basic Engineering Practices Laboratory	0	0	2	1	
8.	ESF	GEC 1104	Computer Programming I	1	0	2	2	23

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC 1281	Advanced Calculus	3	1	0	4	
2.	BS	-	Physics Elective	2	0	2	3	
3.	BS	-	Chemistry Elective	2	0	2	3	
4.	ESF	GEC 1211	Basic Engineering Mechanics	3	1	0	4	
5.	BS	GEC 1212	Environmental Studies	2	0	0	2	
6.	ESF	GEC 1213	Computer Programming II	1	0	2	2	
7.	EC	CSC 1211	Digital System	3	0	2	4	
8.	EC	CSC 1212	Python Programming	2	0	0	2	24

SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	MAC 2181	Partial Differential Equations and Transforms	3	1	0	4
2.	HS	-	Humanities Elective I	2	0	0	2
3.	HS	ENC 2181	Oral Communication	0	0	2	1
4.	EC	CSC 2101	Data Structures	3	1	0	4
5.	EC	CSC 2102	Computer Networks	3	0	2	4
6.	EC	CSC 2103	Software Engineering	3	0	0	3
7.	EC	CSC 2104	Object Oriented Programming	3	0	0	3
8.	EC	CSC 2105	Data structures Lab	0	0	2	1
9.	EC	CSC 2106	Object Oriented Programming Lab	0	0	2	1
							23

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	-	Mathematics Elective I	3	1	0	4
2.	HS	-	Humanities Elective II	2	0	0	2
3.	HS	ENC 2282	Written Communication	0	0	2	1
4.	EC	CSC 2211	Computer architecture and microprocessor	3	0	0	3
5.	EC	CSC 2212	Analysis of Algorithms	3	1	0	4
6.	EC	CSC 2213	Network Security and Cryptography	3	0	0	3
7.	EC	CSC 2214	Database Management Systems	2	0	0	2
8.	EC	CSC 2215	DBMS Lab	0	0	2	1
9.	EC	CSC 2216	Security Lab	0	0	2	1
10.	PE		Programme Elective ^{##1}				3* 24

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181 MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	GE	-	General Elective I	3	0	0	3
3.	HS	ENC3181	Communication & soft skill – I Career Choice	0	0	2	1
4.	EC	CSC 3101	Operating Systems	3	0	0	3
5.	EC	CSC 3102	Web Development using JAVA	3	0	0	3
6.	EC	CSC 3103	Artificial Intelligence and Machine Learning	3	0	0	3
7.	EC	CSC 3104	Web Development Lab	0	0	2	1
8.	PE		Programme Elective ##2				6** 23

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	EF	MSC 3181 MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	BS	-	Mathematics Elective II	2	0	0	2
3.	HS	ENC3281	Communication and soft skill – II Confidence Building	0	0	2	1
4.	EC	CSC 3211	Big Data Analytics	3	0	0	3
5.	EC	CSC 3212	Distributed Computing	3	0	0	3
6.	EC	CSC 3213	Graph Theory and Application	3	1	0	4
7.	EC	CSC 3214	Big Data Analytics Tools Lab	0	0	2	1
8.	EC	CSC 3215	Mobile Application Development Lab	0	0	2	1
9.	PE	-	Programme Elective ##3				6** 24

SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	GE	-	General Elective II	3	0	0	3
2.	EC	CSC 4101	Software project management	3	0	0	3
3.	EC	CSC 4102	Compiler Design	3	0	0	3
4.	EC	CSC 4103	Cloud Computing	3	0	2	4
5.	EC	CSC 4104	Compiler Lab	0	0	2	1
6.	EC	CSC 4105	Internship	0	0	0	1*
7.	PE	-	Programme Elective ^{###4}				9**
							24

SEMESTER VIII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	EC	CSC 4211	Project Work	0	0	24	12
							12

Total credits – 177

* Industrial training will be undertaken during Third year summer vacation for 15 days. The credit will be awarded in the 7th Semester.

ELECTIVE LIST

SEMESTER IV

Programme Elective ^{##1} (3 Credits*)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
43.	PE	CSC X201	Multimedia and Animation	3	0	0	3
44.	PE	CSC X202	VFX	2	0	0	2
45.	PE	CSC X203	Digital Transmission	3	0	0	3
46.	PE	CSC X204	Information Technology in Organization	3	0	0	3
47.	PE	CSC X205	Innovation and New product development	3	0	0	3
48.	PE	CSC X206	Fundamentals of Computer Forensics	2	0	0	2
49.	PE	CSC X207	Open Source Technologies	2	0	2	3
50.	PE	CSC X208	User Interface Design	2	0	0	2
51.	PE	CSC X209	Stress Management	1	0	0	1
52.	PE	CSC X210	Information Ethics	1	0	0	1

SEMESTER V

Programme Elective ^{##2} (3+3 Credits*)

Elective 1 (3 Credits*)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	CSC X101	2D Character Animation	2	0	0	2
2.	PE	CSC X102	Image processing	2	0	2	3
3.	PE	CSC X103	Multimedia Application Development	2	0	0	2
4.	PE	CSC X104	3D Animation	2	0	2	3
5.	PE	CSC X105	Advanced Computer Graphics	2	0	2	3

6.	PE	CSC X106	Multimedia Design program	2	0	2	3
7.	PE	CSC X107	Network Modeling	3	0	0	3
8.	PE	CSC X108	Cellular and Wireless Networks	3	0	0	3
9.	PE	CSC X109	Network Management	2	0	0	2
10.	PE	CSC X110	Network Simulators	1	0	0	1
11.	PE	CSC X111	Agile practices	1	0	0	1
12.	PE	CSC X112	Software testing	2	0	0	2
13.	PE	CSC X113	Software Requirements and Modeling	2	0	0	2
14.	PE	CSC X114	Theory of Computation	3	0	0	3
15.	PE	CSC X115	Internet of Things	2	0	0	2
16.	PE	CSC X116	Cognitive Science	2	0	0	2
17.	PE	CSC X117	Virtualization Techniques	3	0	0	3

Elective 2 (3 Credits*)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	CSC X122	Foundation on Robotics	2	0	0	2
2.	PE	CSC X123	Sound Editing and processing	2	0	2	3
3.	PE	CSC X124	Computer Vision	3	0	0	3
4.	PE	CSC X125	Mobile Multimedia Systems	2	0	0	2
5.	PE	CSC X126	Scripting Languages	2	0	2	3
6.	PE	CSC X127	Online video production	1	0	0	1
7.	PE	CSC X128	Multimedia security	2	0	0	2
8.	PE	CSC X129	Multimedia Communication and Networking	3	0	0	3
9.	PE	CSC X130	Gaming Technology	3	0	2	3

10.	PE	CSC X131	Routing Protocols	3	0	0	3
11.	PE	CSC X132	Queuing Theory	3	0	0	3
12.	PE	CSC X133	Network Trouble shooting tools	1	0	0	1
13.	PE	CSC X134	Design of Computer Network services	3	0	0	3
14.	PE	CSC X135	Software architecture for the internet of things	1	0	0	1
15.	PE	CSC X136	Software metrics	2	0	0	2
16.	PE	CSC X137	Multicore Architecture	2	1	0	3
17.	PE	CSC X138	Statistics and analytics using R programming	2	0	2	3
18.	PE	CSC X139	C# & .NET	2	0	0	2

SEMESTER VI

Programme Elective ^{##3}(3+3)

Elective 1 (3 Credits*)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	CSC X214	Information Retrieval	2	0	2	3
2.	PE	CSC X215	Software Risk Management	1	0	0	1
3.	PE	CSC X216	Social Media Security	2	0	0	2
4.	PE	CSC X217	Web Security	2	0	0	2
5.	PE	CSC X218	Software design and architecture	1	0	0	1
6.	PE	CSC X219	Software Configuration Management	2	0	0	2
7.	PE	CSC X220	Human Computer Interaction	2	0	0	2
8.	PE	CSC X221	Green Computing	3	0	0	3
9.	PE	CSC X222	XML and Web services	2	0	0	2
10.	PE	CSC X223	Business Process Management	3	0	0	3
11.	PE	CSC X224	Advanced Machine Learning	3	0	0	3

Elective 2 (3 Credits*)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	CSC X228	Information Visualization	2	1	0	3
2.	PE	CSC X229	Cyber Forensics	3	0	0	3
3.	PE	CSC X230	Smart Devices Security	1	0	0	1
4.	PE	CSC X231	Biometric Security	3	0	0	3
5.	PE	CSC X232	Database Security	3	0	0	3
6.	PE	CSC X233	Software Quality Assurance	3	0	0	3
7.	PE	CSC X234	Customer Relationship Management	1	0	0	1
8.	PE	CSC X235	Service Oriented Architecture	2	0	0	2
9.	PE	CSC X236	Genetic Algorithm	3	0	0	3
10.	PE	CSC X237	Enterprise Resource Planning	2	1	0	3
11.	PE	CSC X238	API Design	2	0	0	2

SEMESTER VII**Programme Elective ##4 (3+3+3)****Elective 1 (3 Credits*)**

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	CSC X143	Deep Learning	2	0	2	3
2.	PE	CSC X144	Intrusion Detection and Prevention	3	0	0	3
3.	PE	CSC X145	Security Law and Compliance	2	0	0	2
4.	PE	CSC X146	Software Design patterns	1	0	0	1
5.	PE	CSC X147	Writing skills for engineering leaders	1	0	0	1
6.	PE	CSC X148	Software Reliability	2	0	0	2

7.	PE	CSC X149	AAIP – Animation with Portfolio Development	2	0	2	3
8.	PE	CSC X150	Advanced SAS: Macros & SQL	2	0	0	2
9.	PE	CSC X151	Advanced Programming in Data Science with Python	3	0	0	3

Elective 2 (3 Credits*)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	CSC X155	Security Issues in Cloud Computing	3	0	0	3
2.	PE	CSC X156	Security Evaluation and Assessment Methodology	3	0	0	3
3.	PE	CSC X157	Software process and product quality	1	0	0	1
4.	PE	CSC X158	System Integration	2	0	0	2
5.	PE	CSC X159	IBM Websphere	2	0	2	3
6.	PE	CSC X160	AVIP - Broadcast	1	0	0	1
7.	PE	CSC X161	Full Stack mobile application development I (Front end)	2	0	2	3
8.	PE	CSC X162	Advanced Game Design	2	0	0	2

Elective 3 (3 Credits)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	CSC X166	Web Application Security	2	0	0	2
2.	PE	CSC X167	Cyber Laws and Ethics	2	0	0	2
3.	PE	CSC X168	Secure Interconnecting systems	3	0	0	3
4.	PE	CSC X169	Software Maintenance	2	0	0	2
5.	PE	CSC X170	Personal Software Process	1	0	0	1
6.	PE	CSC X171	Web analytics and Social media mining	3	0	0	3
7.	PE	CSC X172	Ruby	1	0	0	1

8.	PE	CSC X173	Full Stack mobile application development II (Back end)	2	0	2	3
9.	PE	CSC X174	5G Wireless communication Techniques	3	0	0	3

*** - Student has to take either 3 Credit or 2+1 Credits**

**** - Sum of the credits of the electives**

Physics Elective Courses
(To be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
37.	PHCX 01	Fundamentals of Engineering Materials	2	0	2	3
38.	PHCX 02	Heat and Thermodynamics	2	0	2	3
39.	PHCX 03	Introduction to Nanoscience and Technology	2	0	2	3
40.	PHCX 04	Lasers and their applications	2	0	2	3
41.	PHCX 05	Materials Science	2	0	2	3
42.	PHCX 06	Non-Destructive Testing	2	0	2	3
43.	PHCX 07	Properties of Matter and Acoustics	2	0	2	3
44.	PHCX 08	Properties of Matter and Nondestructive Testing	2	0	2	3
45.	PHCX 09	Semiconductor Physics and Optoelectronics	2	0	2	3

Chemistry Elective Courses
(To be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
37.	CHCX01	Analytical Instrumentation	2	0	2	3
38.	CHCX02	Corrosion and its Control	2	0	2	3
39.	CHCX03	Electrical Materials and Batteries	2	0	2	3
40.	CHCX04	Engineering Materials	2	0	2	3
41.	CHCX05	Fuels and Combustion	2	0	2	3
42.	CHCX06	Fundamentals of Physical Chemistry	2	0	2	3
43.	CHCX07	Green Technology	2	0	2	3
44.	CHCX08	Organic Chemistry of Biomolecules	2	0	2	3
45.	CHCX09	Polymer Science and Technology	2	0	2	3

Maths Elective Courses
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
17.	MACX 01	Discrete Mathematics And Graph Theory	3	1	0	4
18.	MACX 02	Probability And Statistics	3	1	0	4
19.	MACX 03	Random Processes	3	1	0	4
20.	MACX 04	Applied Numerical Methods	3	1	0	4

Maths Elective Courses
(To be offered in VI Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
21.	MACX 05	Mathematical Programming	2	0	0	2
22.	MACX 06	Statistical Methods for Data Analysis	2	0	0	2
23.	MACX 07	Numerical Methods for Integral and Differential Equations	2	0	0	2
24.	MACX 08	Mathematical Modelling	2	0	0	2
25.	MACX 09	Graph Theory	2	0	0	2

Humanities Elective I
(To be offered in III Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
13.	SSCX01	Fundamentals of Economics	2	0	0	2
14.	SSCX02	Principles of Sociology	2	0	0	2
15.	SSCX03	Sociology of Indian Society	2	0	0	2

Humanities Elective II
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
13.	SSCX04	Economics of Sustainable Development	2	0	0	2
14.	SSCX05	Industrial Sociology	2	0	0	2
15.	SSCX06	Law for Engineers	2	0	0	2

General Elective
Group I Courses
(To be offered in V semester)

Sl. No.	Course Code	Course Title	Offering Department
69.	GECX101	Disaster Management	Civil
70.	GECX102	Total Quality Management	Mechanical
71.	GECX103	Energy Studies	Mechanical
72.	GECX104	Robotics	Mechanical
73.	GECX105	Transport Management	Automobile
74.	GECX106	Control Systems	EEE
75.	GECX107	Introduction to VLSI Design	ECE
76.	GECX108	Plant Engineering	EIE
77.	GECX109	Network Security	CSE
78.	GECX110	Knowledge management	CSE
79.	GECX111	Cyber security	IT
80.	GECX112	Genetic Engineering	LS
81.	GECX113	Fundamentals of Project Management	CBS
82.	GECX114	Operations Research	Mathematics
83.	GECX115	Nano Technology	Physics / Chemistry
84.	GECX116	Vehicle Maintenance	Automobile
85.	GECX117	Fundamentals of Digital Image Processing	ECE

Group II Courses
(To be offered in VII semester)

Sl. No.	Course Code	Course Title	Offering Department
55.	GECX201	Green Design and Sustainability	Civil
56.	GECX202	Appropriate Technology	Civil / Mechanical
57.	GECX203	Engineering System Modelling and Simulation	Mechanical
58.	GECX204	Value Analysis and Engineering	Mechanical
59.	GECX205	Industrial Safety	Mechanical
60.	GECX206	Advanced Optimization Techniques	Mechanical
61.	GECX207	Mat Lab Simulation	EEE
62.	GECX208	Embedded Systems and its Applications	ECE
63.	GECX209	Usability Engineering	CSE
64.	GECX210	Supply Chain Management	CBS
65.	GECX211	System Analysis and Design	CA
66.	GECX212	Advanced Materials	Physics & Chemistry
67.	GECX213	National Service Scheme	School of Humanities
68.	GECX214	Automotive Pollution and Control	Automobile
69.	GECX215	Motor Vehicle Act, Insurance and Policy	Automobile
70.	GECX216	Principles of Communication Systems	ECE
71.	GECX217	Lean Management	Civil
72.	GECX218	Spatial Data Modeling & Analysis	Civil

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SEMESTER I

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC 1181	Differential Calculus and Geometry	3	1	0	4	
2.	HS	ENC 1181/ ISC 1181/ LNC 1181/ LNC 1182 / LNC 1183	English / Arabic / Mandarin / German / Japanese	3	0	0	3	
3.	BS	PHC 1181	Physics	3	0	2	4	
4.	BS	CHC 1181	Chemistry	3	0	2	4	
5.	ESF	GEC 1101	Engineering Graphics	2	0	2	3	
6.	ESF	GEC 1102	Engineering Design	2	0	0	2	
7.	ESF	GEC 1103	Basic Engineering Practices Laboratory	0	0	2	1	
8.	ESF	GEC 1104	Computer Programming I	1	0	2	2	23

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC 1281	Advanced Calculus	3	1	0	4	
2.	BS	-	Physics Elective	2	0	2	3	
3.	BS	-	Chemistry Elective	2	0	2	3	
4.	ESF	GEC 1211	Basic Engineering Mechanics	3	1	0	4	
5.	BS	GEC 1212	Environmental Studies	2	0	0	2	
6.	ESF	GEC 1213	Computer Programming II	1	0	2	2	
7.	EC	EIC 1211	Electrical Circuits	3	0	2	4	
8.	EC	EIC 1212	Electronic Devices	2	0	0	2	24

SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	MAC2181	Partial Differential Equations and Transforms	3	1	0	4
2.	HS	-	Humanities Elective I	2	0	0	2
3.	HS	ENC 2181	Oral Communication	0	0	2	1
4.	EC	EIC 2101	Transducer Engineering	3	0	0	3
5.	EC	EIC 2102	Electrical, Electronic and Physical Measurements	3	0	0	3
6.	EC	EIC 2103	Analog and Linear Integrated Circuits	3	1	0	4
7.	EC	EIC 2104	Electrical Machines	3	0	0	3
8.	EC	EIC 2105	Transducer Engineering Lab	0	0	2	1
9.	EC	EIC 2106	Electrical Machines and Measurements Lab	0	0	3	1
10.	EC	EIC2107	Analog Electronics Lab	0	0	3	1

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	-	Mathematics Elective I	3	1	0	4
2.	HS	-	Humanities Elective II	2	0	0	2
3.	HS	ENC2282	Written Communication	0	0	2	1
4.	EC	EIC 2211	Thermodynamics and Fluid Mechanics	3	0	2	4
5.	EC	EIC 2212	Digital Electronics	3	1	0	4
6.	EC	EIC 2213	Industrial Instrumentation I	3	0	0	3
7.	EC	EIC 2214	Digital Electronics Lab	0	0	2	1
8.	EC	EIC 2215	Industrial Instrumentation I Lab	0	0	2	1
9.	PE		Programme Elective				3

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181 MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	GE	-	General Elective I	3	0	0	3
3.	HS	ENC3181	Communication and soft skill I Career Choice	0	0	2	1
4.	EC	EIC 3101	Control Systems	3	1	0	4
5.	EC	EIC 3102	Microprocessor and Microcontroller	3	0	0	3
6.	EC	EIC 3103	Industrial Instrumentation II	3	0	0	3
7.	EC	EIC 3104	Control Systems Laboratory	0	0	2	1
8.	EC	EIC 3105	Microprocessor and Microcontroller Laboratory	0	0	2	1
9.	EC	EIC 3106	Industrial Instrumentation II Lab	0	0	2	1
10.	PE		Programme Elective				3
							23

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181 MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	BS	-	Mathematics Elective II	2	0	0	2
3.	HS	ENC3281	Communication and soft skill II Confidence Building	0	0	2	1
4.	EC	EIC 3211	Process control	3	0	0	3
5.	EC	EIC 3212	Communication Engineering	3	0	0	3
6.	EC	EIC 3213	Digital signal processing	3	1	0	4
7.	EC	EIC 3214	Process control Lab	0	0	3	1
8.	PE	-	Programme Elective				6

SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	GE	-	General Elective II	3	0	0	3
2.	EC	EIC4101	Digital Process control	3	0	0	3
3.	EC	EIC4102	Industrial Automation	3	0	0	3
4.	EC	-	Programme Elective				9
5.	PE	EIC4103	Industrial Automation Lab	0	0	4	2
6.	EC	EIC4104	Instrumentation System Design	0	0	4	2
7.	EC	EIC4105	Internship	0	0	0	1* 23

SEMESTER VIII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	EC	EIC4211	Project Work	0	0	24	12 12

Total credits – 174

* Industrial training will be undertaken during Third year summer vacation for 15 days. The credit will be awarded in the 7th Semester.

PROGRAMME ELECTIVES**GROUP I**

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
53.	PE	EICX01	Biomedical Instrumentation	3	0	0	3
54.	PE	EICX02	Fiber Optic and Laser Instrumentation	3	0	0	3
55.	PE	EICX03	Micro Electro Mechanical Systems	3	0	0	3
56.	PE	EICX04	Qt software	1	0	4	3
57.	PE	EICX05	Telemetry and Tele control	3	0	0	3

GROUP II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	EICX11	Control system components	3	0	0	3
2.	PE	EICX12	Advanced Sensors	3	0	0	3
3.	PE	EICX13	Ultrasonic Instrumentation	3	0	0	3
4.	PE	EICX14	Computer networks	3	0	0	3

GROUP III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	EICX21	Industrial Data Networks	3	0	0	3
2.	PE	EICX22	Robotics and Automation	3	0	0	3
3.	PE	EICX23	Virtual Instrumentation	3	0	0	3
4.	PE	EICX24	Applied Analytical Instrumentation	3	0	0	3

GROUP IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	EICX31	Modeling and Simulation	3	0	0	3

2.	PE	EICX32	Digital Image Processing	3	0	0	3
3.	PE	EICX33	Instrumentation and Control in Petrochemical Industries	3	0	0	3
4.	PE	EICX34	Wireless sensor networks	3	0	0	3

GROUP V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	EICX41	Nonlinear Control System	3	0	0	3
2.	PE	EICX42	Advanced Digital Signal Processing	3	0	0	3
3.	PE	EICX43	Instrumentation and control in Iron and Steel Industries	3	0	0	3
4.	PE	EICX44	Internet of Things for automation	3	0	0	3

GROUP VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	EICX51	Modern Control System	3	0	0	3
2.	PE	EICX52	Embedded System and RTOS	3	0	0	3
3.	PE	EICX53	System Identification	3	0	0	3
4.	PE	EICX54	Industrial Drives and Control	3	0	0	3

GROUP VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	EICX61	Adaptive Control	3	0	0	3
2.	PE	EICX62	Thermal Power Plant Instrumentation	3	0	0	3
3.	PE	EICX63	Applied soft computing for Instrumentation Engineers	3	0	0	3
4.	PE	EICX64	VLSI Design	3	0	0	3

Physics Elective Courses
(To be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
46.	PHCX 01	Fundamentals of Engineering Materials	2	0	2	3
47.	PHCX 02	Heat and Thermodynamics	2	0	2	3
48.	PHCX 03	Introduction to Nanoscience and Technology	2	0	2	3
49.	PHCX 04	Lasers and their applications	2	0	2	3
50.	PHCX 05	Materials Science	2	0	2	3
51.	PHCX 06	Non-Destructive Testing	2	0	2	3
52.	PHCX 07	Properties of Matter and Acoustics	2	0	2	3
53.	PHCX 08	Properties of Matter and Nondestructive Testing	2	0	2	3
54.	PHCX 09	Semiconductor Physics and Optoelectronics	2	0	2	3

Chemistry Elective Courses
(To be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
46.	CHCX01	Analytical Instrumentation	2	0	2	3
47.	CHCX02	Corrosion and its Control	2	0	2	3
48.	CHCX03	Electrical Materials and Batteries	2	0	2	3
49.	CHCX04	Engineering Materials	2	0	2	3
50.	CHCX05	Fuels and Combustion	2	0	2	3
51.	CHCX06	Fundamentals of Physical Chemistry	2	0	2	3
52.	CHCX07	Green Technology	2	0	2	3
53.	CHCX08	Organic Chemistry of Biomolecules	2	0	2	3
54.	CHCX09	Polymer Science and Technology	2	0	2	3

Maths Elective Courses
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
21.	MACX 01	Discrete Mathematics And Graph Theory	3	1	0	4
22.	MACX 02	Probability And Statistics	3	1	0	4
23.	MACX 03	Random Processes	3	1	0	4
24.	MACX 04	Applied Numerical Methods	3	1	0	4

Maths Elective Courses
(To be offered in VI Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
26.	MACX 05	Mathematical Programming	2	0	0	2
27.	MACX 06	Statistical Methods for Data Analysis	2	0	0	2
28.	MACX 07	Numerical Methods for Integral and Differential Equations	2	0	0	2
29.	MACX 08	Mathematical Modelling	2	0	0	2
30.	MACX 09	Graph Theory	2	0	0	2

Humanities Elective I
(To be offered in III Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
16.	SSCX01	Fundamentals of Economics	2	0	0	2
17.	SSCX02	Principles of Sociology	2	0	0	2
18.	SSCX03	Sociology of Indian Society	2	0	0	2

Humanities Elective II
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
16.	SSCX04	Economics of Sustainable Development	2	0	0	2
17.	SSCX05	Industrial Sociology	2	0	0	2
18.	SSCX06	Law for Engineers	2	0	0	2

General Elective
Group I Courses
(To be offered in V semester)

Sl. No.	Course Code	Course Title	Offering Department
86.	GECX101	Disaster Management	Civil
87.	GECX102	Total Quality Management	Mechanical
88.	GECX103	Energy Studies	Mechanical
89.	GECX104	Robotics	Mechanical
90.	GECX105	Transport Management	Automobile
91.	GECX106	Control Systems	EEE
92.	GECX107	Introduction to VLSI Design	ECE
93.	GECX108	Plant Engineering	EIE
94.	GECX109	Network Security	CSE
95.	GECX110	Knowledge management	CSE
96.	GECX111	Cyber security	IT
97.	GECX112	Genetic Engineering	LS
98.	GECX113	Fundamentals of Project Management	CBS
99.	GECX114	Operations Research	Mathematics
100.	GECX115	Nano Technology	Physics / Chemistry
101.	GECX116	Vehicle Maintenance	Automobile
102.	GECX117	Fundamentals of Digital Image Processing	ECE

Group II Courses
(To be offered in VII semester)

Sl. No.	Course Code	Course Title	Offering Department
73.	GECX201	Green Design and Sustainability	Civil
74.	GECX202	Appropriate Technology	Civil / Mechanical
75.	GECX203	Engineering System Modelling and Simulation	Mechanical
76.	GECX204	Value Analysis and Engineering	Mechanical
77.	GECX205	Industrial Safety	Mechanical
78.	GECX206	Advanced Optimization Techniques	Mechanical
79.	GECX207	Mat Lab Simulation	EEE
80.	GECX208	Embedded Systems and its Applications	ECE
81.	GECX209	Usability Engineering	CSE
82.	GECX210	Supply Chain Management	CBS
83.	GECX211	System Analysis and Design	CA
84.	GECX212	Advanced Materials	Physics & Chemistry
85.	GECX213	National Service Scheme	School of Humanities
86.	GECX214	Automotive Pollution and Control	Automobile
87.	GECX215	Motor Vehicle Act, Insurance and Policy	Automobile
88.	GECX216	Principles of Communication Systems	ECE
89.	GECX217	Lean Management	Civil
90.	GECX218	Spatial Data Modeling & Analysis	Civil

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE AND TECHNOLOGY**B.TECH. POLYMER ENGINEERING****SEMESTER I**

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	MAC 1181	Differential Calculus and Geometry	3	1	0	4
2.	HS	ENC 1181/ ISC 1181/ LNC 1181/ LNC 1182 / LNC 1183	English / Arabic / Mandarin / German / Japanese	3	0	0	3
3.	BS	PHC 1181	Physics	3	0	2	4
4.	BS	CHC 1181	Chemistry	3	0	2	4
5.	ESF	GEC 1101	Engineering Graphics	2	0	2	3
6.	ESF	GEC 1102	Engineering Design	2	0	0	2
7.	ESF	GEC 1103	Basic Engineering Practices Laboratory	0	0	2	1
8.	ESF	GEC 1104	Computer Programming I	1	0	2	2
							23

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	MAC 1281	Advanced Calculus	3	1	0	4
2.	BS	-	Physics Elective	2	0	2	3
3.	BS	-	Chemistry Elective	2	0	2	3
4.	ESF	GEC 1211	Basic Engineering Mechanics	3	1	0	4
5.	BS	GEC 1212	Environmental Studies	2	0	0	2
6.	ESF	GEC 1213	Computer Programming II	1	0	2	2
7.	EC	PEC 1211	Basic Mechanical Operations	2	0	0	2
8.	EC	PEC 1212	Principles of Chemical Engineering	3	0	0	3

9.	EC	PEC 1213	Chemical Engineering Lab	0	0	2	1	24
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SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC2181	Partial Differential Equations and Transforms	3	1	0	4	
2.	HS	-	Humanities Elective I	2	0	0	2	
3.	HS	ENC 2181	Oral Communication	0	0	2	1	
4.	EC	PEC 2101	Chemistry of Macromolecules	3	0	0	3	
5.	EC	PEC 2102	Physics of Macromolecules	3	0	0	3	
6.	EC	PEC 2103	Plastic Materials Technology	3	0	0	3	
7.	ESF	EEC 2181	Introduction to Electrical and Electronics Engineering	3	1	0	4	
8.	EC	PEC 2104	Polymer Synthesis Lab	0	0	2	1	
9.	ESF	EEC 2182	Electrical and Electronics Lab	0	0	2	1	
10.	EC	PEC 2105	Machining Practice Lab	0	0	2	1	23

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	-	Mathematics Elective I	3	1	0	4	
2.	HS	-	Humanities Elective II	2	0	0	2	
3.	HS	ENC2282	Written Communication	0	0	2	1	
4.	EC	PEC 2211	Polymer Rheology	3	0	0	3	
5.	EC	PEC 2212	Science and Technology of Rubbers	3	0	0	3	
6.	EC	PEC 2213	Plastics Compounding Technology	3	0	0	3	
7.	EC	PEC 2214	Polymer Analysis and Characterization	3	0	0	3	
8.	EC	-	Programme Elective –I	3	0	0	3	
9.	PE	PEC 2215	Polymer Characterization Lab	0	0	2	1	23

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181 MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	GE	-	General Elective I	3	0	0	3
3.	HS	ENC3181	Communication and soft skill I Career Choice	0	0	2	1
4.	EC	PEC 3101	Plastics Process Engineering	3	0	0	3
5.	EC	PEC 3102	Strength of Materials	3	0	0	3
6.	EC	PEC 3103	Plastic and Rubber Testing Technology	3	0	0	3
7.	PE	-	Programme Elective – II	1	0	0	1
8.	PE	-	Programme Elective – III	2	0	0	2
9.	PE	-	Programme Elective – IV	3	0	0	3
10.	EC	PEC 3104	Plastics Processing Lab	0	0	2	1
11.	EC	PEC 3105	Rubber Processing Lab	0	0	2	1
							24

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181 MSC 3182	Leadership and CEO Training / Social Entrepreneurship	3	0	0	3
2.	BS	-	Mathematics Elective II	2	0	0	2
3.	HS	ENC3281	Communication & soft skill II Confidence Building	0	0	2	1
4.	EC	PEC 3211	Plastic and Rubber Product Design	3	1	0	4
5.	EC	PEC 3212	Process Control and Instrumentation	3	0	0	3
6.	EC	PEC 3213	Polymer Reaction Engineering	3	0	0	3
7.	PE	-	Programme Elective – V	1	0	0	1

8.	PE	-	Programme Elective – VI	2	0	0	2	
9.	PE	-	Programme Elective – VII	3	0	0	3	
10.	EC	PEC 3214	Plastic Product Design using CAD	0	0	2	1	
11.	EC	PEC 3215	Polymer Testing Lab	0	0	2	1	24

SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	GE	-	General Elective II	3	0	0	3	
2.	EC	PEC 4101	Mould and Die Design	3	1	0	4	
3.	EC	PEC 4102	Polymer Composite Engineering	3	0	0	3	
4.	EC	PEC 4103	Polymer Nanocomposites	3	0	0	3	
5.	PE	-	Programme Elective –VIII	1	0	0	1	
6.	PE	-	Programme Elective – IX	2	0	0	2	
7.	PE	-	Programme Elective –X	3	0	0	3	
8.	PE	-	Programme Elective –XI	3	0	0	3	
9.	EC	PEC 4104	Mould Design and Flow Simulation lab	0	0	2	1	
10.	EC		Internship				1*	24

SEMESTER VIII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	EC	PEC 4211	Project Work	0	0	24	12	12

Total credits – 177

* Industrial training will be undertaken during Third year summer vacation. The credit will be awarded in the 7th Semester.

PROGRAMME ELECTIVES
GROUP I (Materials)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
58.	PE	PECX 001	Thermoplastic Polyesters	1	0	0	1
59.	PE	PECX 002	Thermoplastic Elastomers	1	0	0	1
60.	PE	PECX 003	Electroactive Polymers	1	0	0	1
61.	PE	PECX 004	Heat Resistant Polymers	1	0	0	1
62.	PE	PECX 005	Biodegradable Plastics	2	0	0	2
63.	PE	PECX 006	Bioplastics Technology	2	0	0	2
64.	PE	PECX 007	Biomedical Polymers	2	0	0	2
65.	PE	PECX 008	High Performance Polymers	2	0	0	2
66.	PE	PECX 009	Nanotechnology	3	0	0	3
67.	PE	PECX 010	Nanomaterials Technology	3	0	0	3
68.	PE	PECX 011	Polymers for Electronics	1	0	0	1
69.	PE	PECX 012	Polymers for Energy Technology	2	0	0	2

GROUP II (Process Engineering)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
5.	PE	PECX 013	PVC Technology	1	0	0	1
6.	PE	PECX 014	Nylon Technology	1	0	0	1
7.	PE	PECX 015	Latex Technology	1	0	0	1
8.	PE	PECX 016	Thermoforming Process	1	0	0	1
9.	PE	PECX 017	Injection Moulding Technology	2	0	0	2
10.	PE	PECX 018	Extrusion Moulding Technology	2	0	0	2

11.	PE	PECX 019	Blow Moulding Technology	2	0	0	2
12.	PE	PECX 020	Post Processing Operations	1	0	0	1
13.	PE	PECX 021	Rubber Product Manufacturing Technology	3	0	0	3
14.	PE	PECX 022	Rubber Process Engineering	3	0	0	3
15.	PE	PECX 023	Tyre Manufacturing Technology	3	0	0	3
16.	PE	PECX 024	Plastics Recycling	2	0	0	2
17.	PE	PECX 025	Plastic Waste Management	1	0	0	1

GROUP III (Product & Mould Design)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
5.	PE	PECX 026	Computer Aided Modelling	2	0	0	2
6.	PE	PECX 027	Computer Aided Manufacturing	2	0	0	2
7.	PE	PECX 028	Design of Composite Structures	2	0	0	2
8.	PE	PECX 029	Industrial Hydraulics and Pneumatics	2	0	0	2
9.	PE	PECX 030	Rapid Prototyping - 3D Printing	2	0	0	2
10.	PE	PECX 031	Failure Analysis of Polymers	2	0	0	2
11.	PE	PECX 032	Mould Manufacturing Techniques	2	0	0	2

GROUP IV (Blends, Composites, Adhesives and Coatings)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
5.	PE	PECX 033	Biocomposite Technology	2	0	0	2
6.	PE	PECX 034	Mechanics of Composites	2	0	0	2
7.	PE	PECX 035	Analysis of Composite Structures	2	0	0	2

8.	PE	PECX 036	Polymer Blends and Alloys	2	0	0	2
9.	PE	PECX 037	Paint Technology	1	0	0	1
10.	PE	PECX 038	Adhesives Technology	1	0	0	1
11.	PE	PECX 039	Surface Coating Technology	2	0	0	2
12.	PE	PECX 040	Plastic Packaging Technology	3	0	0	3
13.	PE	PECX 033	Biocomposite Technology	2	0	0	2

Physics Elective Courses
(To be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
55.	PHCX 01	Fundamentals of Engineering Materials	2	0	2	3
56.	PHCX 02	Heat and Thermodynamics	2	0	2	3
57.	PHCX 03	Introduction to Nanoscience and Technology	2	0	2	3
58.	PHCX 04	Lasers and their applications	2	0	2	3
59.	PHCX 05	Materials Science	2	0	2	3
60.	PHCX 06	Non-Destructive Testing	2	0	2	3
61.	PHCX 07	Properties of Matter and Acoustics	2	0	2	3
62.	PHCX 08	Properties of Matter and Nondestructive Testing	2	0	2	3
63.	PHCX 09	Semiconductor Physics and Optoelectronics	2	0	2	3

Chemistry Elective Courses
(To be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
55.	CHCX01	Analytical Instrumentation	2	0	2	3
56.	CHCX02	Corrosion and its Control	2	0	2	3
57.	CHCX03	Electrical Materials and Batteries	2	0	2	3
58.	CHCX04	Engineering Materials	2	0	2	3
59.	CHCX05	Fuels and Combustion	2	0	2	3
60.	CHCX06	Fundamentals of Physical Chemistry	2	0	2	3
61.	CHCX07	Green Technology	2	0	2	3
62.	CHCX08	Organic Chemistry of Biomolecules	2	0	2	3
63.	CHCX09	Polymer Science and Technology	2	0	2	3

Maths Elective Courses
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
25.	MACX 01	Discrete Mathematics And Graph Theory	3	1	0	4
26.	MACX 02	Probability And Statistics	3	1	0	4
27.	MACX 03	Random Processes	3	1	0	4
28.	MACX 04	Applied Numerical Methods	3	1	0	4

Maths Elective Courses
(To be offered in VI Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
31.	MACX 05	Mathematical Programming	2	0	0	2
32.	MACX 06	Statistical Methods for Data Analysis	2	0	0	2
33.	MACX 07	Numerical Methods for Integral and Differential Equations	2	0	0	2
34.	MACX 08	Mathematical Modelling	2	0	0	2
35.	MACX 09	Graph Theory	2	0	0	2

Humanities Elective I
(To be offered in III Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
19.	SSCX01	Fundamentals of Economics	2	0	0	2
20.	SSCX02	Principles of Sociology	2	0	0	2
21.	SSCX03	Sociology of Indian Society	2	0	0	2

Humanities Elective II
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
19.	SSCX04	Economics of Sustainable Development	2	0	0	2
20.	SSCX05	Industrial Sociology	2	0	0	2
21.	SSCX06	Law for Engineers	2	0	0	2

General Elective
Group I Courses
(To be offered in V semester)

Sl. No.	Course Code	Course Title	Offering Department
103.	GECX101	Disaster Management	Civil
104.	GECX102	Total Quality Management	Mechanical
105.	GECX103	Energy Studies	Mechanical
106.	GECX104	Robotics	Mechanical
107.	GECX105	Transport Management	Automobile
108.	GECX106	Control Systems	EEE
109.	GECX107	Introduction to VLSI Design	ECE
110.	GECX108	Plant Engineering	EIE
111.	GECX109	Network Security	CSE
112.	GECX110	Knowledge management	CSE
113.	GECX111	Cyber security	IT
114.	GECX112	Genetic Engineering	LS
115.	GECX113	Fundamentals of Project Management	CBS
116.	GECX114	Operations Research	Mathematics
117.	GECX115	Nano Technology	Physics / Chemistry
118.	GECX116	Vehicle Maintenance	Automobile
119.	GECX117	Fundamentals of Digital Image Processing	ECE

Group II Courses
(To be offered in VII semester)

Sl. No.	Course Code	Course Title	Offering Department
91.	GECX201	Green Design and Sustainability	Civil
92.	GECX202	Appropriate Technology	Civil / Mechanical
93.	GECX203	Engineering System Modelling and Simulation	Mechanical
94.	GECX204	Value Analysis and Engineering	Mechanical
95.	GECX205	Industrial Safety	Mechanical
96.	GECX206	Advanced Optimization Techniques	Mechanical
97.	GECX207	Mat Lab Simulation	EEE
98.	GECX208	Embedded Systems and its Applications	ECE
99.	GECX209	Usability Engineering	CSE
100.	GECX210	Supply Chain Management	CBS
101.	GECX211	System Analysis and Design	CA
102.	GECX212	Advanced Materials	Physics & Chemistry
103.	GECX213	National Service Scheme	School of Humanities
104.	GECX214	Automotive Pollution and Control	Automobile
105.	GECX215	Motor Vehicle Act, Insurance and Policy	Automobile
106.	GECX216	Principles of Communication Systems	ECE
107.	GECX217	Lean Management	Civil
108.	GECX218	Spatial Data Modeling & Analysis	Civil

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE AND TECHNOLOGY
B.TECH. INFORMATION & TECHNOLOGY

SEMESTER I

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC 1181	Differential Calculus and Geometry	3	1	0	4	
2.	HS	ENC 1181/ ISC 1181/ LNC 1181/ LNC 1182 / LNC 1183	English / Arabic / Mandarin / German / Japanese	3	0	0	3	
3.	BS	PHC 1181	Physics	3	0	2	4	
4.	BS	CHC 1181	Chemistry	3	0	2	4	
5.	ESF	GEC 1101	Engineering Graphics	2	0	2	3	
6.	ESF	GEC 1102	Engineering Design	2	0	0	2	
7.	ESF	GEC 1103	Basic Engineering Practices Laboratory	0	0	2	1	
8.	ESF	GEC 1104	Computer Programming I	1	0	2	2	23

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC 1281	Advanced Calculus	3	1	0	4	
2.	BS	-	Physics Elective	2	0	2	3	
3.	BS	-	Chemistry Elective	2	0	2	3	
4.	ESF	GEC 1211	Basic Engineering Mechanics	3	1	0	4	
5.	BS	GEC 1212	Environmental Studies	2	0	0	2	
6.	ESF	GEC 1213	Computer Programming II	1	0	2	2	
7.	EC	ITC 1201	Digital Principles and Applications	2	0	2	3	
8.	EC	ITC1202	Programming in Python	2	0	2	3	24

SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	MAC 2181	Partial Differential Equations and Transforms	3	1	0	4
2.	HS	-	Humanities Elective I	2	0	0	2
3.	HS	ENC 2181	Oral Communication	0	0	2	1
4.	EC	ITC 2101	Data Structures	3	0	2	4
5.	EC	ITC 2102	Computer Architecture	3	0	0	3
6.	EC	ITC 2103	Computer Networks	3	0	0	3
7.	EC	ITC 2104	Advanced Java & Internet Programming	3	0	0	3
8.	EC	EEC2181	Fundamentals of Digital Signal Processing	1	0	0	1
9.	EC	ITC 2106	Network Configuration Lab	0	0	3	1
10.	EC	ITC 2107	Advanced Java & Internet Programming Lab	0	0	3	1

SEMESTER IV

[illegible]

SEMESTER V

[illegible]

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	BS	-	Mathematics Elective II	2	0	0	2
3.	HS	ENC3281	Communication and soft skill II Confidence Building	0	0	2	1
4.	EC	ITC3211	Software Testing	3	0	0	3
5.	EC	ITC3212	Cloud Computing Technologies	3	0	0	3
6.	EC	ITC3213	Distributed Computing	3	0	0	3
7.	EC	ITC3214	Open Source Technologies	1	0	2	2
8.	EC	ITC3215	Software Development Lab	0	0	3	1
9.	PE	-	Programme Elective - III				6
							24

SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	GE	-	General Elective II	3	0	0	3
2.	EC	ITC4101	Internet of Things	3	0	2	4
3.	EC	ITC4102	Information Security	3	0	0	3
4.	EC	ITC4103	Wireless Networks	3	0	0	3
5.	EC	ITC4104	Programming in Hadoop	0	0	2	1
6.	PE	-	Programme Elective - IV	0	0	0	9
7.	EC	ITC4105	Internship				1* 24

SEMESTER VIII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	EC	ITC4211	Project Work	0	0	24	12 12

Total credits – 178

* Industrial training will be undertaken during Third year summer vacation for 15 days.

The credit will be awarded in the 7th Semester.

PROGRAMME ELECTIVES
IV SEMESTER ELECTIVES – I (3 CREDITS)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
	PE	ITCX201	Android Application Development	1	0	2	2
	PE	ITCX202	Nextgen Technologies	3	0	0	3
	PE	ITCX203	Multimedia Tools & Techniques	0	0	2	1
	PE	ITCX204	System Software	3	0	0	3
	PE	ITCX205	Principles of Communication	3	0	0	3
	PE	ITCX206	Principles of Compiler Design	3	0	0	3
	PE	ITCX207	User Interface Design	2	0	0	2
	PE	ITCX208	Semantic Web	2	0	0	2

V SEMESTER ELECTIVES – II (6 CREDITS)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
	PE	ITCX101	iOS Application Development	1	0	2	2
	PE	ITCX102	Oracle Database Programming	0	0	2	1
	PE	ITCX103	Natural Language Processing	3	0	0	3
	PE	ITCX104	Game Theory	3	0	0	3
	PE	ITCX105	Soft Computing	3	0	0	3
	PE	ITCX106	Machine Learning Algorithms	3	0	0	3
	PE	ITCX107	C# and .NET Framework	3	0	2	4
	PE	ITCX108	Artificial Intelligence	3	0	0	3
	PE	ITCX109	Grid Computing	3	0	0	3

PE	ITCX110	Information Coding Techniques	3	0	0	3
PE	ITCX111	Mobile AdHoc Networks	3	0	0	3

VI SEMESTER ELECTIVES – III (6 CREDITS)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
	PE	ITCX212	Hybrid Application Development	1	0	2	2
	PE	ITCX213	Data Warehousing, Data Mining and Data Mining Tools	3	0	2	4
	PE	ITCX214	Cyber Security	3	0	0	3
	PE	ITCX215	Software Requirements Management & Software Project Management	3	0	0	3
	PE	ITCX216	Business and Data Analytics	3	0	0	3
	PE	ITCX217	Web Services	3	0	0	3
	PE	ITCX218	Image Processing	3	0	0	3
	PE	ITCX219	Expert Systems	3	0	0	3
	PE	ITCX220	Service Oriented Architecture	3	0	0	3
	PE	ITCX221	Pervasive Computing	3	0	0	3
	PE	ITCX222	Knowledge Based Decision Support System	3	0	0	3
	PE	ITCX223	Electronics Commerce	3	0	0	3

VII SEMESTER ELECTIVES – IV (9 CREDITS)

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
	PE	ITCX117	Cloud Middleware Tools	1	0	2	2
	PE	ITCX118	Big Data Tools & Technologies	1	0	2	2
	PE	ITCX119	Forensic Tool Kits	0	0	2	1
	PE	ITCX120	MongoDB for Java developers	0	0	2	1

PE	ITCX121	Green Computing	0	0	2	1
PE	ITCX122	XML Technologies	0	0	2	1
PE	ITCX123	Big Data Analytics	3	0	0	3
PE	ITCX124	Programming in Scala	0	0	2	1
PE	ITCX125	R Programming	1	0	2	2
PE	ITCX126	Virtual Reality	2	0	2	3
PE	ITCX127	Software Quality Management	3	0	0	3
PE	ITCX128	Cloud Forensics	3	0	0	3
PE	ITCX129	Networking Simulators	0	0	2	1
PE	ITCX130	Enterprise Resource Planning	3	0	0	3
PE	ITCX131	Embedded System	3	0	0	3

Physics Elective Courses
(To be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	PHCX 01	Fundamentals of Engineering Materials	2	0	2	3
	PHCX 02	Heat and Thermodynamics	2	0	2	3
	PHCX 03	Introduction to Nanoscience and Technology	2	0	2	3
	PHCX 04	Lasers and their applications	2	0	2	3
	PHCX 05	Materials Science	2	0	2	3
	PHCX 06	Non-Destructive Testing	2	0	2	3
	PHCX 07	Properties of Matter and Acoustics	2	0	2	3
	PHCX 08	Properties of Matter and Nondestructive Testing	2	0	2	3
	PHCX 09	Semiconductor Physics and Optoelectronics	2	0	2	3

Chemistry Elective Courses
(To be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	CHCX01	Analytical Instrumentation	2	0	2	3
	CHCX02	Corrosion and its Control	2	0	2	3
	CHCX03	Electrical Materials and Batteries	2	0	2	3
	CHCX04	Engineering Materials	2	0	2	3
	CHCX05	Fuels and Combustion	2	0	2	3
	CHCX06	Fundamentals of Physical Chemistry	2	0	2	3
	CHCX07	Green Technology	2	0	2	3
	CHCX08	Organic Chemistry of Biomolecules	2	0	2	3
	CHCX09	Polymer Science and Technology	2	0	2	3

Maths Elective Courses
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	MACX 01	Discrete Mathematics And Graph Theory	3	1	0	4
	MACX 02	Probability And Statistics	3	1	0	4
	MACX 03	Random Processes	3	1	0	4
	MACX 04	Applied Numerical Methods	3	1	0	4

Maths Elective Courses
(To be offered in VI Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	MACX 05	Mathematical Programming	2	0	0	2
	MACX 06	Statistical Methods for Data Analysis	2	0	0	2
	MACX 07	Numerical Methods for Integral and Differential Equations	2	0	0	2
	MACX 08	Mathematical Modelling	2	0	0	2
	MACX 09	Graph Theory	2	0	0	2

Humanities Elective I
(To be offered in III Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	SSCX01	Fundamentals of Economics	2	0	0	2
	SSCX02	Principles of Sociology	2	0	0	2
	SSCX03	Sociology of Indian Society	2	0	0	2

Humanities Elective II
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	SSCX04	Economics of Sustainable Development	2	0	0	2
	SSCX05	Industrial Sociology	2	0	0	2
	SSCX06	Law for Engineers	2	0	0	2

General Elective
Group I Courses
(To be offered in V semester)

Sl. No.	Course Code	Course Title	Offering Department
	GECX101	Disaster Management	Civil
	GECX102	Total Quality Management	Mechanical
	GECX103	Energy Studies	Mechanical
	GECX104	Robotics	Mechanical
	GECX105	Transport Management	Automobile
	GECX106	Control Systems	EEE
	GECX107	Introduction to VLSI Design	ECE
	GECX108	Plant Engineering	EIE
	GECX109	Network Security	CSE
	GECX110	Knowledge management	CSE
	GECX111	Cyber security	IT
	GECX112	Genetic Engineering	LS
	GECX113	Fundamentals of Project Management	CBS
	GECX114	Operations Research	Mathematics
	GECX115	Nano Technology	Physics / Chemistry
	GECX116	Vehicle Maintenance	Automobile
	GECX117	Fundamentals of Digital Image Processing	ECE

Group II Courses
(To be offered in VII semester)

Sl. No.	Course Code	Course Title	Offering Department
	GECX201	Green Design and Sustainability	Civil
	GECX202	Appropriate Technology	Civil / Mechanical
	GECX203	Engineering System Modelling and Simulation	Mechanical
	GECX204	Value Analysis and Engineering	Mechanical
	GECX205	Industrial Safety	Mechanical
	GECX206	Advanced Optimization Techniques	Mechanical
	GECX207	Mat Lab Simulation	EEE
	GECX208	Embedded Systems and its Applications	ECE
	GECX209	Usability Engineering	CSE
	GECX210	Supply Chain Management	CBS
	GECX211	System Analysis and Design	CA
	GECX212	Advanced Materials	Physics & Chemistry
	GECX213	National Service Scheme	School of Humanities
	GECX214	Automotive Pollution and Control	Automobile
	GECX215	Motor Vehicle Act, Insurance and Policy	Automobile
	GECX216	Principles of Communication Systems	ECE
	GECX217	Lean Management	Civil
	GECX218	Spatial Data Modeling & Analysis	Civil

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE AND TECHNOLOGY**B.TECH. AERONAUTICAL ENGINEERING****SEMESTER I**

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC 1181	Differential Calculus and Geometry	3	1	0	4	
2.	HS	ENC 1181/ ISC 1181/ LNC 1181/ LNC 1182 / LNC 1183	English / Arabic / Mandarin / German / Japanese	3	0	0	3	
3.	BS	PHC 1181	Physics	3	0	2	4	
4.	BS	CHC 1181	Chemistry	3	0	2	4	
5.	ESF	GEC 1101	Engineering Graphics	2	0	2	3	
6.	ESF	GEC 1102	Engineering Design	2	0	0	2	
7.	ESF	GEC 1103	Basic Engineering Practices Laboratory	0	0	2	1	
8.	ESF	GEC 1104	Computer Programming I	1	0	2	2	23

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC 1281	Advanced Calculus	3	1	0	4	
2.	BS	-	Physics Elective	2	0	2	3	
3.	BS	-	Chemistry Elective	2	0	2	3	
4.	ESF	GEC 1211	Basic Engineering Mechanics	3	1	0	4	
5.	BS	GEC 1212	Environmental Studies	2	0	0	2	
6.	ESF	GEC 1213	Computer Programming II	1	0	2	2	
7.	EC	AEC 1211	Introduction to Aeronautical Engineering	3	0	0	3	
8.	EC	AEC 1212	Aircraft Component Modeling & Drafting Lab	0	0	3	1	

9.	EC	AEC 1213	Aircraft Structure Repair Lab	0	0	3	1	23
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SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC 2181	Partial Differential Equations and Transforms	3	1	0	4	
2.	HS	-	Humanities Elective I	2	0	0	2	
3.	HS	ENC 2181	Oral Communication	0	0	3	1	
4.	EC	AEC2101	Solid Mechanics	3	0	0	3	
5.	EC	AEC2102	Thermodynamics	3	1	0	4	
6.	EC	AEC2103	Fluid Mechanics	3	0	0	3	
7.	EC	EIC2181	Basic Electrical & Electronics Engg	3	0	0	3	
8.	EC	AEC2104	Thermodynamics Lab	0	0	3	1	
9.	EC	AEC2105	Fluid Mechanics Lab	0	0	3	1	
10.	EC	EIC2182	Electrical & Electronics Engg Lab	0	0	3	1	23

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	-	Mathematics Elective I	3	1	0	4
2.	HS	-	Humanities Elective II	2	0	0	2
3.	HS	ENC 2282	Written Communication	0	0	3	1
4.	EC	AEC2211	Low Speed Aerodynamics	3	0	0	3
5.	EC	AEC2212	Aircraft Structural Mechanics	3	1	0	4
6.	EC	AEC2213	Air Breathing Propulsion	3	0	0	3
7.	EC	AEC2214	Aircraft Systems & Instruments	3	0	0	3
8.	EC	AEC2215	Solid Mechanics Lab	0	0	3	1
9.	EC	AEC2216	Aircraft Systems & Instruments	0	0	3	1

			Lab					
10.	EC	AEC2217	Propulsion Lab	0	0	3	1	23

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	MS	MSC 3181/ MSC 3182	Leadership & CEO training / Social Entrepreneurship	3	0	0	3	
2.	GE	-	General Elective I	3	0	0	3	
3.	HS	ENC3181	Communication & Soft Skill I Career Choice	0	0	3	1	
4.	EC	AEC3101	Design of Compressors & Turbines	2	0	0	2	
5.	EC	AEC3102	Aircraft Structural Analysis	3	1	0	4	
6.	EC	AEC3103	High Speed Aerodynamics	3	0	0	3	
7.	EC	-	Electives	6	0	0	6	
8.	EC	AEC3104	Aerodynamics Lab	0	0	3	1	
9.	EC	AEC3105	Aircraft Structural Analysis Lab	0	0	3	1	24

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181/ MSC 3182	Leadership & CEO training / Social Entrepreneurship	3	0	0	3
2.	BS	-	Mathematics Elective II	2	0	0	2
3.	HS	ENC3281	Communication and Soft skill II Confidence Building	0	0	3	1
4.	EC	AEC3211	Rocket Propulsion	3	0	0	3
5.	EC	AEC3212	Flight Dynamics	3	0	0	3
6.	EC	AEC3213	Aircraft Materials	3	0	0	3

7.	EC	-	Electives	6	0	0	6	
8.	EC	AEC3214	Aircraft Design Project - I	1	0	3	2	23

SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	GE	-	General Elective II	3	0	0	3	
2.	EC	AEC4101	Avionics (Lab Integrated)	3	0	3	4	
3.	EC	AEC4102	Finite Element Method	3	0	0	3	
4.	EC	AEC4103	UAV/MAV Systems	1	0	3	2	
5.	EC	-	Electives	7	0	0	7	
6.	EC	AEC4104	Aircraft Design Project – II	1	0	3	2	
7.	EC	AEC4105	CFD-Structural Analysis Lab	0	0	3	1	
8.	EC	AEC4106	Internship	0	0	*	1**	23

SEMESTER VIII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	EC	AEC4211	Project Work	0	0	24	12	12

Total credits – 174

* 15 days

** Industrial training will be undertaken during third year summer vacation. The credit will be awarded in the 7th Semester.

PROGRAMME ELECTIVES
ODD SEMESTER ELECTIVES

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
	PE	AECX01	Flow Diagnostics	2	0	0	2
	PE	AECX02	Hypersonic Aerodynamics	3	0	0	3
	PE	AECX03	Industrial Aerodynamics	3	0	0	3
	PE	AECX04	Grid Generation	2	0	0	2
	PE	AECX05	Wind Tunnel Model Design	1	0	0	1
	PE	AECX06	Theory of Elasticity	3	0	0	3
	PE	AECX07	Fatigue and Fracture	3	0	0	3
	PE	AECX08	Structural Analysis Tools (NASTRAN and PATRAN)	2	0	0	2
	PE	AECX09	Smart Structures	1	0	0	1
	PE	AECX10	Aircraft Structural Testing and Qualification	1	0	0	1
	PE	AECX11	Measurement Systems	2	0	0	2
	PE	AECX12	NDT techniques for Aircraft Structures	1	0	0	1
	PE	AECX13	Airframe Repair & Maintenance	3	0	0	3
	PE	AECX14	Behavior of Materials at High Temperatures	3	0	0	3
	PE	AECX15	Heat Transfer	3	0	0	3
	PE	AECX16	Advanced Propulsion Systems	3	0	0	3
	PE	AECX17	Micro Propulsion	1	0	0	1
	PE	AECX18	Cryogenics	2	0	0	2
	PE	AECX19	Air Traffic Control and Aerodrome Design	3	0	0	3
	PE	AECX20	Aviation Rules and Regulation	3	0	0	3

EVEN SEMESTER ELECTIVES

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
	PE	AECX21	Experimental Aerodynamics	3	0	0	3
	PE	AECX22	Introduction to CFD	3	0	0	3
	PE	AECX23	Viscous Flows	3	0	0	3
	PE	AECX24	Aero-acoustics	1	0	0	1
	PE	AECX25	Wind Engineering	3	0	0	3
	PE	AECX26	Composite Materials and Structures	3	0	0	3
	PE	AECX27	Experimental Stress Analysis	3	0	0	3
	PE	AECX28	Theory of Vibration	3	0	0	3
	PE	AECX29	Hyper Mesh	2	0	0	2
	PE	AECX30	Aviation Fuels and Combustion	3	0	0	3
	PE	AECX31	Rockets and Missiles	3	0	0	3
	PE	AECX32	Design of Power Plants for Aircrafts	3	0	0	3
	PE	AECX33	Aircraft Cooling Systems	1	0	0	1
	PE	AECX34	Aircraft General Engineering Maintenance	2	0	0	2
	PE	AECX35	Product Development and 3D Printing Technologies	2	0	0	2
	PE	AECX36	Advanced Manufacturing Technologies	2	0	0	2
	PE	AECX37	Aircraft Navigation & Guidance	3	0	0	3
	PE	AECX38	Control Engineering	3	0	0	3
	PE	AECX39	Microprocessor & Microcontroller for Aircraft Systems	2	0	0	2
	PE	AECX40	Mathematical Modeling & Simulation (MATLAB)	2	0	0	2

Physics Elective Courses
(To be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	PHCX 01	Fundamentals of Engineering Materials	2	0	2	3
	PHCX 02	Heat and Thermodynamics	2	0	2	3
	PHCX 03	Introduction to Nanoscience and Technology	2	0	2	3
	PHCX 04	Lasers and their applications	2	0	2	3
	PHCX 05	Materials Science	2	0	2	3
	PHCX 06	Non-Destructive Testing	2	0	2	3
	PHCX 07	Properties of Matter and Acoustics	2	0	2	3
	PHCX 08	Properties of Matter and Nondestructive Testing	2	0	2	3
	PHCX 09	Semiconductor Physics and Optoelectronics	2	0	2	3

Chemistry Elective Courses
(To be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	CHCX01	Analytical Instrumentation	2	0	2	3
	CHCX02	Corrosion and its Control	2	0	2	3
	CHCX03	Electrical Materials and Batteries	2	0	2	3
	CHCX04	Engineering Materials	2	0	2	3
	CHCX05	Fuels and Combustion	2	0	2	3
	CHCX06	Fundamentals of Physical Chemistry	2	0	2	3
	CHCX07	Green Technology	2	0	2	3
	CHCX08	Organic Chemistry of Biomolecules	2	0	2	3
	CHCX09	Polymer Science and Technology	2	0	2	3

Maths Elective Courses
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	MACX 01	Discrete Mathematics And Graph Theory	3	1	0	4
	MACX 02	Probability And Statistics	3	1	0	4
	MACX 03	Random Processes	3	1	0	4
	MACX 04	Applied Numerical Methods	3	1	0	4

Maths Elective Courses
(To be offered in VI Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	MACX 05	Mathematical Programming	2	0	0	2
	MACX 06	Statistical Methods for Data Analysis	2	0	0	2
	MACX 07	Numerical Methods for Integral and Differential Equations	2	0	0	2
	MACX 08	Mathematical Modelling	2	0	0	2
	MACX 09	Graph Theory	2	0	0	2

Humanities Elective I
(To be offered in III Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	SSCX01	Fundamentals of Economics	2	0	0	2
	SSCX02	Principles of Sociology	2	0	0	2
	SSCX03	Sociology of Indian Society	2	0	0	2

Humanities Elective II
(To be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	SSCX04	Economics of Sustainable Development	2	0	0	2
	SSCX05	Industrial Sociology	2	0	0	2
	SSCX06	Law for Engineers	2	0	0	2

General Elective
Group I Courses
(To be offered in V semester)

Sl. No.	Course Code	Course Title	Offering Department
	GECX101	Disaster Management	Civil
	GECX102	Total Quality Management	Mechanical
	GECX103	Energy Studies	Mechanical
	GECX104	Robotics	Mechanical
	GECX105	Transport Management	Automobile
	GECX106	Control Systems	EEE
	GECX107	Introduction to VLSI Design	ECE
	GECX108	Plant Engineering	EIE
	GECX109	Network Security	CSE
	GECX110	Knowledge management	CSE
	GECX111	Cyber security	IT
	GECX112	Genetic Engineering	LS
	GECX113	Fundamentals of Project Management	CBS
	GECX114	Operations Research	Mathematics
	GECX115	Nano Technology	Physics / Chemistry
	GECX116	Vehicle Maintenance	Automobile
	GECX117	Fundamentals of Digital Image Processing	ECE

Group II Courses
(To be offered in VII semester)

Sl. No.	Course Code	Course Title	Offering Department
	GECX201	Green Design and Sustainability	Civil
	GECX202	Appropriate Technology	Civil / Mechanical
	GECX203	Engineering System Modelling and Simulation	Mechanical
	GECX204	Value Analysis and Engineering	Mechanical
	GECX205	Industrial Safety	Mechanical
	GECX206	Advanced Optimization Techniques	Mechanical
	GECX207	Mat Lab Simulation	EEE
	GECX208	Embedded Systems and its Applications	ECE
	GECX209	Usability Engineering	CSE
	GECX210	Supply Chain Management	CBS
	GECX211	System Analysis and Design	CA
	GECX212	Advanced Materials	Physics & Chemistry
	GECX213	National Service Scheme	School of Humanities
	GECX214	Automotive Pollution and Control	Automobile
	GECX215	Motor Vehicle Act, Insurance and Policy	Automobile
	GECX216	Principles of Communication Systems	ECE
	GECX217	Lean Management	Civil
	GECX218	Spatial Data Modeling & Analysis	Civil

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE AND TECHNOLOGY**B.TECH. AUTOMOBILE ENGINEERING****SEMESTER I**

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	MAC 1181	Differential Calculus and Geometry	3	1	0	4
2.	HS	ENC 1181/ ISC 1181/ LNC 1181/ LNC 1182/ LNC 1183	English / Arabic / Mandarin / German / Japanese	3	0	0	3
3.	BS	PHC 1181	Physics	3	0	2	4
4.	BS	CHC 1181	Chemistry	3	0	2	4
5.	ESF	GEC 1101	Engineering Graphics	2	0	2	3
6.	ESF	GEC 1102	Engineering Design	2	0	0	2
7.	ESF	GEC 1103	Basic Engineering Practices Laboratory	0	0	2	1
8.	ESF	GEC 1104	Computer Programming I	1	0	2	2
							23

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	MAC 1281	Advanced Calculus	3	1	0	4
2.	BS	-	Physics Elective	2	0	2	3
3.	BS	-	Chemistry Elective	2	0	2	3
4.	ESF	GEC1211	Basic Engineering Mechanics	3	1	0	4
5.	BS	GEC 1212	Environmental Studies	2	0	0	2
6.	ESF	GEC 1213	Computer Programming II	1	0	2	2
7.	ESF	EEC 1281	Automobile Electrical Engineering	2	0	0	2
8.	EC	AUC1211	Applied Fluid Mechanics	3	0	0	3

9.	ESF	EEC 1282	Automobile Electrical Engineering Laboratory	0	0	3	1	24
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SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	MAC 2181	Partial Differential Equations and Transforms	3	1	0	4	
2.	HS	-	Humanities Elective I	2	0	0	2	
3.	EC	AUC 2101	Applied Thermal Engineering	3	1	0	4	
4.	EC	AUC 2102	Vehicle Body Engineering	2	0	0	2	
5.	EC	AUC 2103	Basic Manufacturing Process	2	0	2	3	
6.	EC	AUC 2104	Strength of Materials	3	1	0	4	
7.	EC	AUC 2105	Automotive Engines	2	0	2	3	
8.	HS	ENC 2181	Oral Communication	0	0	2	1	
9.	EC	AUC 2106	Applied Fluid Mechanics Laboratory	0	0	3	1	24

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	BS	-	Mathematics Elective I	3	1	0	4	
2.	HS	-	Humanities Elective I	2	0	0	2	
3.	EC	AUC 2211	Automotive Chassis	3	0	0	3	
4.	EC	AUC 2212	Mechanics of Machinery	3	1	0	4	
5.	EC	AUC 2213	Automotive Electrical & Electronics	2	0	0	2	
6.	EC	AUC 2214	Automotive Material & Metallurgy	2	0	2	3	
7.	EC	AUC 2215	Production Process of Automotive Components	2	0	0	2	
8.	HS	ENC 2282	Written Communication	0	0	2	1	
9.	EC	AUC 2216	Automotive Chassis Laboratory	0	0	3	1	
10.	EC	AUC 2217	Automotive Electrical and	0	0	3	1	23

Electronics Laboratory

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181/ MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	EC	AUC 3101	Design of Automotive Components	3	1	0	4
3.	EC	AUC 3102	Automotive Transmission	3	0	0	3
4.	EC	AUC 3103	Measurements and Instrumentation	1	0	2	2
5.	PE	-	Professional Elective I	3	0	0	3
6.	PE	-	Professional Elective II	3	0	0	3
7.	GE	-	General Elective I	3	0	0	3
8.	HS	ENC 3181	Communication and Soft Skills - I Career Choice	0	0	2	1
9.	EC	AUC 3104	Engine Testing and Emission Laboratory	0	0	3	1

23

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181/ MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	BS	-	Mathematics Elective II	2	0	0	2
3.	EC	AUC 3211	Modeling of Simulation of vehicle system	3	0	0	3
4.	EC	AUC 3212	Vehicle Dynamics	3	1	0	4
5.	PE	-	Professional Elective III	3	0	0	3
6.	PE	-	Professional Elective IV	3	0	0	3
7.	HS	ENC 3281	Communication and Soft Skill – II Confidence Building	0	0	2	1

8.	EC	AUC 3213	Vehicle system simulation laboratory	0	0	3	1	
9.	EC	AUC 3214	Vehicle Dynamics Laboratory	0	0	3	1	
10.	EC	AUC 3215	Vehicle design laboratory (C- language, CATIA)	0	0	3	1	22

SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	EC	AUC 4101	Vehicle Management System	2	0	0	2	
2.	EC	AUC 4102	Finite Element Analysis of Automotive Components	2	0	2	3	
3.	EC	AUC 4103	Evaluation of Vehicle Performance	3	0	0	3	
4.	EC	AUC 4104	Two and Three Wheelers Technology	2	0	2	3	
5.	PE	-	Professional Elective V	3	0	0	3	
6.	PE	-	Professional Elective VI	3	0	0	3	
7.	GE	-	General Elective II	3	0	0	3	
8.	EC	AUC 4105	Vehicle Maintenance Laboratory	0	0	3	1	
9.	EC	AUC 4106	Advanced Automobile Components Manufacturing Laboratory	0	0	3	1	
10.	EC	AUC 4107	Internship.	0	0	0	1*	23

SEMESTER VIII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	EC	AUC 4211	Project Work	0	0	24	12	12

Total credits – 174

*Industrial training will be undertaken during Third year summer vacation. The credit will be awarded in the 7th Semester.

PROGRAMME ELECTIVES							
Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
	PE	AUCX01	Homologation and Testing	1	0	0	1
	PE	AUCX02	Geometric Dimensioning and Tolrencing	1	0	0	1
	PE	AUCX03	Vehicle Engineering & Integration	1	0	0	1
	PE	AUCX04	Automotive HVAC(Heating Ventilation Air Conditioning)	2	0	0	2
	PE	AUCX05	Hybrid and electric vehicle	2	0	0	2
	PE	AUCX06	Vehicle diagnostics	2	0	0	2
	PE	AUCX07	Spark Engine Technology	3	0	0	3
	PE	AUCX08	Compression Engine Technology	3	0	0	3
	PE	AUCX09	Automotive Aerodynamics	3	0	0	3
	PE	AUCX10	Vehicle comfort system and ergonomics	3	0	0	3
	PE	AUCX11	Vehicle design data characteristics	3	0	0	3
	PE	AUCX12	Design of transmission system and characteristics analysis	3	0	0	3
	PE	AUCX13	Design of hydraulics pneumatics systems for automotives	3	0	0	3
	PE	AUCX14	Tractor and Agricultural Machineries	3	0	0	3
	PE	AUCX15	Vehicle control system	3	0	0	3
	PE	AUCX16	Modern and intelligent vehicle system	3	0	0	3
	PE	AUCX17	Alternative Fuels and Energy Systems	3	0	0	3
	PE	AUCX18	Vehicle Networking and Internet of things	3	0	0	3
	PE	AUCX19	Automotive Instrumentation & Embedded System	3	0	0	3

PE	AUCX20	Advanced material testing and failure analysis	3	0	0	3
PE	AUCX21	Computer aided design and manufacturing	3	0	0	3
PE	AUCX22	Design of Jigs, Fixtures and Press Tools	3	0	0	3
PE	AUCX23	Simulation of I.C. Engine Processes	3	0	0	3
PE	AUCX24	Combustion thermodynamics and heat transfer	3	0	0	3
PE	AUCX25	Unconventional energy sources	3	0	0	3
PE	AUCX26	Computational flow and heat transfer	3	0	0	3
PE	AUCX27	Motorsport Engineering	3	0	0	3
PE	AUCX28	Composite Materials for Automobiles	3	0	0	3
PE	AUCX29	Traffic Engineering	3	0	0	3
PE	AUCX30	Surface Engineering	3	0	0	3
PE	AUCX31	Advanced IC Engine	3	0	0	3
PE	AUCX32	Fuel Cell Technology	3	0	0	3
PE	AUCX33	Power Plant Engineering	3	0	0	3

Physics Elective Courses
(to be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	PHCX 01	Fundamentals of Engineering Materials	2	0	2	3
	PHCX 02	Heat and Thermodynamics	2	0	2	3
	PHCX 03	Introduction to Nanoscience and Technology	2	0	2	3
	PHCX 04	Lasers and their applications	2	0	2	3
	PHCX 05	Materials Science	2	0	2	3
	PHCX 06	Non-Destructive Testing	2	0	2	3
	PHCX 07	Properties of Matter and Acoustics	2	0	2	3
	PHCX 08	Properties of Matter and Nondestructive Testing	2	0	2	3
	PHCX 09	Semiconductor Physics and Optoelectronics	2	0	2	3

Chemistry Elective Courses
(to be offered in II Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	CHCX01	Analytical Instrumentation	2	0	2	3
	CHCX02	Corrosion and its Control	2	0	2	3
	CHCX03	Electrical Materials and Batteries	2	0	2	3
	CHCX04	Engineering Materials	2	0	2	3
	CHCX05	Fuels and Combustion	2	0	2	3
	CHCX06	Fundamentals of Physical Chemistry	2	0	2	3
	CHCX07	Green Technology	2	0	2	3
	CHCX08	Organic Chemistry of Biomolecules	2	0	2	3
	CHCX09	Polymer Science and Technology	2	0	2	3

Maths Elective Courses
(to be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	MACX 01	Discrete Mathematics And Graph Theory	3	1	0	4
	MACX 02	Probability And Statistics	3	1	0	4
	MACX 03	Random Processes	3	1	0	4
	MACX 04	Applied Numerical Methods	3	1	0	4

Maths Elective Courses
(to be offered in VI Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	MACX 05	Mathematical Programming	2	0	0	2
	MACX 06	Statistical Methods for Data Analysis	2	0	0	2
	MACX 07	Numerical Methods for Integral and Differential Equations	2	0	0	2
	MACX 08	Mathematical Modelling	2	0	0	2
	MACX 09	Graph Theory	2	0	0	2

Humanities Elective I
(to be offered in III Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	SSCX01	Fundamentals of Economics	2	0	0	2
	SSCX02	Principles of Sociology	2	0	0	2
	SSCX03	Sociology of Indian Society	2	0	0	2

Humanities Elective II
(to be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	SSCX04	Economics of Sustainable Development	2	0	0	2
	SSCX05	Industrial Sociology	2	0	0	2
	SSCX06	Law for Engineers	2	0	0	2

General Elective**Group I Courses****(To be offered in V semester)**

Sl. No.	Course Code	Course Title	Offering Department
	GECX101	Disaster Management	Civil
	GECX102	Total Quality Management	Mechanical
	GECX103	Energy Studies	Mechanical
	GECX104	Robotics	Mechanical
	GECX105	Transport Management	Automobile
	GECX106	Control Systems	EEE
	GECX107	Introduction to VLSI Design	ECE
	GECX108	Plant Engineering	EIE
	GECX109	Network Security	CSE
	GECX110	Knowledge management	CSE
	GECX111	Cyber security	IT
	GECX112	Genetic Engineering	LS
	GECX113	Fundamentals of Project Management	CBS
	GECX114	Operations Research	Mathematics
	GECX115	Nano Technology	Physics / Chemistry
	GECX116	Vehicle Maintenance	Automobile
	GECX117	Fundamentals of Digital Image Processing	ECE

Group II Courses

(To be offered in VII semester)

Sl. No.	Course Code	Course Title	Offering Department
	GECX201	Green Design and Sustainability	Civil
	GECX202	Appropriate Technology	Civil / Mechanical
	GECX203	Engineering System Modelling and Simulation	Mechanical
	GECX204	Value Analysis and Engineering	Mechanical
	GECX205	Industrial Safety	Mechanical
	GECX206	Advanced Optimization Techniques	Mechanical
	GECX207	Mat Lab Simulation	EEE
	GECX208	Embedded Systems and its Applications	ECE
	GECX209	Usability Engineering	CSE
	GECX210	Supply Chain Management	CBS
	GECX211	System Analysis and Design	CA
	GECX212	Advanced Materials	Physics & Chemistry
	GECX213	National Service Scheme	School of Humanities
	GECX214	Automotive Pollution and Control	Automobile
	GECX215	Motor Vehicle Act, Insurance and Policy	Automobile
	GECX216	Principles of Communication Systems	ECE
	GECX217	Lean Management	Civil
	GECX218	Spatial Data Modeling & Analysis	Civil

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE AND TECHNOLOGY
B.TECH. BIOTECHNOLOGY

SEMESTER I

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	MAC 1182	Mathematics for biotechnology	3	1	0	4
2.	HS	ENC 1181/	English /				
		ISC 1181/	Arabic /				
		LNC 1181/	Mandarin /	3	0	0	3
		LNC 1182/	German /				
		LNC 1183	Japanese				
3.	BS	PHC 1181	Physics	3	0	2	4
4.	BS	CHC 1181	Chemistry	3	0	2	4
5.	ESF	BTC 1101	Fundamentals in Biotechnology	3	0	0	4
6.	ESF	GEC 1102	Engineering Design	2	0	0	2
7.	ESF	BTC 1102	Fundamentals in Biotechnology Laboratory	0	0	2	1
8.	ESF	GEC 1104	Computer Programming I	1	0	2	2

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	BS	MAC 1282	Biostatistics	3	1	0	4
2.	BS	-	Physics Elective	2	0	2	3
3.	BS	-	Chemistry Elective	2	0	2	3
4.	ESF	BTC1211	Biochemistry	4	0	0	3
5.	ESF	BTC1212	Cell Biology	4	0	0	3
7.	ESF	BTC1213	Microbiology	4	0	0	4
8.	ESF	BTC1214	Biochemistry – Lab	0	0	4	2
9.	ESF	BTC1215	Cell Biology– Lab	0	0	3	1
10.	ESF	BTC1216	Microbiology – Lab	0	0	2	1

SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	HS	-	Humanities Elective I	2	0	0	2
2.	HS	ENC 2181	Oral Communication	0	0	2	1
3.	EC	BTC2101	Enzyme Technology	4	0	0	4
4.	EC	BTC2102	Fundamentals of Chemical Engineering	4	0	0	4
5.	EC	BTC2103	Molecular Biology	4	0	0	4
6.	EC	BTC2104	Basic Bioanalytical Techniques	3	0	0	3
7.	EC	BTC2105	Molecular Biology – Lab	0	0	3	1
8.	EC	BTC2106	Enzyme Technology –Lab	0	0	3	1
9.	EC	BTC2107	Bioanalytical Techniques – Lab	0	0	3	1

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	HS	-	Humanities Elective I	2	0	0	2
2.	HS	ENC 2282	Written Communication	0	0	2	1
3.	EC	BTC2211	Genetic Engineering	4	0	0	4
4.	EC	BTC2212	Immunotechnology	4	0	0	4
5.	EC	BTC2213	Chemical and Bio Thermodynamics	4	0	0	4
6.	EC	BTC2214	Plant and Animal Biotechnology	3	0	0	3
7.	EC	BTC2215	Genetic Engineering Lab	0	0	3	1
8.	EC	BTC2216	Immunology Lab	0	0	3	1
9.	EC	BTC2217	Animal and Plant cell culture Lab	0	0	3	1
							21

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181/ MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	GE	-	General Elective I	3	0	0	3
3.	HS	ENC3181	Communication & soft skill I	0	0	2	1
4.	EC	BTC3101	Protein Engineering	4	0	0	4
6.	EC	BTC3102	Chemical Reaction Engineering	4	0	0	4
7.	EC	-	Programme Elective I	3	0	0	3
8.	EC	-	Programme Elective II	3	0	0	3
9.	EC	BTC3107	Chemical Reaction Engineering Lab	0	0	3	1
10.	PE	BTC3109	Protein Engineering Lab	0	0	3	1
							23

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	MS	MSC 3181/ MSC 3182	Leadership and CEO Training/ Social Entrepreneurship	3	0	0	3
2.	BS	-	Mathematics Elective II	3	0	0	3
3.	HS	-	Communication & soft skill IV	0	0	2	1
4.	EC	BTC3211	Structural Biology	3	1	0	4
5.	EC	BTC3212	Bioprocess Engineering	4	0	0	4
6.	EC	BTC3213	Bioinformatics	4	0	0	3
7.	PE	-	Programme Elective III	3	0	0	3
8.	PE	-	Programme Elective IV	3	0	0	3
9.	EC	BTC3214	Bioprocess Lab	0	0	3	1

10.	EC	BTC3102	Bioinformatics- Lab	0	0	3	1	26
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SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	EC	BTC4101	Bioreactor Design and Analysis	4	0	0	4	
2.	EC	BTC4102	Fermentation Technology	4	0	0	4	
3.	EC	BTC4103	Nano Biotechnology	4	0	0	4	
4.	EC	-	Programme Elective V	3	0	0	3	
5.	PE	-	Programme Elective VI	3	0	0	3	
6.	PE	-	Programme Elective VII	3	0	0	3	
7.	PE	-	General Elective II	3	0	0	3	
8.	GE	BTC4104	Fermentation- Lab	0	0	3	1	
9.	EC	BTC4105	Bioreactor Design and Drawing laboratory	0	0	3	1	
10.	EC	BTC4106	Internship.	0	0	0	1*	27

SEMESTER VIII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	EC	BTC 4211	Project Work	0	0	24	12	12

Total credits – 174

*Industrial training will be undertaken during Third year summer vacation. The credit will be awarded in the 7th Semester.

PROGRAMME ELECTIVES							
Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
	PE	BTCX01	Biophysics	3	0	0	3
	PE	BTCX02	Industrial Biotechnology	3	0	0	0
	PE	BTCX03	Bio-Organic Chemistry	3	0	0	3
	PE	BTCX04	Molecular Pathology	3	0	0	3
	PE	BTCX05	Food Biotechnology	3	0	0	3
	PE	BTCX06	Cancer Biology	3	0	0	3
	PE	BTCX07	Tissue Engineering	3	0	0	3
	PE	BTCX08	Developmental Biology	3	0	0	3
		BTCX09	Bioseparation Technology				
	PE	BTCX10	Proteomics & Genomics	3	0	0	3
	PE	BTCX11	Biomedical Instrumentation	3	0	0	3
	PE	BTCX12	Pharmaceutical Biotechnology	3	0	0	3
	PE	BTCX13	Medical Biotechnology	3	0	0	3
	PE	BTCX14	Drug Design and Development	3	0	0	3
	PE	BTCX15	Intellectual Property Rights	3	0	0	3
	PE	BTCX16	Recombinant DNA Technology	3	0	0	3
	PE	BTCX17	Material science	3	0	0	3
	PE	BTCX18	Molecular & Cellular Diagnostics	3	0	0	3
	PE	BTCX19	Biomedical Engineering	3	0	0	3
	PE	BTCX20	Biosafety and Bioethics	3	0	0	3
	PE	BTCX21	Healthcare Biotechnology	3	0	0	3
	PE	BTCX22	Molecular Farming	3	0	0	3

PE	BTCX23	Stem Cells in Health Care	3	0	0	3
PE	BTCX24	Transport phenomena in Bioprocess	3	0	0	3

**Physics Elective Courses
(to be offered in II Semester)**

Sl. No.	Course Code	Course Title	L	T	P	C
	PHCX 01	Fundamentals of Engineering Materials	2	0	2	3
	PHCX 02	Heat and Thermodynamics	2	0	2	3
	PHCX 03	Introduction to Nanoscience and Technology	2	0	2	3
	PHCX 04	Lasers and their applications	2	0	2	3
	PHCX 05	Materials Science	2	0	2	3
	PHCX 06	Non-Destructive Testing	2	0	2	3
	PHCX 07	Properties of Matter and Acoustics	2	0	2	3
	PHCX 08	Properties of Matter and Nondestructive Testing	2	0	2	3
	PHCX 09	Semiconductor Physics and Optoelectronics	2	0	2	3

**Chemistry Elective Courses
(to be offered in II Semester)**

Sl. No.	Course Code	Course Title	L	T	P	C
	CHCX01	Analytical Instrumentation	2	0	2	3
	CHCX02	Corrosion and its Control	2	0	2	3
	CHCX03	Electrical Materials and Batteries	2	0	2	3
	CHCX04	Engineering Materials	2	0	2	3
	CHCX05	Fuels and Combustion	2	0	2	3
	CHCX06	Fundamentals of Physical Chemistry	2	0	2	3

CHCX07	Green Technology	2	0	2	3
CHCX08	Organic Chemistry of Biomolecules	2	0	2	3
CHCX09	Polymer Science and Technology	2	0	2	3

Humanities Elective I
(to be offered in III Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	SSCX01	Fundamentals of Economics	2	0	0	2
	SSCX02	Principles of Sociology	2	0	0	2
	SSCX03	Sociology of Indian Society	2	0	0	2

Humanities Elective II
(to be offered in IV Semester)

Sl. No.	Course Code	Course Title	L	T	P	C
	SSCX04	Economics of Sustainable Development	2	0	0	2
	SSCX05	Industrial Sociology	2	0	0	2
	SSCX06	Law for Engineers	2	0	0	2

General Elective
Group I Courses
(To be offered in V semester)

Sl. No.	Course Code	Course Title	Offering Department
	GECX101	Disaster Management	Civil
	GECX102	Total Quality Management	Mechanical
	GECX103	Energy Studies	Mechanical
	GECX104	Robotics	Mechanical
	GECX105	Transport Management	Automobile
	GECX106	Control Systems	EEE
	GECX107	Introduction to VLSI Design	ECE
	GECX108	Plant Engineering	EIE
	GECX109	Network Security	CSE
	GECX110	Knowledge management	CSE
	GECX111	Cyber security	IT
	GECX112	Genetic Engineering	LS
	GECX113	Fundamentals of Project Management	CBS
	GECX114	Operations Research	Mathematics
	GECX115	Nano Technology	Physics / Chemistry
	GECX116	Vehicle Maintenance	Automobile
	GECX117	Fundamentals of Digital Image Processing	ECE

Group II Courses

(To be offered in VII semester)

Sl. No.	Course Code	Course Title	Offering Department
	GECX201	Green Design and Sustainability	Civil
	GECX202	Appropriate Technology	Civil / Mechanical
	GECX203	Engineering System Modelling and Simulation	Mechanical
	GECX204	Value Analysis and Engineering	Mechanical
	GECX205	Industrial Safety	Mechanical
	GECX206	Advanced Optimization Techniques	Mechanical
	GECX207	Mat Lab Simulation	EEE
	GECX208	Embedded Systems and its Applications	ECE
	GECX209	Usability Engineering	CSE
	GECX210	Supply Chain Management	CBS
	GECX211	System Analysis and Design	CA
	GECX212	Advanced Materials	Physics & Chemistry
	GECX213	National Service Scheme	School of Humanities
	GECX214	Automotive Pollution and Control	Automobile
	GECX215	Motor Vehicle Act, Insurance and Policy	Automobile
	GECX216	Principles of Communication Systems	ECE
	GECX217	Lean Management	Civil
	GECX218	Spatial Data Modeling & Analysis	Civil

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE & TECHNOLOGY
B.A (BACHELOR OF ARTS) - ISLAMIC STUDIES

SEMESTER I

Sl. No.	Course Code	Course Title	L	T	P	C	
1	ISC1101	Functional Arabic	3	0	0	3	
2	ISC1102	Communicative English	3	0	0	3	
3	ISC1103	Introduction to Quranic Studies	4	0	0	4	
4	ISC1104	Guidance of Prophet (PBUH) - Ibadaath	4	0	0	4	
5	ISC1105	Islamic Fiqh: Ibadaath	3	0	0	3	
6	ISC1106	Modern Arabic Grammar	4	0	0	4	
7	ISC 1107	Arabic Comprehension	2	0	0	2	23

SEMESTER II

Sl. No.	Course Code	Course Title	L	T	P	C	
1	ISC1211	Communicative Arabic	3	0	0	3	
2	ISC1212	Advanced Communicative English	3	0	0	3	
3	ISC1213	Quran Exegesis: Al Baqara	4	0	0	4	
4	ISC1214	Hadeeth - Teachings of Prophet	4	0	0	4	
5	ISC1215	Islamic Doctrine - Aqeedah	3	0	0	3	
6	ISC1216	Advanced Arabic Grammar	3	0	0	3	
7	ISC 1217	Islamic History: Seerah & Caliphate Period	3	0	0	3	23

SEMESTER III

Sl. No.	Course Code	Course Title	L	T	P	C	
1	ISC2101	Advanced Communicative Arabic	2	0	0	2	
2	ISC2102	Quran Exegesis: An Nisa	4	0	0	4	
3	ISC2103	A Study on Abu Dawood	4	0	0	4	
4	ISC2104	Islamic Fiqh: Al Muamalah	4	0	0	4	
5	ISC2105	Principles of Jurisprudence: Al Adillah	3	0	0	3	
6	ISC2106	Arabic Language: Grammar & Rhetoric	3	0	0	3	
7	ISC2107	Islamic History: Umayyad & Abbasids Period	3	0	0	3	23

SEMESTER IV

Sl. No.	Course Code	Course Title	L	T	P	C	
1	ISC2211	Quran Exegesis: Selected Chapters	4	0	0	4	
2	ISC2212	Hadeeth: Sunan At Tirmidhi	3	0	0	3	
3	ISC2213	Muslim Personal Law: Inheritance & Waqf	3	0	0	3	
4	ISC2214	Principles of Jurisprudence: Al Qawayid	3	0	0	3	
5	ISC2215	Development of Islamic Religious Sciences: Tafseer & Hadeeth	4	0	0	4	
6	ISC2216	Arabic Literature: Prose & Poetry	4	0	0	4	
7	ISC2217	Modern Islamic Finance	2	0	0	2	23

SEMESTER V

Sl. No.	Course Code	Course Title	L	T	P	C	
1	ISC3101	Shariah Rulings in Chapter Al Maidah	4	0	0	4	
2	ISC3102	A Special Study on Saheeh Muslim	4	0	0	4	
3	ISC3103	Muslim Family Law	4	0	0	4	
4	ISC3104	Comparative Fiqh	3	0	0	3	
5	ISC3105	History of Islamic Thought	3	0	0	3	
6	ISC3106	Muslims in India & Plural Society	2	0	0	2	
7	ISC3107	Business Arabic & Secretarial Practice	2	0	0	2	
8	ISC3108	Indian Constitution: Minority Rights	1	0	0	1	23

SEMESTER VI

Sl. No.	Course Code	Course Title	L	T	P	C	
1	ISC3211	Thematic Study of Quran	3	0	0	3	
2	ISC3212	A Special Study on Saheeh Al Bukhari	4	0	0	4	
3	ISC3213	Quran and Modern Issues	3	0	0	3	
4	ISC3214	Da'wah & Comparative Religion	3	0	0	3	
5	ISC3215	History of Modern Arabic Literature	3	0	0	3	
6	ISC3216	Islamic Philosophy	4	0	0	4	
7	ISC3217	Journal Arabic, Web Translation & SAP	1	0	0	1	
8	ISC3218	Project	2	0	0	2	23

TOTAL CREDITS – 138

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE & TECHNOLOGY
B.COM GENERAL

SEMESTER I

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	AECC	ENC 1183	General English-I	3	1	0	3	
2	AECC	LNC1182/ LNC 1184	German - I Tamil - I	3	1	0	3	
3	PC	COC 1101	Financial accounting	4	2	0	5	
4	PC	COC 1102	Business management	4	1	0	5	
5	PA	COC 1103	Quantitative techniques & reasoning	4	1	0	5	
6	SS	COC 1104	Excel modeling	0	0	4	2	
7	AECC	COC 1105	Environmental studies	2	0	0	2	25

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	AECC	ENC 1284	General English II	3	0	1	3	
2	AECC	LNC1283/ LNC 1284	German - II/ Tamil - II	3	1	0	3	
3	PC	COC 1201	Advanced Financial Accounting	4	2	0	5	
4	PC	COC 1202	Business Law	4	1	0	5	
5	PA	COC 1203	Business Economics	4	1	0	5	
6	SS	COC 1204	Database Management System	0	0	4	2	
7	AECC	COC 1205/ LNC 1285/ LNC 1286	Business Environment/ Basic Tamil I/ Advanced Tamil I	2	0	0	2	25

SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	PC	COC 2101	Corporate Accounting I	4	2	0	5	
2	PC	COC 2102	Company Law	4	1	0	5	
3	PC	COC 2103	Marketing Management	4	2	0	5	
4	PC	COC 2104	Entrepreneurship Development	4	0	1	5	
5	PA	COC 2105	Business Statistics I	4	2	0	5	
6	SS	COC 2106/ LNC 2185/ LNC 2186	Office Management/ Basic Tamil II/ Advance Tamil II	2	0	0	2	27

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1	PC	COC 2201	Advanced Corporate Accounting	4	2	0	5
2	PC	COC 2202	Financial Management	5	1	0	5
3	PC	COC 2203	Banking Theory, Law& Practice	4	0	1	5
4	PC	COC 2204	Financial Markets & Services	4	0	1	5
5	PA	COC 2205	Business Statistics II	5	1	0	5
6	SS	COC 2206	Communication Skills	2	0	0	2
				27			

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1	PC	COC 3101	Income Tax Law & Practice I	4	2	0	5
2	PC	COC 3102	Cost Accounting	4	2	0	5
3	PC	COC 3103	Human Resource Management	4	0	2	5
4	PE		Elective I	4	2	0	5

5	PE	COC 3104	Project Work/Elective II	4	2	0	5	
6	GE	COC 3105	Personality Development	2	0	0	2	27

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	PC	COC 3201	Income Tax Law & Practice II	4	2	0	5	
2	PC	COC 3202	Management Accounting	4	2	0	5	
3	PC	COC 3203	Auditing	4	0	2	5	
4	PE		Elective III	4	2	0	5	
5	PE		Elective IV	0	2	4	5	
6	GE	COC 3204	Value Education	2	0	0	2	27

Total Credits: 158 Credits

LIST OF PROGRAMME ELECTIVE

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1	PE	COCX01	Investment management	4	1	0	5
2	PE	COCX02	International finance	4	1	0	5
3	PE	COCX03	Business strategies	4	1	0	5
4	PE	COCX04	Insurance	4	1	0	5
5	PE	COCX05	Research methodology	3	2	0	5
6	PE	COCX06	E-commerce	4	1	0	5

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE & TECHNOLOGY
B.COM ACCOUNTS AND FINANCE

SEMESTER I

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	AECC	ENC 1183	General English-I	3	1	0	3	
2	AECC	LNC1182/ LNC 1184	German - I Tamil - I	3	1	0	3	
3	PC	COC 1101	Financial Accounting	4	2	0	5	
4	PC	COC 1102	Business Management	4	1	0	5	
5	PA	COC 1103	Quantitative Techniques & Reasoning	4	1	0	5	
6	SS	COC 1104	Excel Modeling	0	0	4	2	
7	AECC	COC 1105	Environmental Studies	2	0	0	2	25

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	AECC	ENC 1284	General English II	3	1	0	3	
2	AECC	LNC1282/ LNC 1284	German - II/ Tamil - II	3	1	0	3	
3	PC	COC 1201	Advanced Financial Accounting	4	2	0	5	
4	PC	COC 1211	Banking & Financial System	4	1	0	5	
5	PA	COC 1212	Business Statistics & Operations Research	4	1	0	5	
6	SS	COC 1204	Database Management System	0	0	4	2	
7	AECC	COC 1205/ LNC 1285/ LNC 1286	Business Environment/ Basic Tamil I/ Advanced Tamil I	2	0	0	2	25

SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	PC	COC 2101	Corporate Accounting I	4	2	0	5	
2	PC	COC 2111	Management Accounting	4	1	0	5	
3	PC	COC 2112	Foreign Exchange	4	2	0	5	
4	PC	COC 2113	Practical Auditing	4	0	1	5	
5	PA	COC 2114	Business Economics	4	2	0	5	
6	SS	COC 2106/ LNC 2185/ LNC 2186	Office Management/ Basic Tamil II/ Advance Tamil II	2	0	0	2	27

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	PC	COC 2201	Advanced Corporate Accounting	4	2	0	5	
2	PC	COC 2211	Principles of Financial Management	5	1	0	5	
3	PC	COC 2212	Business & Corporate Laws	4	0	1	5	
4	PC	COC 2213	Goods & Service Tax(gst)	4	0	1	5	
5	PA	COC 2214	Fiscal Economics	5	1	0	5	
6	SS	COC 2206	Communication Skills	2	0	0	2	27

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	PC	COC 3111	Principles of Cost Accounting	4	2	0	5	
2	PC	COC 3112	Advanced Financial Management	4	2	0	5	
3	PC	COC 3113	Taxation I	4	0	2	5	
4	PE		Elective I	4	2	0	5	

5	PE	COC 3104	Project work/Elective II	4	2	0	5	
6	GE	COC 3105	Personality Development	2	0	0	2	27

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	PC	COC 3211	Advanced Cost Accounting	4	2	0	5	
2	PC	COC 3212	Computerized Accounting	4	2	0	5	
3	PC	COC 3213	Taxation II	4	0	2	5	
4	PE		Elective III	4	2	0	5	
5	PE		Elective IV	0	2	4	5	
6	GE	COC 3204	Value Education	2	0	0	2	27

Total Credits: 158 Credits

LIST OF PROGRAMME ELECTIVE

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1	PE	COCX01	Investment management	4	1	0	5
2	PE	COCX02	International finance	4	1	0	5
3	PE	COCX04	Insurance	4	1	0	5
4	PE	COCX05	Research methodology	3	2	0	5
5	PE	COCX21	Accounting Standards	4	1	0	5
6	PE	COCX22	Financial Markets and Services	4	1	0	5

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE & TECHNOLOGY
B.B.A LL.B (Hons.)

SEMESTER I

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	SSC1184	Principles of Economics	4	1	0	5	
2.	BLC1101	Principles of Management and Human Resources Management	4	1	0	5	
3.	BLC1102	Introduction to Law	4	1	0	5	
4.	BLC1103	General Principles of Contract	4	1	0	5	
5.	BLC1104	Law and Language	4	1	0	5	25

SEMESTER II

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	ENC1286	Business English	4	1	0	5	
2.	SSC1281	Introduction to Sociology	4	1	0	5	
3.	BLC1201	Law of Torts & Accident Claims under The Motor Vehicles Act	4	1	0	5	
4.	BLC1202	Special Contracts	4	1	0	5	
5.	BLC1211	Organizational Behaviour	4	1	0	5	
6.	GEC1212	Environmental Studies	2	0	0	2	27

SEMESTER III

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	BLC2101	Law of Crimes-I(IPC)	4	1	0	5	
2.	BLC2102	Family Law-I	4	1	0	5	
3.	BLC2103	Constitutional Law-I	4	1	0	5	
4.	SSC2181	Introduction to Political Science	4	1	0	5	

5.	BLC2111	Financial and Management Accounting	4	1	0	5	25
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SEMESTER IV

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	BLC2201	Managerial Economics	4	1	0	5	
2.	BLC2202	Jurisprudence	4	1	0	5	
3.	BLC2203	Family Law-II	4	1	0	5	
4.	BLC2204	Constitutional Law-II	4	1	0	5	
5.	BLC2211	Computer for Management and Management Information System(MIS)	4	1	0	5	
6.	BLC2205	Internship – I	0	0	2	2	27

SEMESTER V

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	BLC3101	Public International Law	4	1	0	5	
2.	BLC3102	Company Law	4	1	0	5	
3.	BLC3103	Human Rights	4	1	0	5	
4.	BLC3111	Strategic Management	4	1	0	5	
5.	BLC3112	Fundamentals of Marketing Management	4	1	0	5	25

SEMESTER VI

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	BLC3201	Law of Evidence	4	1	0	5	
2.	BLC3202	Labour Law – I	4	1	0	5	
3.	BLC3203	Administrative Law	4	1	0	5	
4.	BLC3204	Professional Ethics, Bar Bench Relations and Accountancy for Lawyers (Practical Training Paper)	2	0	3	5	

5.	BLC3211	Public Finance	4	1	0	5	
6.	BLC3205	Internship – II	0	0	2	2	27

SEMESTER VII

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	BLC4101	Transfer of Property Law	4	1	0	5	
2.	BLC4102	Intellectual Property Law	4	1	0	5	
3.	BLC4103	Labour Law – II	4	1	0	5	
4.	BLC4104	Law of Banking	4	1	0	5	
5.	BLC4105	Alternative Dispute Resolution Mechanism (Practical Training Paper)	2	0	3	5	25

SEMESTER VIII

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	BLC4201	Competition law	4	1	0	5	
2.	BLC4202	Environmental law	4	1	0	5	
3.	BLC4203	Maritime Law	4	1	0	5	
4.	BLC4204	International Trade Law	4	1	0	5	
5.	BLC4205	Cyber Law and Forensic	4	1	0	5	
6.	BLC4206	Internship – III	0	0	2	2	27

SEMESTER IX

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	BLC5101	Law of Insurance	4	1	0	5	
2.	BLC5102	Law of Crimes-II(Cr.P.C.)	4	1	0	5	
3.	BLC5103	Civil Procedure Code & Limitation Act	4	1	0	5	
4.	BLC5104	Land Laws	4	1	0	5	
5.	BLC5105	Moot Court (Practical Training Paper)	0	0	5	5	25

SEMESTER X						
Sl. No.	Course Code	Course Title	L	T	P	C
1.	BLC5201	Interpretation of Statutes	4	1	0	5
2.	BLC5202	Right to Information Act	4	1	0	5
3.	BLC5203	Law of Taxation	4	1	0	5
4.	BLC5204	Criminology and Penology	4	1	0	5
5.	BLC5205	Drafting Pleading and Conveyancing. (Practical Training Paper)	2	0	3	5
6.	BLC5206	Internship – IV	0	0	2	2
						27

TOTAL CREDITS : 260

**B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE & TECHNOLOGY,
B.COM LL.B (Hons.)**

SEMESTER I

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	SSC1184	Principles of Economics	4	1	0	5	
2.	BLC1101	Principles of Management and Human Resources Management	4	1	0	5	
3.	BLC1102	Introduction to Law	4	1	0	5	
4.	BLC1103	General Principles of Contract	4	1	0	5	
5.	BLC1104	Law and Language	4	1	0	5	25

SEMESTER II

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	ENC1286	Business English	4	1	0	5	
2.	SSC1281	Introduction to Sociology	4	1	0	5	
3.	BLC1201	Law of Torts & Accident Claims under The Motor Vehicles Act	4	1	0	5	
4.	BLC1202	Special Contracts	4	1	0	5	
5.	BLC1203	Principles of Accounting	4	1	0	5	
6.	GEC1212	Environmental Studies	2	0	0	2	27

SEMESTER III

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	BLC2101	Law of Crimes-I(IPC)	4	1	0	5	
2.	BLC2102	Family Law-I	4	1	0	5	
3.	BLC2103	Constitutional Law-I	4	1	0	5	
4.	SSC2181	Introduction to Political Science	4	1	0	5	

5	BLC2105	Financial Market and Marketing Management	4	1	0	5	25
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SEMESTER IV

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	BLC2201	Managerial Economics	4	1	0	5	
2.	BLC2202	Jurisprudence	4	1	0	5	
3.	BLC2203	Family Law-II	4	1	0	5	
4.	BLC2204	Constitutional Law-II	4	1	0	5	
5.	BLC2206	Cost and Management Accounting	4	1	0	5	
6.	BLC2205	Internship – I	0	0	2	2	27

SEMESTER V

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	BLC3101	Public International Law	4	1	0	5	
2.	BLC3102	Company Law	4	1	0	5	
3.	BLC3103	Human Rights	4	1	0	5	
4.	BLC3104	Financial Management	4	1	0	5	
5.	BLC3105	Practical Auditing	4	1	0	5	25

SEMESTER VI

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	BLC3201	Law of Evidence	4	1	0	5	
2.	BLC3202	Labour Law – I	4	1	0	5	
3.	BLC3203	Administrative Law	4	1	0	5	
4.	BLC3204	Professional Ethics, Bar Bench Relations and Accountancy for Lawyers. (Practical Training Paper)	2	0	3	5	
5.	BLC3206	Customs and Excise Laws	4	1	0	5	
6.	BLC3205	Internship – II	0	0	2	2	27

SEMESTER VII

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	BLC4101	Transfer of Property Law	4	1	0	5	
2.	BLC4102	Intellectual Property Law	4	1	0	5	
3.	BLC4103	Labour Law – II	4	1	0	5	
4.	BLC4104	Law of Banking	4	1	0	5	
5.	BLC4105	Alternative Dispute Resolution Mechanism (Practical Training Paper)	2	0	3	5	25

SEMESTER VIII

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	BLC4201	Competition law	4	1	0	5	
2.	BLC4202	Environmental law	4	1	0	5	
3.	BLC4203	Maritime Law	4	1	0	5	
4.	BLC4204	International Trade Law	4	1	0	5	
5.	BLC4205	Cyber Law and Forensic	4	1	0	5	
6.	BLC4206	Internship – III	0	0	2	2	27

SEMESTER IX

Sl. No.	Course Code	Course Title	L	T	P	C	
1.	BLC5101	Law of Insurance	4	1	0	5	
2.	BLC5102	Law of Crimes-II(Cr.P.C.)	4	1	0	5	
3.	BLC5103	Civil Procedure Code & Limitation Act	4	1	0	5	
4.	BLC5104	Land Laws	4	1	0	5	
5.	BLC5105	Moot Court (Practical Training Paper)	0	0	5	5	25

SEMESTER X						
Sl. No.	Course Code	Course Title	L	T	P	C
1.	BLC5201	Interpretation of Statutes	4	1	0	5
2.	BLC5202	Right to Information Act	4	1	0	5
3.	BLC5203	Law of Taxation	4	1	0	5
4.	BLC5204	Criminology and Penology	4	1	0	5
5.	BLC5205	Drafting Pleading and Conveyancing.(Practical Training Paper)	2	0	3	5
6.	BLC5206	Internship – IV	0	0	2	2
						27

TOTAL CREDITS : 260

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE & TECHNOLOGY**B.B.A FINANCIAL SERVICES****SEMESTER I**

Sl. No.	Course Code	Course Title	L	T	P	C	
1	MSC 1101	Accounting – I	5	0	0	4	
2	MSC 1102	Management Concepts and Practices	4	0	0	3	
3	MSC 1103	Fundamentals of Managerial Economics	4	0	0	3	
4	MSC 1104	Basic Statistics	5	0	0	4	
5	MSC 1105	Corporate Communication	4	0	0	3	
6	MSC 1106	IT for Business and Accounting	3	0	0	2	
7	MSC 1107	IT for Business and Accounting - Lab	0	0	2	2	21

SEMESTER II

Sl. No.	Course Code	Course Title	L	T	P	C	
1	MSC 1201	Accounting - II	5	0	0	4	
2	MSC 1202	Business Mathematics	5	0	0	4	
3	MSC 1203	Organizational Behaviour	4	0	0	3	
4	MSC 1204	Applied Macro Economics	4	0	0	3	
5	MSC 1205	Indian Financial System	4	0	0	3	
6	MSC 1206	Excel Applications for Business	2	0	0	2	
7	MSC 1207	Excel Applications for Business - Lab	0	0	4	2	

8	GEC1212	Environmental Studies	2	0	0	2	23
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SEMESTER III

Sl. No.	Course Code	Course Title	L	T	P	C	
1	MSC 2101	Marketing Management	4	0	0	3	
2	MSC 2102	Introduction to Financial Management	5	0	0	4	
3	MSC 2103	Human Resource Management	4	0	0	3	
4	MSC 2104	Introduction to Entrepreneurship	4	0	0	3	
5	MSC 2105	Fundamentals of Foreign Trade	4	0	0	3	
6	MSC 2106	Cost Accounting	5	0	0	4	20

SEMESTER IV

Sl. No.	Course Code	Course Title	L	T	P	C	
1	MSC 2201	Business Research Methods	5	0	0	4	
2	MSC 2202	Derivatives and Commodity Market	4	0	0	3	
3	MSC 2203	Enterprise Resource Planning	5	0	0	4	
4	MSC 2204	Financial Services	4	0	0	3	
5	MSC 2205	Managerial Accounting	4	0	0	3	
6	MSC 2206	Introduction to Financial Planning	4	0	0	3	20

SEMESTER V

Sl. No.	Course Code	Course Title	L	T	P	C	
1	MSC 3101	Introduction to Business Law	5	0	0	4	
2	MSC 3102	Business Ethics and Values	5	0	0	4	
3	MSC 3103	Insurance and Risk Management	5	0	0	4	
4	MSC 3104	Cost Management	5	0	0	4	
5	MSC 3105	Fundamental Analysis and Technical Analysis	5	0	0	4	20

SEMESTER VI

Sl. No.	Course Code	Course Title	L	T	P	C	
1	MSC 3201	Fundamentals of Business Taxation	5	0	0	4	
2	MSC 3202	Introduction to International Finance	4	0	0	3	
3	MSC 3203	Investment Banking and Operations	4	0	0	3	
4	MSC 3204	Financial Modelling Using Excel	2	1	0	3	
5	MSC 3205	Financial Modelling Using Excel – Lab	0	0	4	2	
6	MSC 3206	Project Work	-	-	-	5	20

TOTAL CREDITS – 124

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE & TECHNOLOGY**B.SC BIO TECHNOLOGY****SEMESTER I**

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	AEC	LNC1184	Tamil – I	4	1	0	4	
2	AEC	ENC1182	General English – I	4	1	0	4	
3	PC	LSC1101	Cell Biology	4	0	0	4	
4	PC	LSC1102	Cell Biology Lab	0	0	3	2	
5	PC	LSC1103	Microbiology	4	0	0	4	
6	PC	LSC1104	Microbiology Lab	0	0	3	2	
7	PC	CHB1182	Chemistry	3	0	0	3	
8	AEC	ENC1183	Communication Skills	2	0	0	2	25

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	AEC	ENC1283	English - II	4	1	0	4	
2	AEC	LNC1284	Tamil – II	4	1	0	4	
3	PC	LSC1201	Molecular Biology	4	0	0	4	
4	PC	LSC1202	Molecular Biology Lab	0	0	3	2	
5	PC	LSC1203	Bioinstrumentation	4	0	0	4	
6	PC	LSC1204	Bioinstrumentation Lab	0	0	3	2	
7	AC	LSC1205	Basics of Computers	4	0	0	4	
8	AEC	LSC1206	Confidence building behavioural skills	2	0	0	2	26

SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	PC	LSC2101	Biochemistry	3	2	0	4	
2	PC	LSC2102	Biochemistry Lab	0	0	3	2	
3	PC	LSC2103	Basics of Genetics	4	0	0	4	
4	PC	LSC2104	Basics of Genetics Lab	0	0	3	2	
5	AC	LSC2105	Biostatistics	3	2	0	4	
6	GE		Skill Based Electives - I	3	0	0	3	
7	PE		Open Elective -I	4	0	0	4	
8	AEC	LSC2106	Quantitative Aptitude & Reasoning -I	2	0	0	2	25

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	PC	LSC2201	Bioprocess Technology	3	2	0	4	
2	PC	LSC2202	Enzymology	3	2	0	4	
3	PC	LSC2203	Bioprocess Technology Lab	0	0	3	2	
4	PC	LSC2204	Enzymology Lab	0	0	3	2	
5	AC	LSC2205	Biophysics	3	2	0	4	
6	GE		Skill Based Electives - II	3	0	0	3	
7	PE		Open Elective -II	4	0	0	4	
8	AEC	LSC2206	Quantitative Aptitude & Reasoning -II	2	0	0	2	23

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	PC	LSC3101	Plant Biotechnology	4	1	0	4	
2	PC	LSC3102	Animal Biotechnology	4	1	0	4	
3	PC	LSC3103	Plant Biotechnology Lab	0	0	3	2	
4	PC	LSC3104	Animal Biotechnology Lab	0	0	3	2	
5	PE		Core Elective -I	4	1	0	4	
6	PE		Core Elective -II	4	1	0	4	
7	GE	LSC3105	Environmental Science	3	0	0	3	23

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1	PC	LSC3201	Immunotechnology	4	1	0	4	
2	PC	LSC3202	Genomics and Proteomics	4	1	0	4	
3	PC	LSC3203	Immunology Lab	0	0	3	2	
4	PC	LSC3204	Genomics and Proteomics lab	0	0	3	2	
5	PE		Core Elective -III	4	1	0	4	
6	PE		Core Elective -IV	4	0	0	4	
7		LSC3205	Mini Project (Review/Research)	0	0	5	3	23

Total Credits: 145 Credits

LIST OF PROGRAMME ELECTIVES

SKILL BASED ELECTIVES

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
70.	GE	LSCX101	Disease Management	3	0	0	3
71.	GE	LSCX102	Cytogenetics	3	0	0	3
72.	GE	LSCX103	Biofertilizer Technology	3	0	0	3
73.	GE	LSCX201	Agricultural Biotechnology	3	0	0	3
74.	GE	LSCX202	Herbal Technology	3	0	0	3

CORE ELECTIVES

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	LSCX111	Human physiology	4	0	0	4
2.	PE	LSCX112	Medical Biotechnology	4	0	0	4
3.	PE	LSCX113	Bioinformatics	4	0	0	4
4.	PE	LSCX114	Bioethics, IPR and Biosafety	4	0	0	4
5.	PE	LSCX115	Environmental Biotechnology	4	0	0	4
6.	PE	LSCX211	Nanobiotechnonology	4	0	0	4
7.	PE	LSCX212	Cancer Biology	4	0	0	4
8.	PE	LSCX213	Pharmacology	4	0	0	4
9.	PE	LSCX214	Regenerative Medicine	4	0	0	4
10.	PE	LSCX215	rDNA Technology	4	0	0	4

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE & TECHNOLOGY
B.SC BIO TECHNOLOGY

SEMESTER I

Course code	Name of the course	No. of	Tutorial	Credit points
	Human Anatomy and Physiology I–			
BP101T	Theory	3	1	4
BP102T	Pharmaceutical Analysis I – Theory	3	1	4
BP103T	Pharmaceutics I – Theory	3	1	4
BP104T	Pharmaceutical Inorganic Chemistry –			
	Theory	3	1	4
BP105T	Communication skills – Theory *	2	-	2
BP106RBT	Remedial Biology/			
BP106RMT	Remedial Mathematics – Theory*	2	-	2
	Human Anatomy and Physiology –			
BP107P	Practical	4	-	2
BP108P	Pharmaceutical Analysis I – Practical	4	-	2
BP109P	Pharmaceutics I – Practical	4	-	2
BP110P	Pharmaceutical Inorganic Chemistry –			
#	Practical	4	-	2
BP111P	Communication skills – Practical*	2	-	1
BP112RBP	Remedial Biology – Practical*	2	-	1

Applicable ONLY for the students who have studied Mathematics / Physics / Chemistry at HSC and appearing for Remedial Biology (RB)course.

\$Applicable ONLY for the students who have studied Physics / Chemistry / Botany / Zoology at HSC and appearing for Remedial Mathematics (RM)course.

*Non University examination
(NUE)

SEMESTER II

Course Code	Name of the course	No. of hours	Tutorial	Credit points
BP201T	Human Anatomy and Physiology II – Theory	3	1	4
BP202T	Pharmaceutical Organic Chemistry I – Theory	3	1	4
BP203T	Biochemistry – Theory	3	1	4
BP204T	Pathophysiology – Theory	3	1	4
BP205T	Computer Applications in Pharmacy – Theory *	3	-	3
BP206T	Environmental sciences – Theory *	3	-	3
BP207P	Human Anatomy and Physiology II – Practical	4	-	2
BP208P	Pharmaceutical Organic Chemistry I– Practical	4	-	2
BP209P	Biochemistry – Practical	4	-	2
BP210P	Computer Applications in Pharmacy – Practical*	2	-	1
Total		32	4	29

*Non University Examination (NUE)

SEMESTER III

Course code	Name of the course	No. of hours	Tutorial	Credit points
BP301T	Pharmaceutical Organic Chemistry II – Theory	3	1	4
BP302T	Physical Pharmaceutics I – Theory	3	1	4
BP303T	Pharmaceutical Microbiology – Theory	3	1	4
BP304T	Pharmaceutical Engineering – Theory	3	1	4
BP305P	Pharmaceutical Organic Chemistry II – Practical	4	-	2
BP306P	Physical Pharmaceutics I – Practical	4	-	2
BP307P	Pharmaceutical Microbiology – Practical	4	-	2
BP 308P	Pharmaceutical Engineering –Practical	4	-	2
Total		28	4	24

SEMESTER IV

Course code	Name of the course	No. of hours	Tutorial	Credit points
BP401T	Pharmaceutical Organic Chemistry III– Theory	3	1	4
BP402T	Medicinal Chemistry I – Theory	3	1	4
BP403T	Physical Pharmaceutics II – Theory	3	1	4
BP404T	Pharmacology I – Theory	3	1	4
BP405T	Pharmacognosy and Phytochemistry I– Theory	3	1	4
BP406P	Medicinal Chemistry I – Practical	4	-	2
BP407P	Physical Pharmaceutics II – Practical	4	-	2
BP408P	Pharmacology I – Practical	4	-	2
BP409P	Pharmacognosy and Phytochemistry I – Practical	4	-	2
Total		31	5	28

SEMESTER V

Course code	Name of the course	No. of hours	Tutorial	Credit points
BP501T	Medicinal Chemistry II – Theory	3	1	4
BP502T	Industrial PharmacyI– Theory	3	1	4
BP503T	Pharmacology II – Theory	3	1	4
BP504T	Pharmacognosy and Phytochemistry II– Theory	3	1	4
BP505T	Pharmaceutical Jurisprudence – Theory	3	1	4
BP506P	Industrial PharmacyI – Practical	4	-	2
BP507P	Pharmacology II – Practical	4	-	2
BP508P	Pharmacognosy and Phytochemistry II – Practical	4	-	2
Total		27	5	26

SEMESTER VI

Course code	Name of the course	No. of hours	Tutorial	Credit points
BP601T	Medicinal Chemistry III – Theory	3	1	4
BP602T	Pharmacology III – Theory	3	1	4
BP603T	Herbal Drug Technology – Theory	3	1	4
BP604T	Biopharmaceutics and Pharmacokinetics – Theory	3	1	4
BP605T	Pharmaceutical Biotechnology – Theory	3	1	4
BP606T	Quality Assurance –Theory	3	1	4
BP607P	Medicinal chemistry III – Practical	4	-	2
BP608P	Pharmacology III – Practical	4	-	2
BP609P	Herbal Drug Technology – Practical	4	-	2
Total		30	6	30

SEMESTER VII

Course code	Name of the course	No. of hours	Tutorial	Credit points
BP701T	Instrumental Methods of Analysis – Theory	3	1	4
BP702T	Industrial PharmacyII – Theory	3	1	4
BP703T	Pharmacy Practice – Theory	3	1	4
BP704T	Novel Drug Delivery System – Theory	3	1	4
BP705P	Instrumental Methods of Analysis –	4	-	2
BP706PS	Practice School*	12	-	6
Total		28	5	24

* Non University Examination (NUE)

SEMESTER VIII

Course code	Name of the course	No. of hours	Tutorial	Credit points
BP801T	Biostatistics and Research Methodology	3	1	4
BP802T	Social and Preventive Pharmacy	3	1	4
BP803ET	Pharma Marketing Management			
BP804ET	Pharmaceutical Regulatory Science			
BP805ET	Pharmacovigilance			
	Quality Control and Standardization of			
BP806ET	Herbals			
BP807ET	Computer Aided Drug Design			
BP808ET	Cell and Molecular Biology			
BP809ET	Cosmetic Science	3 + 3 =		4 + 4 =
BP810ET	Experimental Pharmacology		1 + 1 = 2	
BP811ET	Advanced Instrumentation Techniques	6		8
BP812ET	Dietary Supplements and			
BP813PW	Project Work	12	-	6
	Total	24	4	22

SEMESTER WISE CREDITS DISTRIBUTION

Semester	Credit Points
I	27/29 ³ /30 #
II	29
III	26
IV	28
V	26
VI	26
VII	24
VIII	22
Extracurricular/ Co curricular activities	01*
Total credit points for the program	209/211³/212 #

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE & TECHNOLOGY
B.C.A (CLOUD TECHNOLOGY AND INFORMATION SECURITY)

SEMESTER I

Sl. No.	Course Code	Course Title	L	T	P	C	
1	ENC1182	General English – I	3	0	0	3	
2	MAC1187	Algebra ,Calculus and Trigonometry	3	1	0	4	
3	CAC1101	Communication Skills	3	0	0	3	
4	CAC1102	Computer Fundamentals & Organization	3	0	0	3	
5	CAC1103	Programming in C	3	0	0	3	
6	CAC1104	Introduction to Linux	3	0	0	3	
7	CAC1105	Programming in C Lab	0	0	3	1	
8	CAC1106	Linux Lab	0	0	3	1	21

SEMESTER II

Sl. No.	Course Code	Course Title	L	T	P	C	
1	ENC1283	General English – II	3	0	0	3	
2	MAC1288	Probability and Statistics	3	1	0	4	
3	CAC1201	Fundamentals of Storage Management	3	0	0	3	
4	CAC1202	Operating System	3	0	0	3	
5	CAC1203	OOPS with C++	3	0	0	3	
6	CAC1204	Data Structures using C	3	0	0	3	
7	CAC1205	OOPS with C++ Lab	0	0	3	1	
8	CAC1206	Data Structures Lab	0	0	3	1	
9	CAC1256	Environmental Studies	2	0	0	2	23

SEMESTER III

Sl. No.	Course Code	Course Title	L	T	P	C
1	CAC2101	Reasoning and Thinking	3	0	0	3
2	CAC2102	Information Security Fundamentals	3	0	0	3
3	CAC2103	Software Engineering	3	0	0	3
4	CAC2104	RDBMS	3	0	0	3
5	CAC2105	Computer Networks	3	0	0	3
6	CAC2106	Programming in JAVA	3	0	0	3
7	CAC2107	RDBMS Lab	0	0	3	1
8	CAC2108	Programming in JAVA Lab	0	0	3	1

20

SEMESTER IV

Sl. No.	Course Code	Course Title	L	T	P	C
1	CAC2211	Server Operating System	3	0	0	3
2	CAC2202	Ethical Hacking Fundamentals	3	0	0	3
3	CAC2203	Cryptography Fundamentals	3	0	0	3
4	CAC2212	Introduction to Cloud Technology	3	0	0	3
5	CAC2213	Fundamentals of Datacenter	3	0	0	3
6	CAC2214	Introduction to Public Speaking	2	0	0	2
7	CAC2207	Ethical Hacking – Lab	0	0	4	2
8	CAC2215	Server Operating System -Lab	0	0	4	2

21

SEMESTER V

Sl. No.	Course Code	Course Title	L	T	P	C
1	CAC3101	Computer Forensics and Investigation	3	0	0	3
2	CAC3102	Virtualization and Cloud Security	3	0	0	3
3	CAC3103	IT Governance, Risk, & Information Security Management	3	0	0	3
4	CAC3111	Linux Administration	3	0	0	3
5	CAC3112	Introduction to Cloud Computing Solution	3	0	0	3
6	CAC3113	Principles of Virtualization	3	0	0	3
7	CAC3107	Computer Forensics and Investigation – lab	0	0	4	2
8	CAC3114	Principles of Virtualization – Lab	0	0	4	2
						22

SEMESTER VI

Sl. No.	Course Code	Course Title	L	T	P	C
1	CAC3211	Mobile, wireless and VOIP Security	3	0	0	3
2	CAC3202	Project and Viva-Voce	0	0	24	16
						19

TOTAL CREDITS – 126

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE & TECHNOLOGY**B.C.A DATA SCIENCE****SEMESTER I**

Sl. No.	Course Code	Course Title	L	T	P	C	
1	ENC1182	General English – I	3	0	0	3	
2	MAC1187	Algebra, Calculus and Trigonometry	3	1	0	4	
3	CAC1102	Computer Fundamentals & Organization	3	0	0	3	
4	CAC1103	Programming in C	3	0	0	3	
5	CAC1121	Operating System	3	1	0	4	
6	CAC1122	Programming in C Lab	0	0	4	2	
7	CAC1123	Linux& Shell Programming Lab	0	0	4	2	21

SEMESTER II

Sl. No.	Course Code	Course Title	L	T	P	C	
1	ENC1283	General English – II	3	0	0	3	
2	MAC1288	Probability and Statistics	3	1	0	4	
3	CAC1221	Introduction to Data Science	3	1	0	4	
4	CAC1203	OOPS with C++	3	0	0	3	
5	CAC1204	Data Structures using C	3	0	0	3	
6	CAC1222	Programming in C++ Lab	0	0	4	2	

7	CAC1223	Data Structures and Algorithms Lab	0	0	4	2	
8	CAC1256	Environmental Studies	2	0	0	2	
9	CAC1224	Communication Skills	3	0	0	-	23

SEMESTER III

Sl. No.	Course Code	Course Title	L	T	P	C	
1	CAC2101	Reasoning and Thinking	3	0	0	3	
2	CAC2121	Database Management Systems	3	0	0	3	
3	CAC2105	Computer Networks	3	0	0	3	
4	CAC2122	Linear Algebra	3	1	0	4	
5	CAC2123	Scientific Programming Using R	3	0	0	3	
6	CAC2124	Database Management Systems Lab	0	0	4	2	
7	CAC2125	Scientific Programming Using R Lab	0	0	4	2	20

SEMESTER IV

Sl. No.	Course Code	Course Title	L	T	P	C	
1	CAC2221	Inferential Statistics	3	0	0	3	
2	CAC2222	NoSQL Databases	3	0	0	3	
3	CAC2223	Big Data Analytics	3	0	0	3	
4	CAC2224	Cloud Computing	3	1	0	4	

5	CAC2225	Exploratory Data Analysis	3	1	0	4	
6	CAC2226	Inferential Statistics Lab	0	0	4	2	
7	CAC2227	Big Data Analytics Lab	0	0	4	2	
8	CAC2228	Employability Skills	3	0	0	-	21

SEMESTER V

Sl. No.	Course Code	Course Title	L	T	P	C	
1	CAC3121	Times Series Analysis	3	0	0	3	
2	CAC3122	Data Analytics using MS Excel and SQL	3	0	2	4	
3	CAC3123	Machine Learning Algorithms	3	0	0	3	
4		Elective – I	3	0	0	3	
5		Elective – II	3	1	0	4	
6	CAC3124	Times Series Analysis Lab	0	0	4	2	
7	CAC3125	Machine Learning Algorithms Lab	0	0	4	2	21

SEMESTER VI

Sl. No.	Course Code	Course Title	L	T	P	C	
1		Elective – III	3	0	0	3	
2	CAC3221	Project and Viva-Voce	0	0	0	16	19

TOTAL CREDITS – 125

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE & TECHNOLOGY**B.C.A. MOBILE APPLICATIONS AND INFORMATION SECURITY****SEMESTER I**

Sl. No.	Course Code	Course Title	L	T	P	C	
1	ENC1182	General English – I	3	0	0	3	
2	MAC1187	Algebra. Calculus and Trigonometry	3	1	0	4	
3	CAC1101	Communication Skills	3	0	0	3	
4	CAC1102	Computer Fundamentals & Organization	3	0	0	3	
5	CAC1103	Programming in C	3	0	0	3	
6	CAC1104	Introduction to Linux	3	0	0	3	
7	CAC1105	Programming in C Lab	0	0	3	1	
8	CAC1106	Linux Lab	0	0	3	1	21

SEMESTER II

Sl. No.	Course Code	Course Title	L	T	P	C	
1	ENC1283	General English – II	3	0	0	3	
2	MAC1288	Probability and Statistics	3	1	0	4	
3	CAC1201	Fundamentals of Storage Management	3	0	0	3	
4	CAC1202	Operating System	3	0	0	3	
5	CAC1203	OOPS with C++	3	0	0	3	
6	CAC1204	Data Structures using C	3	0	0	3	
7	CAC1205	OOPS with C++ Lab	0	0	3	1	
8	CAC1206	Data Structures Lab	0	0	3	1	
9	CAC1256	Environmental Studies	2	0	0	2	23

SEMESTER III

Sl. No.	Course Code	Course Title	L	T	P	C	
1	CAC2101	Reasoning and Thinking	3	0	0	3	
2	CAC2102	Information Security Fundamentals	3	0	0	3	
3	CAC2103	Software Engineering	3	0	0	3	
4	CAC2104	RDBMS	3	0	0	3	
5	CAC2105	Computer Networks	3	0	0	3	
6	CAC2106	Programming in JAVA	3	0	0	3	
7	CAC2107	RDBMS Lab	0	0	3	1	
8	CAC2108	Programming in JAVA Lab	0	0	3	1	20

SEMESTER IV

Sl. No.	Course Code	Course Title	L	T	P	C	
1	CAC2201	Employability Skills	2	0	0	2	
2	CAC2202	Ethical Hacking Fundamentals	3	0	0	3	
3	CAC2203	Cryptography Fundamentals	3	0	0	3	
4	CAC2204	Basic Android	3	0	0	3	
5	CAC2205	Mobile Architecture and App Development	3	0	0	3	
6	CAC2206	Mobile, Wireless and VOIP Security	3	0	0	3	
7	CAC2207	Ethical Hacking Lab	0	0	4	2	
8	CAC2208	Basic Android Lab	0	0	4	2	22

SEMESTER V

Sl. No.	Course Code	Course Title	L	T	P	C	
1	CAC3101	Computer Forensics and Investigation	3	0	0	3	
2	CAC3102	Virtualization and Cloud Security	3	0	0	3	
3	CAC3103	IT Governance, Risk, & Information Security Management	3	0	0	3	

4	CAC3104	Advanced Android	3	0	0	3	
5	CAC3105	Mobile Ecosystem, Business Analysis and Models	3	0	0	3	
6	CAC3106	Enterprise Application Development	3	0	0	3	
7	CAC3107	Computer Forensics and Investigation - Lab	0	0	4	2	
8	CAC3108	Advanced Android – Lab	0	0	4	2	22

SEMESTER VI

Sl. No.	Course Code	Course Title	L	T	P	C	
1	CAC3201	Mobile Testing	3	0	0	3	
2	CAC3202	Project and Viva-Voce	0	0	24	16	19

TOTAL CREDITS – 126

LIST OF PROGRAMME ELECTIVES

ELECTIVE I

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
75.	PE	CACX40	Internet of Things	3	0	0	3
76.	PE	CACX10	Artificial Intelligence	3	0	0	3
77.	PE	CACX16	Mobile Computing	3	0	0	3

ELECTIVE II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	CACX41	Python Programming	3	1	0	4
2.	PE	CACX42	Java Programming	3	1	0	4
3.	PE	CACX43	Data Visualization	3	1	0	4

ELECTIVE III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	CACX44	Data Science Project Management	3	0	0	3
2.	PE	CACX45	Software Engineering	3	0	0	3
3.	PE	CACX46	Agile Methodology	3	0	0	3

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE & TECHNOLOGY
B. Arch
(Ten Semesters / Full Time)
CURRICULUM

SEMESTER I

S.No	Course Code	Course Name	Lecture	Studio/ Practical	Credit
Theory					
1	ARC B 1101	History of Built Environment – I	3	0	3
2	ARC B 1102	Theory of Architecture	3	0	3
Theory cum Studio					
3	ARC B 1103	Architectural Representations	1	3	3
4	ARC B 1104	Building Materials & Construction - I	2	3	4
5	ARC B 1105	Skill Development – I	1	1	2
Studio					
6	ARC B 1106	Architectural Design Studio – I	0	9	6
7	ARC B 1107	Allied Design Studio - I	1	5	4
Total Credit					25

SEMESTER II

S.No	Course Code	Course Name	Lecture	Studio/ Practical	Credit
Theory					
1	ARC B 1201	Theory & Principles of Design	3	0	3
2	ARC B 1202	Environmental Studies	3	0	3
3	ARC B 1203	Theory and Design of Structures - I	3	0	3
Theory cum Studio					
5	ARC B 1204	Building Materials & Construction - II	1	4	4
6	ARC B 1205	Skill Development – II	1	1	2
Studio					
6	ARC B 1206	Architectural Design Studio– II	0	12 to 14	6
7	ARC B 1207	Allied Design Studio - II	0	3	4
Total Credit					25

SEMESTER III

S.No	Course Code	Course Name	Lecture	Studio/ Practical	Credit
Theory					
1	ARC B 2301	History of Built Environment - II	3	0	3
2	ARC B 2302	Climatic Design	3	0	3
3	ARC B 2303	Theory and Design of Structures - II	3	0	3
Theory cum Studio					
4	ARCB 2304	Building Materials & Construction - III	2	4	4
Studio					
5	ARC B 2306	Architectural Design Studio – III	0	14 to16	8
6	ARC B 2307	Allied Design Studio – III & Computer Graphics	0	3	4
Total Credit					25

SEMESTER IV

S.No	Course Code	Course Name	Lecture	Studio/ Practical	Credit
Theory					
1	ARC B 2401	History of Built Environment - III	3	0	3
2	ARC B 2402	Building Services - I	3	0	3
3	ARC B 2403	Structural &Construction Systems –I	3	0	3
4	ARC B 2405	Site Planning &Analysis	3	0	3
Theory cum Studio					
5	ARC B 2404	Building Materials & Construction -IV	2	4	4
Studio					
6	ARC B 2406	Architectural Design Studio – IV	0	14 to 16	8
Total Credit					24

SEMESTER V

S.No	Course Code	Course Name	Lecture	Studio/ Practical	Credit
Theory					
1	ARC B 3501	History of Built Environment - IV	3	0	3
2	ARC B 3502	Building Services – II	3	0	3
3	ARC B 3503	Structural &Construction Systems II	3	0	3

4	ARC B XXXX	Elective – I	3	0	3
Theory cum Studio					
5	ARC B 3504	Building Materials & Construction -V	2	4	4
Studio					
6	ARC B 3506	Architectural Design Studio – V	0	16	8
	LEED GA	Stage I (Non Credit Course)			
Total Credit					24

SEMESTER VI

S.No	Course Code	Course Name	Lecture	Studio/ Practical	Credit
Theory					
1	AR B 3601	Contemporary Built Environment	3	0	3
2	AR B 3602	Structural &Construction Systems III	3	0	3
3	AR B 3603	Estimation and Specification	3	0	3
4	AR B XXXX	Elective – II	3	0	3
Theory cum Studio					
5	AR B 3605	Architectural Acoustics & Detailing	2	4	4
Studio					
6	AR B 3606	Architectural Design – VI	0	16	8
Total Credit					24

SEMESTER VII PRACTICAL TRAINING

S.No	Course Code	Course Name	Lecture	Studio/ Practical	Credit
1	AR B 4701	Practical Training –I	-	-	10
Total Credit					10

SEMESTER VIII

S.No	Course Code	Course Name	Lecture	Studio/ Practical	Credit
Theory					
1	AR B 4801	Professional Ethics & Practice - I	3	0	3
2	AR B 4802	Advanced Structures	3	0	3
3	AR B 4803	Green & Sustainable Design	3	0	3

4	AR B 4804	Research Methods	3	0	3
5	AR B XXXX	Elective – III	3	0	3
Studio					
6	AR B 4806	Architectural Design -VII	0	16	9
Total Credit					24

SEMESTER IX

S.No	Course Code	Course Name	Lecture	Studio/ Practical	Credit
Theory					
1	AR B 5901	Professional Ethics & Practice-II	3	0	3
2	AR B 5902	Human Settlement Planning	3	0	3
3	AR B 5903	Urban Design	3	0	3
4	AR B XXXX	Elective -IV	3	0	3
Studio					
5	AR B 5904	Dissertation	3	0	3
6	AR B 5906	Urban Design Studio - VIII	0	16	9
Total Credit					24

SEMESTER X

S.No	Course Code	Course Name	Lecture	Studio/ Practical	Credit
Theory					
1	AR B 5101	LEED AP – STAGE II	3	0	3
2	AR B XXXX	Elective -V	3	0	3
Studio					
3	AR B 5106	Architectural Thesis	0	28	14
Total Credit					20

LIST OF ELECTIVES

LIST OF BASIC ELECTIVES:

1. Interior Design (Theory)
2. Art Appreciation
3. Landscape Architecture
4. Energy Efficient Architecture
5. Digital Art
6. Vernacular Architecture

LIST OF ADVANCED ELECTIVES:

1. Urban Housing
2. Architectural Conservation
3. Earthquake resistant Buildings
4. Architectural Management
5. Construction Technology
6. Safety systems and Building Automation

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE & TECHNOLOGY**B.Sc. (COMPUTER SCIENCE)****SEMESTER I**

Sl. No.	Course Code	Course Title	L	T	P	C	
1	ENC1182	General English - I	3	0	0	3	
2	MAC1187	Algebra, Calculus and Trigonometry	3	1	0	4	
3	CAC1151	Computer Fundamentals & Organization	3	0	0	3	
4	CAC1152	Problem Solving Techniques and Programming in C	3	1	0	4	
5	CAC1153	Introduction to Linux	3	0	0	3	
6	CAC1154	Problem Solving Techniques and Programming in C Lab	0	0	4	2	
7	CAC1155	Linux Lab	0	0	4	2	23

SEMESTER II

I. No.	Course Code	Course Title	L	T	P	C	
1	ENC1283	General English - II	3	0	0	3	
2	MAC1288	Probability and Statistics	3	1	0	4	
3	CAC1251	Operating Systems	3	0	0	3	
4	CAC1252	OOPS with C++	3	1	0	4	
5	CAC1253	Data Structures Using C	3	0	0	3	

6	CAC1254	OOPS with C++ Lab	0	0	4	2	
7	CAC1255	Data Structures Using C Lab	0	0	4	2	
8	CAC1256	Environmental Studies	2	0	0	2	21

SEMESTER III

Sl. No.	Course Code	Course Title	L	T	P	C	
1	CAC2151	Database Management Systems	3	1	0	4	
2	CAC2152	Digital Electronics	3	0	0	3	
3	CAC2153	Computer Networks	3	0	0	3	
4	CAC2154	Fundamentals of Algorithms	3	0	0	3	
5	CAC2155	Multimedia and its Applications	3	0	0	3	
6	CAC2156	Database Management Systems Lab	0	0	4	2	
7	CAC2157	Multimedia Lab	0	0	4	2	
8	CAC2158	Communication Skills	0	0	4	2	22

SEMESTER IV

Sl. No.	Course Code	Course Title	L	T	P	C	
1	CAC2251	Programming in Java	3	1	0	4	
2	CAC2252	Computer Security	3	0	0	3	
3	CAC2253	Software Engineering	3	0	0	3	

4	CAC2254	Information Storage and Management	3	0	0	3	
5		Elective I	3	0	0	3	
6	CAC2255	Programming in Java Lab	0	0	4	2	20
7	CAC2256	Soft Skills and Personality Development	0	0	4	2	

SEMESTER V

Sl. No.	Course Code	Course Title	L	T	P	C	
1	CAC3151	C# and .NET Programming	3	1	0	4	
2	CAC3152	Web Designing	3	0	0	3	
3	CAC3153	Software Testing	3	0	0	3	
4		Elective 2	3	0	0	3	
5		Elective 3	3	0	0	3	
6	CAC3154	C# and .NET Programming Lab	0	0	4	2	
7	CAC3155	Web Designing Lab	0	0	4	2	20

SEMESTER VI

Sl. No.	Course Code	Course Title	L	T	P	C	
1	CAC3251	Network Security	3	1	0	4	
2	CAC3252	Data Mining and Warehousing	3	1	0	4	
3		Elective 4	3	0	0	3	

4		Elective 5	3	0	0	3	
5	CAC3253	Project	0	0	12	6	20

TOTAL CREDITS – 124

LIST OF PROGRAMME ELECTIVES

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
78.	PE	CACX01	Advanced Computer Architecture	3	0	0	3
79.	PE	CACX02	Compiler Design	3	0	0	3
80.	PE	CACX03	Database Design	3	0	0	3
81.	PE	CACX04	E-Commerce	3	0	0	3
82.	PE	CACX05	High Performance Computer Systems	3	0	0	3
83.	PE	CACX06	Internet and Web Programming	3	0	0	3
84.	PE	CACX07	Management Information Systems	3	0	0	3
85.	PE	CACX08	Parallel Algorithms	3	0	0	3
86.	PE	CACX09	Wireless Networks	3	0	0	3

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
4.	PE	CACX10	Artificial Intelligence	3	0	0	3
5.	PE	CACX11	Client / Server Technology	3	0	0	3
6.	PE	CACX12	Distributed Operating System	3	0	0	3
7.	PE	CACX13	Embedded Systems	3	0	0	3
8.	PE	CACX14	Enterprise Resource Planning	3	0	0	3
9.	PE	CACX15	Mobile Commerce	3	0	0	3
10.	PE	CACX16	Mobile Computing	3	0	0	3

11.	PE	CACX17	Software Project Management	3	0	0	3
12.	PE	CACX18	Web Technology	3	0	0	3

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
4.	PE	CACX19	Big Data Analytics	3	0	0	3
5.	PE	CACX20	Human Computer Interaction	3	0	0	3
6.	PE	CACX21	Software Quality Assurance	3	0	0	3
7.	PE	CACX22	Web Services and Service Oriented Architecture	3	0	0	3

B.S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE & TECHNOLOGY**B.DES. INTERIOR ARCHITECTURE****SEMESTER I**

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	Studio	IA C 1101	Basic Studio- I	0	0	11	6	
2.	Theory cum studio	IA C 1102	Allied Design Studio- I	2	0	4	4	
3.	Theory cum studio	IA C 1103	Workshop- I	1	0	4	3	
4.	Theory cum studio	IA C 1104	Integrated Design Studio- I	1	0	4	3	
5.	Theory	IA C 1105	Skill Development-I	2	0	0	2	
6.	Theory	IA C 1106	Theory of Design	3	0	0	3	21

SEMESTER II

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	Studio	IA C 1201	Interior Design Studio- I	0	-	11	6	
2.	Theory cum studio	IA C 1202	Allied Design Studio- II	2	0	3	4	
3.	Theory cum studio	IA C 1203	Workshop- II	1	0	3	3	
4.	Theory cum studio	IA C 1204	Integrated Design Studio- II	1	0	4	3	
5.	Theory	IA C 1205	History of Interior Design- I	3	0	0	3	
6.	Theory		Theory- Open Elective I	3	0	0	3	22

SEMESTER III

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	Studio	IA C 2301	Interior Design Studio- II	0	0	11	6	
2.	Theory cum studio	IA C 2302	Allied Design Studio- II	2	0	3	4	
3.	Theory cum studio	IA C 2303	Workshop- III	1	0	3	3	
4.	Theory cum studio	IA C 2304	Integrated Design Studio- II	1	0	4	3	
5.	Theory	IA C 2305	Estimation & Costing	3	0	0	3	
6.	Theory		Theory- Open Elective II	3	0	0	3	22

SEMESTER IV

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	Studio	IA C2401	Interior Design Studio- III	0	0	12	6	
2.	Theory cum studio	IA C2402	Allied Design Studio- IV	2	0	3	4	
3.	Theory cum studio	IA C 2403	Workshop- IV	1	0	3	3	
4.	Theory cum studio	IA B2404	Integrated Design Studio- III	1	0	3	3	
5.	Theory	IA C 2405	History of Interior Architecture- II	3	0	0	3	
6.	Theory		Theory- Open Elective III	3	0	0	3	22

SEMESTER V

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	Studio	IA C 3501	Interior Design Studio- IV	0	0	17	8	
2.	Theory cum studio	IA C 3502	Allied Design Studio- V	2	0	4	4	
3.	Theory cum studio	IA C 3504	Integrated Design Studio- IV	1	0	4	3	
4.	Theory	IA C 3505	Furniture Design	3	0	0	3	
5.	Theory		Theory- Open Elective IV	3	0	0	3	21

SEMESTER VI

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.	Studio	IA C 3601	Interior Design Studio- VI	0	0	22	9	
2.	Theory cum studio	IA C 3602	Allied Design Studio- V	2	0	6	5	
3.	Theory	IA C3603	Professional Practice : Contracts, Tenders, Competitions Arbitrations	3	0	0	3	
4.	Theory		Theory- Open Elective V	3	0	0	3	20

SEMESTER VII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C	
1.		IA C4701	Practical Training	0	0	0	10	10

SEMESTER VIII

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.		IA C4801	Interior Architectural Thesis	0	0	0	20 20

Total credits – 158

LIST OF PROFESSIONAL ELECTIVES

Sl. No.	Course Group	Course Code	Course Title	L	T	P	C
1.	PE	IA C X01	Vernacular Architecture	3	0	0	3
2.	PE	IA C X02	Energy efficient design	3	0	0	3
3.	PE	IA C X03	Strategies for Innovation in Design	3	0	0	3
4.	PE	IA C X04	Landscape Design	3	0	0	3
5.	PE	IA C X05	Digital Technology	3	0	0	3
6.	PE	IA C X06	Advanced Building Energy Efficiency	3	0	0	3
7.	PE	IA C X07	Advanced Product design	3	0	0	3
8.	PE	IA C X08	Advanced Lighting design	3	0	0	3
9.	PE	IA C X09	Interior Architecture Photography	3	0	0	3
10.	PE	IA C X10	Advanced Furniture design	3	0	0	3



PG PROGRAMMES

CURRICULUM

M.TECH PROGRAMMES M.ARCH & ARTS AND SCIENCE PROGRAMMES

DEPARTMENT OF MANAGEMENT STUDIES

MBA CURRICULUM 2016

Trimester I					
Course Code	Course Name	L	T	P	C
MSC 6101	Management Concepts	3	0	0	2
MSC 6102	Entrepreneurship Development	2	0	2	2
MSC 6103	Managerial Economics	3	0	0	2
MSC 6104	Quantitative Methods in Business I	3	0	0	2
MSC 6105	Communication Skills	2	0	2	2
MSC 6106	Financial Accounting	3	0	0	2
Career Oriented Practicals					
MSC 6107	Computers and Business Application Lab	0	0	3	1
Total Credit (Trimester I)					13

Trimester II					
Course Code	Course Name	L	T	P	C
MSC 6201	Organizational Behaviour	3	0	0	2
MSC 6202	Legal Aspects of Business	3	0	0	2
MSC 6203	Economic Environment of Business	3	0	0	2
MSC 6204	Quantitative Methods in Business II	3	0	0	2
MSC 6205	Management and Cost Accounting	3	0	0	2
MSC 6206	Marketing Management – I	3	0	0	2
Career Oriented Practicals					
MSC 6207	Written Analysis and Communication	0	0	3	1
MSC 6208	Current Affairs in Business	0	0	3	1
Total Credit (Trimester II)					14

Trimester III					
Course Code	Course Name	L	T	P	C
MSC 6301	Management Information System	3	0	0	2
MSC 6302	Financial Management	3	0	0	2
MSC 6303	Marketing Management – II	3	0	0	2
MSC 6304	Human Resources Management	3	0	0	2
MSC 6305	Production and Operations Management	3	0	0	2
MSC 6306	Research Methodology for Management	3	0	0	2
Career Oriented Practicals					
MSC 6307	Psychometrics and Corporate Connect Lab	0	0	3	1
MSC 6308	Data Analysis Lab	0	0	3	1
Total Credit (Trimester III)					14

Students are to undergo summer internship after the end-term examination of III trimester. Details are given in regulation.

Trimester IV					
Course Code	Course Name	L	T	P	C
MSC7101	International Business Management	2	0	0	1
MSC7102	Introduction to Enterprise Resources Planning	2	0	0	1
	Program Electives	15	0	0	10
Career Oriented Practicals					
MSC7103	Integrated Management Cases	0	0	3	1
MSC7104	Summer Internship*	-	-	-	2
	Total Credit (Trimester IV)				15

*Though summer internship is mentioned in the IV trimester, students have to undergo summer internship in a business organisation after the end-term examination of III trimester. Details are given in regulation.

Trimester V					
Course Code	Course Name	L	T	P	C
MSC7201	Strategic Management	3	0	0	2
	Program Electives	15	0	0	10
Career Oriented Practicals					
MSC7202	Sales Negotiation Lab	0	0	3	1
	Total Credit (Trimester V)				13

Trimester VI					
Course Code	Course Name	L	T	P	C
MSC 7301	Project Work	-	-	10	10
	Total Credit (Trimester VI)				10

TOTAL CREDITS: 79

LIST OF ELECTIVES IN FUNCTIONAL AREAS

Code	Course	L	T	P	C
1. MARKETING MANAGEMENT					
MSCY 001	Retail Marketing	3	0	0	2
MSCY 002	Services Marketing	3	0	0	2
MSCY 003	Marketing Communications and Media Management	3	0	0	2
MSCY 004	Consumer Behaviour	3	0	0	2

MSCY 005	Product and Brand Management	3	0	0	2
MSCY 006	Industrial Marketing	3	0	0	2
MSCY 007	Customer Relationship Management and Marketing	3	0	0	2
MSCY 008	Rural Marketing	3	0	0	2
MSCY 009	Sales and Distribution Management	3	0	0	2
MSCY 010	Digital Marketing	3	0	0	2
2. FINANCIAL MANAGEMENT					
MSCY 021	Security market and analysis	3	0	0	2
MSCY 022	Financial Services	3	0	0	2
MSCY 023	International Financial Management	3	0	0	2
MSCY 024	Strategic Financial Management	3	0	0	2
MSCY 025	Financial Reengineering	3	0	0	2
MSCY 026	Derivatives Management	3	0	0	2
MSCY 027	Risk Management and Insurance	3	0	0	2
MSCY 028	Banking Management	3	0	0	2
MSCY 029	Portfolio Management	3	0	0	2
3. HUMAN RESOURCES MANAGEMENT					
MSCY 041	Managerial Behaviour and Effectiveness	3	0	0	2
MSCY 042	Human Resources Accounting & Audit	3	0	0	2
MSCY 043	Performance Management and Training	3	0	0	2
MSCY 044	Industrial Relations and Labour Legislation	3	0	0	2
MSCY 045	Organisational Structure and Design	3	0	0	2
MSCY 046	Strategic Human Resource Management	3	0	0	2
MSCY 047	International Human Resources Management and Organisational Behaviour	3	0	0	2
MSCY 048	Negotiation Skills and Conflict Management	3	0	0	2
4. SYSTEMS MANAGEMENT					
MSCY 061	Business Analysis and Systems Development	3	0	0	2
MSCY 062	Database Management Systems	3	0	0	2
MSCY 063	E-Commerce Technology	3	0	0	2
MSCY 064	Strategic Information Technologies for Management	3	0	0	2
MSCY 065	Project Management	3	0	0	2
MSCY 066	Web Technologies	3	0	0	2
MSCY 067	Big Data in Management	3	0	0	2
MSCY 068	Application of Enterprise Resource Planning	3	0	0	2
5. OPERATIONS MANAGEMENT					
MSCY 081	Supply Chain Management	3	0	0	2
MSCY 082	Total Quality Management	3	0	0	2
MSCY 083	Integrated Materials Management	3	0	0	2

MSCY 084	Lean Six Sigma	3	0	0	2
MSCY 085	Logistics Management	3	0	0	2
MSCY 086	Services Operations Management	3	0	0	2
MSCY 087	Product Design	3	0	0	2
MSCY 088	System Simulation and Modelling	3	0	0	2
6. ISLAMIC BANKING AND FINANCE					
MSCY 101	Introduction of Islamic Economics	3	0	0	2
MSCY 102	Fundamentals of Islamic Finance	3	0	0	2
MSCY 103	Islamic Insurance (Takaful)	3	0	0	2
MSCY 104	Islamic Bonds (Sukuk)	3	0	0	2
MSCY 105	Islamic Asset and Fund Management	3	0	0	2
MSCY 106	Islamic Banking Products and Services	3	0	0	2
7. ENTREPRENEURSHIP					
MSCY 121	Entrepreneurship, Family Enterprises & Successful Ventures	3	0	0	2
MSCY 122	Financial Institutions and Funding for Entrepreneurs	3	0	0	2
MSCY 123	Marketing for Entrepreneurs	3	0	0	2
MSCY 124	New Venture and Project Preparation	3	0	0	2

MASTER OF COMPUTER APPLICATIONS
(SIX SEMESTERS / FULL TIME)

Sl. No.	Course Code	Course Title	L	T	P	C
SEMESTER I						
1	MAC6185	Discrete mathematics	3	1	0	4
2	CAC6101	Programming in C and C++	3	0	0	3
3	CAC6102	Computer Organization	3	0	0	3
4	CAC6103	Database Management Systems	3	0	0	3
5	CAC6104	Computer Networks	3	0	0	3
6	CAC6105	C and C++ Programming Lab	0	0	4	2
7	CAC6106	DBMS Lab	0	0	4	2
						20
SEMESTER II						
1	CAC6201	Computer Graphics and Multimedia Systems	3	0	1	4
2	CAC6202	Software Engineering	3	0	0	3
3	CAC6203	Data Structures and Design and Analysis of Algorithms	3	0	0	3
4	CAC6204	Operating Systems	3	0	0	3
5		Elective – 1	3	0	0	3
6	CAC6205	Data Structures and Algorithms Lab	0	0	4	2
7	CAC6206	Communication Skills Laboratory	0	0	2	2
8	CAC6206	Seminar	0	0	0	1
						21
SEMESTER III						
1	CAC7101	Object Oriented Analysis and Design	3	0	0	3
2	CAC7102	Internet and Java Programming	3	0	0	3
3	MAC6186	Resource Management Techniques	3	1	0	4

Sl. No.	Course Code	Course Title	L	T	P	C
4		Elective – 2	3	0	0	3
5		Elective –3	3	0	0	3
6		Elective – 4	3	0	0	3
7	CAC7103	Case tools Lab/OOAD Lab	0	0	4	2
8	CAC7104	Programming in JAVA	0	0	4	2
9	CAC7105	Programming Languages				1
						24

SEMESTER IV

1	CAC7201	Advanced Web Design and Development	3	0	0	3
2	CAC7202	XML and Web Services	3	0	0	3
3	CAC7203	Mobile Computing	3	0	0	3
4		Elective – 5	3	0	0	3
5		Elective – 6	3	0	0	3
6		Elective – 7	3	0	0	3
7	CAC7204	Web design and development Lab	0	0	4	2
8	CAC7205	XML and Web Services Lab	0	0	4	2
9	CAC7206	Soft Skills and Personality Development	0	0	2	2
						24

Sl. No.	Course Code	Course Title	L	T	P	C
SEMESTER V						
1	CAC8101	Mobile Application Development	3	0	0	3
2	MSC8181	Essentials of management for computer professionals	3	0	0	3
3	CAC8102	Internet of Things	3	0	0	3
4		Elective – 8	3	0	0	3

5		Elective – 9	3	0	0	3
6	CAC8103	Mobile Application Development Lab	0	0	4	2
7	CAC8104	Mini Project	0	0	4	2
						19

SEMESTER VI

1	CAC8201	Project					12
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LIST OF ELECTIVES

Sl. No.	Course Code	Course Title	L	T	P	C
Semester II						
1	CACY001	Management Information System	3	0	0	3
2	CACY002	Accounting and Financial Management	3	1	0	4
3	CACY003	E-commerce	3	0	0	3
4	CACY004	Business Processes	3	0	0	3
Semester III						
5	CACY005	Grid Computing	3	0	0	3
6	CACY006	Cloud computing	3	0	0	3
7	CACY007	Unix and Network Programming	3	0	0	3
8	CACY008	Multimedia systems and Algorithms	3	0	0	3
9	CACY009	Network Security	3	0	0	3
10	CACY010	Microprocessor and its applications	3	0	0	3
11	CACY011	TCP/IP Protocol Suite	3	0	0	3
12	MSCY181	Management functional areas and information Systems	3	0	0	3
13	CACY012	Adhoc Networks	3	0	0	3
Semester IV						
14	CACY013	Digital Image Processing	3	0	0	3
15	CACY014	Probability and Statistics	3	1	0	4
16	CACY015	Software Quality Management	3	0	0	3
17	CACY016	Embedded Systems	3	0	0	3
18	CACY017	Business Intelligence	3	0	0	3
19	CACY018	Software Testing	3	0	0	3
20	CACY019	Content Management System	3	0	0	3
21	CACY020	Advanced Programming Techniques	3	0	0	3
22	CACY021	Information Storage and Management	3	0	0	3
23	CACY022	Semantic Web	3	0	0	3

Sl. No.	Course Code	Course Title	L	T	P	C
24	CACY023	Healthcare Analytics	3	0	0	3
25	CACY024	Python Programming	3	0	0	3
26	CACY025	Enterprise Resource Planning	3	0	0	3
27	CACY026	Software Project Management	3	0	0	3
28	CACY027	Data Mining and Data warehousing	3	1	0	3
Semester V						
29	CACY028	Unix Internals	3	0	0	3
30	CACY029	Advanced Databases	3	0	0	3
31	CACY030	Software Quality Assurance	3	0	0	3
32	CACY031	Service Oriented Architecture	3	1	0	3
33	CACY032	C# and .NET Framework	3	0	0	3
34	CACY033	PHP Programming	3	0	0	3
35	CACY034	Online Computer Advertising	3	0	0	3
36	CACY035	Web Mining	3	0	0	3
37	CACY036	Digital marketing	3	0	0	3
38	CACY037	Big data and its Analytics	3	0	0	3
39	CACY038	Information Retrieval	3	0	0	3
40	CACY039	Social Media Analysis	3	0	0	3
41	CACY040	Human Computer Interaction	3	0	0	3
42	CACY041	Bio-Informatics	3	0	0	3

S. No.	Course Code	Course Name	L	T	P	C	TC
SEMESTER I							
1.	MAC6182	Differential Equations and Numerical Methods	3	1	0	4	
2	MEC6101	Applied Materials Engineering	3	0	0	3	
3	MEC6102	Computer Graphics and Geometric Modeling (Integrated Lab)	3	0	2	4	
4	MEC6103	Advanced Finite Element Analysis (Integrated Lab)	2	1	2	4	
5	MEC6104	Rapid Prototyping	2	0	0	2	
6		Professional Electives #				3	
		Total credit (Semester I)					20
SEMESTER II							
1	GEC6201	Research Methodology	3	0	0	3	
2	MEC6205	Integrated Product Design	3	0	0	3	
3	MEC6206	Advanced Metrology and NDT (Integrated Lab)	3	0	2	4	
4	MEC6207	Flexible Manufacturing Systems	3	0	0	3	
5		Professional Electives ##				6	
6	MEC6208	Design Project	0	0	2	1	
		Total credit (Semester II)					20
SEMESTER III							
1		General Elective*	3	0	0	3	
2	MEC7109	Non Linear Finite Element Analysis	1	0	0	1	
3		Professional Electives ###				9	
4	MEC7110	Industry Internship	0	0	0	1	
5	MEC7101	Project Work - Phase I	0	0	12	6**	
		Total credit (Semester III)					14
SEMESTER IV							
1	MEC7101	Project Work - Phase II	0	0	36	18**	
		Total credit (Semester IV)			18 + 6		24
Grand Total credit							78

- # Professional Electives can be chosen from the list, provided that the cumulative credits should not be less than 3.
- ## Professional Electives can be chosen from the list, provided that the cumulative credits should not be less than 6.
- ### Professional Electives can be chosen from the list, provided that the cumulative credits should not be less than 9.
- * General Electives can be chosen from the list, provided that the cumulative credits should not be less than 3.
- ** Credits for project work phase I to be accounted along with project work phase II in IV semester

Details of Credits		
S. No.	Details of curriculum	Total Credits
1	Program Core	24
2	Professional Electives	18
3	General Electives	3
4	Project	25
5	Internship	1
6	Mathematics and Research Methodology	7
	Total	78

LIST OF PROFESSIONAL ELECTIVES
PROFESSIONAL ELECTIVES FOR SEMESTER I

S. No.	Course Code	Course Title	L	T	P	C
1	MECY01	Automotive Manufacturing	1	0	0	1
2	MECY02	Design of Hydraulic & Pneumatic Systems	3	0	0	3
3	MECY03	Mechanics of composite materials	2	0	0	2
4	MECY04	Mechatronics for Manufacturing Systems	2	0	0	2
5	MECY05	Precision Engineering and Nano Technology	3	0	0	3

PROFESSIONAL ELECTIVES FOR SEMESTER II

S. No.	Course Code	Course Title	L	T	P	C
6	MECY06	Advanced Strength of Materials	3	0	0	3
7	MECY07	Advances in Manufacturing Technology	3	0	0	3
8	MECY08	CNC Machines and Computer Aided Manufacturing	3	0	0	3
9	MECY09	Computer Aided Process Planning	3	0	0	3
10	MECY10	Integrated Manufacturing Systems and Management	2	0	0	2
11	MECY11	Manufacturing Information Systems	3	0	0	3
12	MECY12	Mechanical Vibrations	3	0	0	3
13	MECY13	Optimization Techniques in Design	3	0	0	3
14	MECY14	Virtual Manufacturing	1	0	0	1

PROFESSIONAL ELECTIVES FOR SEMESTER III

S. No.	Course Code	Course Title	L	T	P	C
15	MECY15	Advanced Mechanisms Design and Simulation	3	0	0	3
16	MECY16	Advanced Tool Design	3	0	0	3
17	MECY17	Computational Fluid Dynamics	3	0	0	3
18	MECY18	Data Communication in CAD/CAM	3	0	0	3
19	MECY19	Design of Material Handling Equipment	3	0	0	3
20	MECY20	Geometric Dimensioning and Tolerancing	1	0	0	1
21	MECY21	Industrial Robotics and Flexible Automation	3	0	0	3
22	MECY22	Industrial Safety Management	3	0	0	3
23	MECY23	Newer Materials	2	0	0	2
24	MECY24	Polymers and Composites	3	0	0	3
25	MECY25	Product Life Cycle Management	1	0	0	1
26	MECY26	Reliability and Total Productive Maintenance	2	0	0	2
27	MECY27	Tribology	3	0	0	3

M.TECH. (COMPUTER SCIENCE AND ENGINEERING)

S.No	Semester	Course Code	Course Title	L	T	P	C
1	I	MAC6181	Applied Algebra and Discrete Algorithms	3	1	0	4
2		CSC6101	Advanced Computer Architecture	3	1	0	4
3		CSC6102	Algorithmic Design and Implementation	3	0	2	4
4		CSC6103	Computer Networks and Management	2	0	2	3
5		CSC6104	Distributed Operating System	3	0	0	3
6		CSC6105	Advanced Software Engineering	3	0	0	3
7			Case Study	0	0	2	1
							22
8	II	CSC6201	Machine Learning Techniques	2	0	2	3
9			Research Methodology	3	0	0	3
10		CSC6202	Applied Cryptography & Network Security	3	0	2	4
11			Professional Electives#				9
							19
12	III	CSC7101	Interns / Term paper*	0	0	2	1
13			Professional Electives##				9
14			General Elective	3	0	0	3
15		CSC7102	Project – Phase I	0	0	12	6
							13
16	IV	CSC7102	Project – Phase II	0	0	36	18
							6+18=24
Total							78

- Student has to take a minimum of 9 credits from the list of Electives.

* - Interns must be taken after the completion on II semester examinations.

- Student has to take a minimum of 9 credits from the list of Electives.

Semester	Credits				Total Credits
	Core Courses	Professional electives	General Electives	Lab/ Case Study/ Project	
I	21	-	-	1	22
II	10	9	-	0	19
III	-	9	3	1 +6*	13
IV	-	-	-	18 + 6 = 24	24
Total Credits	31	18	3	26	78

PROFESSIONAL ELECTIVES

S.No	Course Code	Course Title	L	T	P	C
1.	CSCY01	Mobile Computing	1	0	2	2
2.	CSCY02	Web Technology	1	0	2	2
3.	CSCY03	XML and Web Services	1	0	2	2
4.	CSCY04	Multimedia Systems	1	0	2	2
5.	CSCY05	Software Testing & Quality Assurance	3	0	0	3
6.	ECC6121	Advanced Embedded Systems	3	0	0	3
7.	CSCY06	Mobile Ad hoc Networks	3	0	0	3
8.	CSCY07	Data warehousing and Data mining	3	0	0	3
9.	CSCY08	Performance evaluation of Computer systems and Networks	3	0	0	3
10.	CSCY09	Agent Based Intelligent Systems	1	0	2	2
11.	CSCY10	Advanced Databases	3	0	0	3
12.	CSCY11	Statistical Natural Language Processing	2	0	2	3
13.	CSCY12	Component Based Technology	2	0	2	3
14.	CSCY13	Real Time Systems	3	0	0	3
15.	CSCY14	Multi-core Programming	3	0	0	3
16.	CSCY15	Digital Image Processing	3	0	0	3
17.	CSCY16	Human Computer Interaction	3	0	0	3
18.	CSCY17	Software Project Management	3	0	0	3
19.	CSCY18	Dependable Computing	3	0	0	3

20.	CSCY19	Cognitive Science	3	0	0	3
21.	CSCY20	Deep Learning	3	0	0	3
22.	CSCY21	Pervasive Computing	3	0	0	3
23.	ITC6201	Cloud Computing Technologies	3	0	0	3
24.	ECCY030	Internet of Things	3	0	0	3
25.	CSCY23	Advanced Graphics Programming	3	0	0	3
26.	CSCY24	Computer Vision	3	0	0	3

General Electives

S. No.	Course code	Course Title	L	T	P	C
1	SSB7181	Society, Technology and Sustainability	3	0	0	3
2	CSCZ01	Green Computing	3	0	0	3
3	CSCZ02	Gaming Design	3	0	0	3
4	CSCZ03	Social Computing	3	0	0	3

Single Elective Course

S. No.	Course code	Course Title
1	CSCZ04	Wireless Network Programming
2	CSCZ05	Cloud Computing Programming
3	CSCZ06	Big Data Programming
4	CSCZ07	Visual Systems Programming
5	CSCZ08	Open Source tools
6	CSCZ09	Other Industry related Courses

M.TECH. POWER SYSTEMS ENGINEERING
FOUR SEMESTERS / FULL TIME
SEMESTER I

S. No	Course Code	Course Name	L	T	P	C
Semester I						
1.		Applied Mathematics	3	1	0	4
2.		System Theory	3	0	0	3
3.	EEC6101	Advanced Power System Analysis	3	1	0	4
4.	EEC6102	Power System Protection	3	0	0	3
5.		Professional Electives [#]				6 [#]
6.	EEC6103	Power System Analysis & Protection Lab	0	0	2	1
7.	EEC6104	Design Simulation Project	0	0	2	1
		Total Credit (semester I)				22

[#] Minimum of 6 credits

Semester II						
1.	EEC6211	Power system Operation and Control	3	0	0	3
2.	EEC6212	Power System Dynamics	3	0	0	3
3.	EEC6213	Flexible AC Transmission systems	3	0	0	3
4.	EEC6214	Restructured Power Systems	3	0	0	3
5.		Research Methodology	3	0	0	3
6		Professional Elective ^{##}				3 ^{##}
7	EEC6215	Power System Dynamics Control & Operation Laboratory	0	0	2	1
8	EEC6216	Self Learning	0	2	0	2
9	EEC6217	Industrial Internship	0	0	*	**
		Total Credit (semester II)				21

^{##} Minimum of 3 credits

* Minimum of 30 days Industrial Internship

** Industrial internship will be undertaken during first year summer vacation. The credit will be awarded in the 3rd semester.

S. No	Course Code	Course Name	L	T	P	C
Semester III						
		General Elective	3	0	0	3
		Professional Electives###				9###
	EEC6217	Industrial Internship	0	0	*	2
	EEC7101	Project Phase I #####	0	0	12	6#####
		Total Credit (semester III)				14
### Minimum of 9 credits						
* Minimum of 30 days Industrial Internship						
Semester IV						
	EEC7101	Project Phase II #####	0	0	36	18#####
		Total Credit (semester IV)	18+6 =			24
##### Credits for Project Work Phase I to be accounted along with Project Work Phase II in IV Semester.						
		Grand Total of credits				81

LIST OF ELECTIVES

General Elective (GE1) (to be appended) (3 credits)

1. Project management (3 credits)
2. Society, Technology & Sustainability (3 credits)
3. Artificial Intelligence (3 credits)
4. Programming using MATLAB & SIMULINK (3 credits)
5. Intellectual Property Rights (3 credits)

List of Professional Electives :

S.No	Course Code	Course
1.	EECY01	Electro Magnetic Field Computation and Modeling
2.	EECY02	Advanced Power System Dynamics
3.	EECY03	EHV Power Transmission
4.	EECY04	Special Electrical Machines and Controllers

5.	EECY05	Power Quality
6.	EECY06	Power system Planning and Reliability
7.	EECY07	Advanced Digital Signal Processing
8.	EECY08	Control System Design
9.	EECY09	High Voltage Switch Gear
10.	EECY10	Optimal Control and Filtering
11.	EECY11	Industrial Power System Analysis and Design
12.	EECY12	High Voltage Direct Current Transmission
13.	EECY13	Wind energy Conversion Systems
14.	EECY14	Application of MEMS Technology
15.	EECY15	Outdoor Insulators
16.	EECY16	Power Distribution Systems
17.	EECY17	Electrical Transients in Power Systems
18.	EECY18	High voltage Pulse Generation, Measurement and Testing for Life Sciences
19.	EECY19	Smart Power Grid
20.	EECY20	Distributed Generation and Microgrid
21.		Soft Computing Techniques
22.	EECY21	Reactive Power Management in Power System
23.	EECY22	Power Electronics in Wind and Solar Power Conversion
24.	EECY23	Power Electronics Applications to Power Systems
25.	EECY24	SCADA and DCS
26.		Advanced Power Semiconductor Devices
27.		Modeling and Analysis of Electrical Machines
28.		Analysis of Converters & Inverters
29.		Solid State Drives
30.	EECY25	Electrical Insulation in Power Equipments and Systems
31.	EECY26	Energy Auditing
32.	EECY27	Wide Area Measurement Systems
33.	EECY28	Power System Simulation Software
34.	EECY29	Simulation of Power System Circuits
35.	EECY30	Electric Vehicles

M.TECH. (COMMUNICATION SYSTEMS)
(FOUR SEMESTERS / FULL TIME)
SEMESTER I

Sl. No	Couse Category	Course Code	Course Title	L	T	P	C
1.	BS	MAC61XX	Applied Mathematics	3	1	0	4
2.	EC	ECC6101	Mobile Communication Networks and Modeling	3	0	2	4
3.	EC	ECC6102	Satellite Communication	3	0	0	3
4.	EC	ECC6103	Advanced Digital Communication Techniques	3	0	2	4
5.	EC	ECC6104	Advanced Digital Signal Processing	3	1	0	4
6.	PE		Professional Elective – Minimum of '3' credits to be earned				3
							22
			SEMESTER II				
Sl. No		CourseCode	Course Title	L	T	P	C
1.		GEC6201	Research Methodology	3	0	0	3
2.	EC	ECC6201	Optical Networks	3	0	0	3
3.	EC	ECC6202	Advanced Radiation Systems and practice	3	0	2	4
4.	PE		Professional Electives – Minimum of '9' credits to be earned				9
			SEMESTER III				19
Sl. No		Course Code	Course Title	L	T	P	C
1.		GEC	General Elective	3	0	0	3
2.	PE		Professional Electives – Minimum of '6' credits to be earned				6
3.		ECC7101	Mini Project /internship**	0	0	3	1
4.		ECC7102	Project Work - Phase I*	0	0	12	6*
							16
			SEMESTER IV				
Sl. No		Course Code	Course Title	L	T	P	C
1.		ECC7103	Project Work - Phase II	0	0	36	18*
						18 + 6 = 24	

* Credits for Project Work Phase I to be accounted along with Project Work Phase II in IV Semester

** Internship has to be carried out at the end of second semester during summer vacation

TOTAL CREDITS: 75

LIST OF PROFESSIONAL ELECTIVES

Sl.No	Course Code	Course	L	T	P	C
1.	ECCY001	Digital Image Processing	3	0	0	3
2.	ECCY002	Simulation of Communication Systems and Networks	2	0	0	2
3.	ECCY003	Global Tracking and Positioning Systems	3	0	0	3
4.	ECCY004	Electromagnetic Interference and Compatibility in System Design	3	0	0	3
5.	ECCY005	High Performance Communication Networks	3	0	0	3
6.	ECCY006	Digital Communication Receivers	3	0	0	3
7.	ECCY007	Advanced Microwave systems	3	0	0	3
8.	ECCY008	Speech and Audio Signal Processing	3	0	0	3
9.	ECCY009	Network Security	3	0	0	3
10.	ECCY010	Wireless Communication	3	0	0	3
11.	ECCY011	Medical Image Processing	3	0	0	3
12.	ECCY012	Internet Working Multimedia	3	0	0	3
13.	ECCY013	QoS in Ad Hoc Wireless Networks	3	0	0	3
14.	ECCY014	Wireless sensor Networks	3	0	0	3
15.	ECCY015	RF System Design	3	0	0	3
16.	ECCY016	MIMO systems	3	0	0	3
17.	ECCY017	Cognitive and Co-operative Radio Communications	3	0	0	3
18.	ECCY018	RF Wireless Systems And Standards	3	0	0	3
19.	ECCY019	Communication System Design and Analysis	2	0	2	3
20.	ECCY020	Quantum Computing	3	0	0	3
21.	ECCY021	Error Control Coding	3	0	0	3
22.	ECCY022	Microstrip Antennas	3	0	0	3

23.	ECCY023	Vehicular AD HOC Networks	3	0	0	3
24.	ECCY024	Microwave Integrated Circuits	3	0	0	3
25.	ECCY025	Radiation systems for personal area network	2	0	0	2
26.	ECCY026	Bio signal Processing	1	0	0	1
27.	ECCY027	Ultrasonic Principles and Applications	1	0	0	1
28.	ECCY028	Network Routing Algorithms	2	0	0	2
29.	ECCY029	Adaptive Signal Processing	2	0	0	2
30.	ECCY030	Internet of Things	3	0	0	3
31.	ECCY031	Multimedia Compression Techniques	3	0	0	3
32.	ECCY032	Statistical Signal Processing	3	0	0	3
33.	ECCY033	Classical and Advanced Techniques For Optimization	3	0	0	3
34.	ECCY034	Pattern Recognition Techniques and Applications	2	0	0	2
35.	MACXXX	Applied Mathematics on Transforms for Signal Processing	3	0	0	3

M.TECH. (VLSI & EMBEDDED SYSTEMS)
(FOUR SEMESTERS / FULL TIME)

SEMESTER I

Sl. No	Course Code	Course Title	L	T	P	C
1.	MAC61XX	Applied Mathematics	3	1	0	4
2.	ECC6121	Advanced Embedded System	3	0	0	3
3.	ECC6122	Digital VLSI Design	3	0	2	4
4.	ECC6123	Advanced Microcontroller architectures and Programming	3	0	2	4
5.	ECC6124	Embedded System Lab	0	0	2	1
6.	--	Professional Elective (Minimum of 3 credits to be earned)				3
						19

SEMESTER II

Sl. No	Course Code	Course Title	L	T	P	C
1.	GEC6201	Research Methodology	3	0	0	3
2.	ECC6221	ASIC Design	3	0	0	3
3.	ECC6222	Real Time Operating Systems	3	0	0	3
4.	ECC6223	Analog Integrated Circuit Design	3	0	2	4
5.	ECC6223	VLSI Physical Design Automation Lab	0	0	2	1
6.	--	Professional Elective (Minimum of 9 credits to be earned)				9
						23

SEMESTER III

Sl. No	Course Code	Course Title	L	T	P	C
1.	--	General Elective	3	0	0	3
2.	ECC7121	Mini Project/Internship	0	0	2	1
3.	ECC7122	Project Work - Phase I	0	0	12	6*
4.	--	Professional Elective (Minimum of 6 credits to be earned)				6
						16

SEMESTER IV

Sl. No	Course Code	Course Title	L	T	P	C
1.	ECC7122	Project Work - Phase II	0	0	36	18*
						18 + 6 = 24

* Credits for Project Work Phase I to be accounted along with Project Work Phase II in IV Semester

Total Credits: 76

LIST OF PROFESSIONALELECTIVES

Sl.No.	Course Code	TITLE	L	T	P	C
1.	ECCY051	Advanced Digital System Design	3	0	0	3
2.	ECCY052	Low Power VLSI Design	3	0	0	3
3.	ECCY053	Testing of VLSI Circuits	3	0	0	3
4.	ECCY054	VLSI Digital Signal Processing	3	0	0	3
5.	ECCY055	RF Integrated Circuits Design	3	0	0	3
6.	ECCY056	Reconfigurable Computing	3	0	0	3
7.	ECCY057	CAD for VLSI Circuits	3	0	0	3
8.	ECCY058	CMOS Mixed Signal Circuit Design	3	0	0	3
9.	ECCY059	Semiconductor Memories	3	0	0	3
10.	ECCY060	SoC design and Verification	3	0	0	3
11.	ECCY061	Signal integrity for high speed design	3	0	0	3
12.	ECCY062	Optimization Techniques and their applications in VLSI design	3	0	0	3
13.	ECCY063	Electronic Design Automation Tools	2	0	0	2
14.	ECCY064	Programming System Verilog	2	0	0	2
15.	ECCY065	Programming Verilog HDL	2	0	2	3
16.	ECCY066	Scripting Languages for VLSI Design Automation	1	0	0	1
17.	ECCY067	Embedded Networking	3	0	0	3
18.	ECCY068	Control Area Network	3	0	0	3
19.	ECCY069	RISC Processor Architecture and Programming	3	0	0	3
20.	ECCY070	Distributed Embedded Computing	3	0	0	3
21.	ECCY071	Hardware-software co-design	3	0	0	3
22.	ECCY072	Real Time Systems	3	0	0	3
23.	ECCY073	Software for Embedded Systems	2	0	2	3
24.	ECCY074	Sensors Lab	1	0	0	1
25.	ECCY075	DSP System Design	3	0	0	3
26.	ECCY076	IC packaging and Interconnects	3	0	0	3
27.	ECCY077	MEMS System Design	3	0	0	3
28.	ECCY078	Embedded LINUX	3	0	0	3
29.	ECCY079	WSN Architecture and Programming	3	0	0	3
30.	ECCY080	Internet of Things	3	0	0	3

**M. TECH. (CONSTRUCTION ENGINEERING & PROJECT
MANAGEMENT)**

Sl. No.	Course Code	Course Title	L	T	P	C
SEMESTER I						
1	MAC6183	Probability and Statistics	3	1	0	4
2	CEC6121	Modern Construction Materials and Practices	3	0	0	3
3	CEC6122	Construction Equipment Management	2	0	0	2
4	CEC6123	Project Management in Construction	3	0	0	3
5	CEC6124	Construction Planning, Scheduling and Control	3	0	2	4
6	CEC6104	Advanced Concrete Technology	3	0	2	4
						20
SEMESTER II						
1	GEC6201	Research Methodology for Engineers	3	0	0	3
2	CEC6231	Construction Economics and Finance	4	0	0	4
3	CEC6232	Contract Laws and Regulations	3	0	0	3
4		Professional Electives #				9
5	CEC6233	Mini Project	0	0	2	1
						20
SEMESTER III						
1		General Elective###	3	0	0	3
2		Professional Electives #				9
3	CEC7122	Industrial Internship* / Seminar**	0	0	2	1
4	CEC7121	Project Work - Phase I***	0	0	12	6
						13

SEMESTER IV

1	CEC7121	Project Phase II	0	0	36	18
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6+18=24

Total	78
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Student has to take courses for a minimum of 9 credits from list of odd and even semester professional electives.

Student has to take minimum of 3 credits from the list of general electives.

* 15 days of Industrial training will be undertaken by regular candidates during first year summer vacation and the credit will be awarded in the 3rd semester.

** Seminar will be taken by working Professionals only.

*** Credits for Project Work Phase I to be accounted along with Project Work Phase II in IV Semester

PROFESSIONAL ELECTIVE

Sl. No.	Course Code	Course Title	L	T	P	C
1	CECY121	Construction Personnel Management	3	0	0	3
2	CECY122	Financing and Management of Infrastructure Projects	3	0	0	3
3	CECY123	Intelligent Building Management System	3	0	0	3
4	CECY124	Management Information System	3	0	0	3
5	CECY103	Maintenance and Rehabilitation of Structures	3	0	0	3
6	CECY125	Remote Sensing and GIS in Infrastructure Engineering	3	0	0	3
7	CECY126	Urban Planning and Design	3	0	0	3
8	CECY127	Material Management	2	0	0	2
9	CECY128	Construction Safety management	2	0	0	2
10	CECY129	Transportation Planning and Management	2	0	0	2
11	CECY130	Lean Construction	1	0	0	1
12	CECY131	Site Investigation and Soil Exploration Techniques	1	0	0	1

LIST OF EVEN SEMESTER ELECTIVES

1	CECY221	Construction Quality Management	3	0	0	3
2	CECY222	Environmental Impact Assessment of Infrastructure Projects	3	0	0	3
3	CECY223	Financial Accounting and Management	3	0	0	3
4	CECY224	Green Building and Energy Efficient Structures	3	0	0	3
5	CECY225	Green Concepts in Building Environment	3	0	0	3
6	CECY226	Infrastructure Planning and Management	3	0	0	3
7	CECY227	Shoring Scaffolding and Formwork	3	0	0	3
8	CECY228	Affordable Housing	2	0	0	2
9	CECY229	Application of Information Technology in Infrastructure Management	1	0	2	2
10	CECY230	Cost Estimation and Control	2	0	0	2
11	CECY231	Construction Risk Management	1	0	0	1

M.TECH. POWER ELECTRONICS AND DRIVES

(FOUR SEMESTERS / FULL TIME)

		CURRICULUM				
		SEMESTER I				
Sl. No.	Course Code	Course Title	L	T	P	C
1		Applied Mathematics	3	1	0	4
2	EEC6125	Modeling and Analysis of Electrical Machines	3	0	0	3
3	EEC6126	Advanced Power Semiconductor Devices	3	0	0	3
4	EEC6127	Analysis of Power Converters	3	0	0	3
7		Professional Electives [#]				6 [#]
8	EEC6128	Simulation of Power Electronic Systems - Laboratory	0	0	3	1
9	EEC6129	Design / Fabrication Lab	0	0	3	1
# Minimum of 6 credits						21
		SEMESTER II				
Sl. No.	Course Code	Course Title	L	T	P	C
1	EEC6235	Solid State Drives	3	0	0	3
2	EEC6236	Analysis of Power Inverters	3	0	0	3
3	EEC6237	Switched Mode Power Conversion Systems	3	0	0	3
4		Research Methodology	3	0	0	3
5		Professional Electives [#]				6 [#]
6	EEC6238	Power Electronics and Drives Laboratory	0	0	3	1
7	EEC6239	Self Learning	0	0	2	2
8	EEC6240	Industrial Internship	0	0	*	**
						21
	* Minimum 30 days of industrial internship ** Industrial internship will be undertaken during first year summer vacation. The credit will be awarded in the 3rd semester.					

		SEMESTER III				
Sl.	Course	Course Title	L	T	P	C
No.	Code					
1	EEC7110	Embedded control of Electrical Drives	3	0	2	4
2		Professional Electives [#]				6 [#]
3	EEC6240	Industrial Internship	0	0	*	2
4		General Elective	3	0	0	3
5	EEC7111	Project Work - Phase I ^{##}	0	0	12	6 ^{##}
						15
		SEMESTER IV				
Sl.	Course	Course Title	L	T	P	C
No.	Code					
1	EEC7111	Project work - Phase II ^{##}	0	0	36	18 ^{##}
						18 + 6 = 24

^{##} Credits for Project Work Phase I to be accounted along with Project Work Phase II in IV Semester.

TOTAL CREDITS : 81

		LIST OF ELECTIVES
Sl.	Course	Course Title
No.	Code	
1.	EECY40	Advanced Digital Signal Processing
2.	EECY41	Digital Signal Processors
3.	EECY42	Robotics And Factory Automation
4.	EECY43	SCADA and DCS
5.	EECY44	Spacecraft Power Systems
6.	EECY45	Modern Electric, Hybrid Electric And Fuel Cell Vehicles
7.	EECY46	Transformers and Inductors for Power Electronics
8.	EECY47	Special Electrical Machines and Controllers

9.	EECY48	Vector Control of AC Drives
10.	EECY49	Solar and Energy storage systems
11.	EECY50	Fundamentals of grid connected photo voltaic power electronic converter design
12.	EECY51	Solar power system design
13.	EECY04	Power Quality
14.	EECY	System theory
15.	EECY	Soft Computing Techniques
16.	EECY12	High Voltage Direct Current Transmission
17.	EEC6213	Flexible A.C. Transmission Systems
18.	EECY13	Wind Energy Conversion Systems

M.TECH. (MANUFACTURING)
(FOUR SEMESTERS / FULL TIME)

S. No.	Course Code	Course Name	L	T	P	C	TC
SEMESTER I							
1	MAC6183	Probability and Statistics	3	1	0	4	
2	MEC6121	Advances in Manufacturing Technology	3	0	0	3	
3	MEC6101	Applied Materials Engineering	3	0	0	3	
4	MEC6122	Mechatronics in Manufacturing Systems (Integrated Lab)	3	0	2	4	
5	MEC6103	Advanced Finite Element Analysis (Integrated Lab)	2	1	2	4	
6		Professional Electives #				3	
		Total Credit (semester I)					21
SEMESTER II							
1	GEC6201	Research Methodology	3	0	0	3	
2	MEC6223	Automated and Computer Integrated Manufacturing (Integrated Lab)	3	0	2	4	
3	MEC6224	Metal Forming Techniques	2	0	0	2	
4	MEC6206	Advanced Metrology and NDT (Integrated Lab)	3	0	2	4	
5		Professional Electives ##				6	
6	MEC6225	Manufacturing and Measurement Project	0	0	2	1	
		Total Credit (semester II)					20
SEMESTER III							
1		General Electives*				3	
2		Professional Electives###				9	
3	MEC7126	Industry Internship				1	
4	MEC7101	Project Phase I	0	0	12	6**	
		Total Credit (semester III)					13
SEMESTER IV							
1	MEC7101	Project Phase II	0	0	36	18**	
		Total Credit (semester IV)			18 + 6		24
Grand Total of credits							78

Professional Electives can be chosen from the list, provided that the cumulative credits should not be less than 3

Professional Electives can be chosen from the list, provided that the cumulative credits should not be less than 6

Professional Electives can be chosen from the list, provided that the cumulative credits should not be less than 9

* General Electives can be chosen from the list, provided that the cumulative credits

should not be less than 3

** Credits for project work phase I to be accounted along with project work phase II in IV semester

Details of Credits

S. No.	Details of curriculum	Total Credits
1	Program Core	24
2	Professional Electives	18
3	General Electives	3
4	Project	25
5	Internship	1
6	Mathematics and Research Methodology	7
Total		78

LIST OF PROFESSIONAL ELECTIVES

PROFESSIONAL ELECTIVES FOR SEMESTER I

S. No.	Course Code	Course Title	L	T	P	C
1	MECY28	Manufacturing System Simulation	3	0	0	3
2	MECY29	Productivity Management and Re-engineering	3	0	0	3
3	MECY30	Advances in Casting and Welding	3	0	0	3
4	MECY31	Cellular Manufacturing System	3	0	0	3
5	MECY32	Advanced Joining Process	2	0	0	2
6	MECY01	Automotive Manufacturing	1	0	0	1

PROFESSIONAL ELECTIVES FOR SEMESTER II

7	MECY33	Advanced Tool Design	3	0	0	3
8	MECY34	Mechatronics in CNC Machines	3	0	0	3
9	MECY35	Total Quality System and Management	3	0	0	3
10	MECY36	Manufacturing Management	3	0	0	3
11	MECY37	Tribology of Machines	3	0	0	3
12	MECY38	Supply Chain Management	3	0	0	3
13	MECY39	Plant Layout and Material Handling	2	0	0	2
14	MECY40	Additive Manufacturing	2	0	0	2
15	MECY41	AI based Optimization Techniques	1	0	0	1
16	MECY20	Geometric Dimensioning and Tolerancing	1	0	0	1
17	MECY14	Virtual Manufacturing	1	0	0	1

PROFESSIONAL ELECTIVES FOR SEMESTER III

18	MECY42	Reverse Engineering	3	0	0	3
19	MECY43	MEMS & Nano Technology	3	0	0	3
20	MECY44	Production and Inventory Management	3	0	0	3
21	MECY45	Design for Manufacturing	3	0	0	3
22	MECY46	Polymers and Composite Materials	3	0	0	3
23	MECY47	Industrial Robotics and Expert Systems	3	0	0	3
24	MECY48	Technology Management	3	0	0	3
25	MECY49	Lean Six sigma	3	0	0	3
26	MECY50	Lean Manufacturing System and Implementation	3	0	0	3
27	MECY51	Corrosion and Surface Engineering	3	0	0	3
28	MECY52	Laser Material Processing	2	0	0	2
29	MECY53	Hydraulics and Pneumatics	2	0	0	2
30	MECY54	Entrepreneurship	1	0	0	1
31	MECY55	Reliability Engineering	1	0	0	1
32	MECY56	Failure Mode and Effect Analysis	1	0	0	1

M.TECH. (ELECTRONICS AND INSTRUMENTATION ENGG)

(FOUR SEMESTERS / FULL TIME)

S. No	Course Code	Course Name	L	T	P	C
Semester I						
8.		Applied Mathematics	3	1	0	4
9.		System Theory	3	0	0	3
10.	EIC6101	Microcontroller based system design	3	0	0	3
11.	EIC6102	Frequency domain based Control System	3	0	2	4
12.	EIC6103	Applied Industrial Instrumentation	3	0	2	4
13.		Professional Elective				2/3
		Total Credit (semester I)				20/21
Semester II						
14.		Research Methodology	3	0	0	3
15.	EIC6211	Advanced Process Control	3	0	2	4
16.	EIC6212	Industrial Automation using PLC, SCADA and DCS	2	0	2	3
17.		Professional Elective (min credit of 9)				9
18.	EIC6213	Mini project/Seminar	0	0	2	1
		Total Credit (semester II)				20
Semester III						
19.		Professional Elective				6/7
20.		General Elective (min credit of 3)				3
21.	EIC7102	Industry Internship	0	0	#	1
22.	EIC7101	Project Phase I	0	0	12	6*
		Total Credit (semester III)				16/17
		# - Two week summer internship				
Semester IV						
23.	EIC7101	Project Phase II	0	0	36	18
		Total Credit (semester IV)				18
*Credit for Project Phase I to be accounted along with Project work phase II in IV semester						
**Value added course by industrial person / Academician with 1 credit						
		Grand Total of credits				75

LIST OF ELECTIVES

General Elective (GE1) (to be appended) (3 credits)

1. Project management (3 credits)
2. Society, Technology & Sustainability (3 credits)
3. Artificial Intelligence (3 credits)
4. Programming using Matlab & Simulink (3 credits)
5. Intellectual Property Rights (3 credits)

Professional Elective (Semester 1 Electives) (2/3 credits)

1. EICY101 Piping & Instrumentation for Process Industries (1T+2P) (2 credits)
2. EICY102 Applied soft computing techniques for Process Modeling, Control and Optimization (2T+2P) (3 credits)
3. EICY103 Instrumentation System Design (1T+2P)(2 credits)

Professional Elective - (Semester 2 Electives) (9 credits)

Professional Electives from Parent Department

1. EICY211 System Identification and Modeling (2T+2P)(3 credits)
2. EICY212 Adaptive Control (3 credits)
3. EICY213 Optimal Control (3 credits)
4. EICY214 Virtual Instrumentation (2L+2P)(3 credits)
5. EICY215 Advanced Fiber optics and Laser Instrumentation (3 credits)
6. EICY216 Applied Biomedical Instrumentation (3 credits)

Professional Electives from Other Department

1. ECCY001 Digital Image Processing (3 credits)
2. ECC6104 Advanced Digital Signal Processing (4 credits)
3. ECCY008 Speech and Audio Signal Processing (3 credits)
4. ECCY009 Network Security (3 credits)
5. MACXXX Applied Mathematics on Transforms for Signal Processing (3 credits)

Professional Elective (Semester 3 Electives) (6/7 credits)

1. EICY131 Fault Tolerant Control (3L+2P) (4 credits)
2. EICY132 Robust Control (3 credits)
3. EICY133 Robotics and Automation (3 credits)
4. EICY134 Instrumentation in Power Plant (3 credits)
5. EICY135 Instrumentation Petrochemical Industry (3 credits)
6. EICY136 Mems and Nano Sensors (3 credits)
7. EICY136 Value Added Course (1 credit)

Professional Elective from Other Department

1. ECCY010 Wireless Communication (3 credits)
2. ECCY011 Medical Image Processing (3 credits)
3. ECCY014 Wireless sensor Networks (3 credits)

M.TECH. (INFORMATION TECHNOLOGY)**SEMESTER I**

S. No	Course Code	Course	L	T	P	C
1.	ITC6101	Computer Forensics and Information Security	3	0	0	3
2.	CSC	Advanced Computer Architecture	3	1	0	4
3.	MAC 6181	Applied Algebra and Discrete Algorithms	3	1	0	4
4.	ITC6102	Advanced Data Structures	3	0	2	4
5.	CSC	Computer Networks and Management	3	0	0	3
6.	ITC6103	Software Development Methodologies	2	0	0	2
7.	ITC6104	Industry Oriented Case Study	0	0	2	1

Total Credits: 21**SEMESTER II**

S. No	Course Code	Course	L	T	P	C
1.	ITC6201	Cloud Computing Technologies	3	0	0	3
2.	MAC	Research Methodology	3	0	0	3
3.	ITCY	Elective #1	9	0	0	9
4.	ITC6202	Mobile Application Development	1	0	2	2
5.	ITC6203	Software Development and Testing Lab	0	0	2	1

Total Credits: 18

SEMESTER III

S. No	Course Code	Course	L	T	P	C
1	ITC7301	Internet of Things	3	0	0	3
2		Elective # 2	9	0	0	9
3		General Elective	3	0	0	3
4	ITC7302	Project - Phase I	0	0	12	6*

Total Credits: 15**SEMESTER IV**

S. No	Course Code	Course	L	T	P	C
1	ITC7302	Project - Phase II	0	0	36	18

Total Credits: 6 + 18 = 24

***Credits for Project Work – Phase I to be accounted along with Project Work – Phase II in IV Semester**

Total Credits: 78**SEMESTER II ELECTIVES**

S. No	Course Code	Course	L	T	P	C
1.	ITCY01	Multicore Programming	3	0	0	3
2.	ITCY02	Software Requirements Management	3	0	0	3
3.		Service Oriented Architecture	3	0	0	3
4.	ITCY03	Programming with Hadoop	0	0	2	1
5.		Mobile Technologies	3	0	0	3
6.	ITCY04	Social Network Analysis	3	0	0	3
7.		Advanced Database Technology	3	0	0	3
8.	ITCY05	Multimedia Technology & Applications	3	0	0	3
9.	ITCY06	Distributed Systems	3	0	0	3
10.	ITCY07	IT Infrastructure Management	3	0	0	3
11.	ITCY08	Applied Cryptography	3	0	0	3
12.	ITCY09	Video Processing	3	0	0	3

13.	ITCY10	High Speed Networks	3	0	0	3
14.	ITCY11	Virtualization Techniques	2	0	0	2
15.		Society, Technology & Sustainability	3	0	0	3

SEMESTER III ELECTIVES

S. No	Course Code	Course	L	T	P	C
1.	ITCY12	Data Science Analytics	2	0	0	2
2.	ITCY13	Big Data Analytics Tool	1	0	0	1
3.	ITCY14	Machine Learning	3	0	0	3
4.	ITCY15	Green Computing	1	0	0	1
5.	ITCY16	Wireless Networks	3	0	0	3
6.	ITCY17	Enterprise Resource Planning	3	0	0	3
7.	ITCY18	Soft Computing	3	0	0	3
8.	ITCY19	Web Design & Management	3	0	0	3
9.	ITCY20	Design Patterns	3	0	0	3
10.	ITCY21	Data Warehousing and Data Mining	3	0	0	3
11.	ITCY22	Wireless Mobile Communication	3	0	0	3
12.	ITCY23	Digital Image Processing	3	0	0	3
13.	ITCY24	Software Metrics	3	0	0	3
14.	ITCY25	Security in BigData	3	0	0	3
15.	ITCY26	Multimedia Communication Networks	3	0	0	3
16.	ITCY27	Adhoc and Sensor Networks	3	0	0	3
17.	ITCY28	Ontology and Semantic Web	3	0	0	3
18.	ITCY29	Human Computer Interface	2	0	0	2
19.	ITCY30	Software Project Management	3	0	0	3

M.TECH. (STRUCTURAL ENGINEERING)**CURRICULUM**

Sl. No.	Course Code	Course Title	L	T	P	C
SEMESTER I						
1	MAC6184	Probability, Matrix Theory and Linear Programming	3	1	0	4
2	CEC6101	Theory of Elasticity and Plasticity	3	0	0	3
3	CEC6102	Advanced Concrete Design	3	1	0	4
4	CEC6103	Structural Dynamics	3	0	0	3
5	CEC6104	Advanced Concrete Technology	3	0	2	4
6	CEC6105	Design of Substructures	3	0	0	3
						21
SEMESTER II						
1	GEC6201	Research Methodology for Engineers	3	0	0	3
2	CEC6211	Finite Element Analysis in Structural Engineering	3	0	0	3
3	CEC6212	Advanced Steel Structures	3	1	0	4
4	CEC6213	Experimental Techniques and Instrumentation	2	0	2	3
5		Professional Electives #				6
6	CEC6214	Seminar	0	0	2	1
						20
SEMESTER III						
1		General Elective###	3	0	0	3
2		Professional Electives##				9
3	CEC7101	Project Phase I***	0	0	12	6
4	CEC7102	Internship*/Mini Project**	0	0	2	1
						13

SEMETER IV

1	CEC7101	Project Phase II	0	0
			36	18
				6+18=24
		Total		78

Student has to take a minimum of 6 credits from even semester professional elective courses

Student has to take a minimum of 9 credits from odd semester professional elective courses

Student has to take a minimum of 3 credits from general elective courses

* 15 days of Industrial training will be undertaken by regular candidates during first year summer vacation and the credit will be awarded in the 3rd semester.

** Mini project will be taken by working Professionals only.

*** Credits for Project Work Phase I to be accounted along with Project Work

Phase II in IV Semester

**M.SC. (ACTUARIAL SCIENCE)
(FOUR SEMESTERS / FULL TIME)**

Semester 1					
Course Code	Course Title	L	T	P	C
MAC 6101	Fundamentals of Financial Mathematics	3	1	0	4
MAC 6102	Probability and Distributions	3	1	0	4
MAC 6103	Business Economics	2	1	0	3
MAC 6104	Financial reporting and Accounting	2	1	0	3
MAC 6105	Principles of Insurance	2	0	0	2
MAC 6106	R Programming	0	0	4	2
Total		12	4	4	18

Semester 2					
Course Code	Course Title	L	T	P	C
MAC 6211	Applications of Financial Mathematics	3	1	0	4
MAC 6212	Fundamentals of Life Contingencies	3	1	0	4
MAC 6213	Descriptive Statistical Methods in Actuarial Science	3	1	0	4
MAC 6214	Stochastic Models	2	1	0	3
MAC 6215	Mathematical Statistics	3	1	0	4
ENB 6281	Communication Skills	1	0	2	2
MAC 6216	Actuarial Computational Laboratory (MS-EXCEL)	0	0	4	2
Total		15	5	6	23

Semester 3					
Course Code	Course Title	L	T	P	C
MAC 7101	Applications of Life Contingencies	3	1	0	4
MAC 7102	Financial Economics	3	1	0	4
MAC 7103	Applied Statistical Methods in Actuarial Science	3	1	0	4
MAC 7104	Survival Models	2	1	0	3
MAC 7105	Actuarial Computational Laboratory (SPSS)	0	0	4	2
	Elective – I	3	0	0	3
	Elective- II	3	0	0	3
MAC7106	Project Phase-I	0	0	8	4*
	Total	17	4	12	23

Semester 4					
Course Code	Course Title	L	T	P	C
MAC 7106	Project Phase –II	0	0	24	12*
Total		0	0	24	16

* Credits for Project Work Phase I to be accounted along with Project Work Phase II in IV Semester

LIST OF ELECTIVES

Elective – I					
Course Code	Course Title	L	T	P	C
MACY01	Financial Management of Life Office	3	0	0	3
MACY02	Islamic Investments	3	0	0	3
MACY03	Corporate Finance	3	0	0	3

Elective – II					
Course Code	Course Title	L	T	P	C
MACY04	Financial Markets	3	0	0	3
MACY05	Islamic Asset and Fund Management	3	0	0	3
MACY06	Portfolio and Credit Risk Management	3	0	0	3

Total Credit (18+23+23+16) = 80

M.Sc.CHEMISTRY
(Four Semesters / Full Time)
CURRICULUM

		SEMESTER I				
Sl. No.	Course Code	Course Title	L	T	P	C
1.	CHC6101	Organic Chemistry-I	3	0	0	3
2.	CHC6102	Thermodynamics and Electrochemistry	3	0	0	3
3.	CHC6103	Inorganic Chemistry	3	0	0	3
4.	CHC6104	Analytical Chemistry	3	0	2	4
5.		Elective -I	3	0	0	3
6.	CHC6105	Organic Chemistry Practical-I	0	0	4	2
7.	CHC106	Inorganic Chemistry Practical-I	0	0	4	2
8.	CHC6107	Physical Chemistry Practical -1	0	0	4	2
	CHC6108	seminar		2		1
						23
		Core				19
		Elective				03

		SEMESTER II				
Sl. No.	Course Code	Course Title	L	T	P	C
1		Research methodology	3	0	0	3
2.	CHC6211	Synthetic Organic Chemistry	3	2	0	4
3.	CHC6212	Quantum Chemistry	3	1	0	4
4.	CHC6213	Coordination Chemistry	3	0	0	3
5.		Elective-II	3	1	0	4
7	CHC6214	Synthetic Organic Chemistry Practical	0	0	4	2
8.	CHC6215	Inorganic Chemistry Practical-II	0	0	4	2
9	CHC6216	Physical Chemistry Practical -II	0	0	4	2
		Core				20
		Elective				04

		SEMESTER III				
Sl. No.	Course Code	Course Title	L	T	P	C
		Internship (during summer vacation)				1
1.	CHC7101	Advanced Organic Chemistry	3	0	0	3
2.	CHC7102	Physical Chemistry	3	0	0	3
3.	CHC7103	Advanced Inorganic Chemistry	3	0	0	3
		Elective III	3	0	2	4
4.		Elective IV	3	0	2	4
5.		Elective V	3	0	0	3
		Electives with total minimum of 11 credits				
7	CHC7104	Project Phase -1				2
						23
		core				09
		elective				11

		SEMESTER IV				
Sl. No.	Couse Code	Course Title	L	T	P	C
2.	CHC7201	Project Phase -II	0	0		10
3.		Total Credits				10

Total
Credits =
79

		Total credit				
1		Core (Including Lab)				48
2		Elective				18
		Seminar				01
		Project				12
		Internship				01

80

LIST OF ELECTIVES

Course Code	Course Title	L	T	P	C
Basic Chemistry					
CHCY01	Green chemistry	3	0	2	4
CHCY02	Photophysics and photochemistry	3	0	0	3
CHCY03	Molecular spectroscopy	3	1	0	4
CHCY04	Photochemistry	3	0	0	3
Medicinal ,Pharmaceutical and biological					
CHCY05	Medicinal and Pharmaceutical Chemistry	3	0	0	3
CHCY06	Pharmaceutical Technology	3	0	0	3
CHCY07	GMP, Quality Assurance and Validation	3	0	0	3
CHCY08	Biochemistry	3	0	2	4
Materials and Technology					
CHCY09	Polymer Chemistry	3	0	2	4
CHCY10	Polymer Technology	3	0	0	3
CHCY11	Electrical Properties of Polymeric Materials	3	0	0	3
CHCY12	Polymer Structure and Property Relationship	3	0	0	3
CHCY13	Concepts and Techniques in Catalysis	3	0	0	3
CHCY14	Advanced materials	3	0	0	3
CHCY15	Nanotechnology	3	0	2	4
CHCY16	Inorganic Chemical Technology	3	0	0	3
CHCY17	Organic Chemical Technology	3	0	0	3
CHCY18	Chlor-alkali Technology	3	0	0	3
CHCY19	Module Operations and module Processes	3	0	0	3
Energy ,Water and Environment					

CHCY20	Water and Waste Water Treatment	3	0	0	3
CHCY21	Solid Waste Management and Air Pollution	3	0	0	3
CHCY22	Industrial Electrochemistry	3	0	0	3
CHCY23	Corrosion and Corrosion Control	3	0	0	3
CHCY24	Electrochemical Protection Systems	3	0	0	3
CHCY25	Metal Coating Technology	3	0	0	3
CHCY26	Protective Coatings	3	0	0	3
CHCY27	Fuel Cells and Applications	3	0	0	3
CHCY28	Advanced Batteries and Systems	3	0	0	3
CHCY29	Electrochemical Material Science	3	0	0	3
CHCY30	Electrochemical Energy Conversion and Storage	3	0	0	3
CHCY31	Solar energy	3	0	0	3

M.Sc. Physics

First year (Semester I)

Sl.No.	Course Code	Course Title	L	T	P	C
1	PHC6101	Classical Mechanics and Relativity	4	0	0	4
2	PHC6102	Mathematical Physics	3	1	0	4
3	PHC6103	Solid state Physics	3	0	0	3
4	PHC6104	Physics of semiconductor devices	3	0	0	3
5		Elective – I	3	0	0	3
6		Elective – II	3	0	0	3
7	PHC6105	Materials Science lab	0	0	4	2
8	PHC6106	Semiconductor devices and circuits lab	0	0	4	2
Total			24			

First year (Semester II)

Sl.No.	Course Code	Course Title	L	T	P	C
1	PHC6201	Quantum Mechanics	3	0	0	3
2	PHC6202	Thermodynamics and Statistical Physics	4	0	0	4
3	PHC6203	Atomic Physics and Molecular Spectroscopy	4	0	0	4
4	GEC6202	Research Methodology	3	0	0	3
5		Elective – III	3	0	0	3
6		Elective – IV	3	0	0	3
7	PHC6204	Thermal Physics lab	0	0	4	2
8	PHC6205	Seminar / Mini project – I				1
Total			23			

Second year (Semester III)

Sl.No.	Course Code	Course Title	L	T	P	C
1	PHC7101	Digital Electronics and Microprocessors	4	0	0	4
2	PHC7102	Electromagnetic Theory & Electrodynamics	3	1	0	4
3	PHC7103	Nuclear and Particle Physics	3	0	0	3
4		Elective –V	3	0	0	3
5		Elective –VI	3	0	0	3
6	PHC7104	Advanced Electronics Lab	0	0	4	2
7	PHC7105	Advanced optics lab	0	0	4	2
8	PHC7106	Seminar / Mini Project - II				1
9	PHC7107	Project phase – I				2
Total			24			

Second year (Semester IV)

Sl.No.	Course Code	Course Title	L	T	P	C
1	PHC7107	Project Phase – II				10
Total			10			

Total number of credits: 81

LIST OF ELECTIVES

First year (Semester I)

S.No.	Course Code	Course Name	L	T	P	C
1	PHCY101	Crystal Growth Techniques	3	0	0	3
2	PHCY102	Materials processing	3	0	0	3
3	PHCY103	Characterization of materials	3	0	0	3
4	PHCY104	Smart materials and structures	3	0	0	3
5	PHCY105	Advanced Optics & Laser Technology	3	0	0	3
6	PHCY106	Nonlinear optics	3	0	0	3
7	PHCY107	Optical Fiber communication	3	0	0	3
8	PHCY108	Optical computing	3	0	0	3

First year (Semester II)

S.No.	Course Code	Course Name	L	T	P	C
1	PHCY201	Electro-Optic materials and devices	3	0	0	3
2	PHCY202	Ferroelectric materials and devices	3	0	0	3
3	PHCY203	Structure and properties of alloys	3	0	0	3
4	PHCY204	Photonic materials and devices	3	0	0	3
5	PHCY205	Numerical methods and programming	3	0	0	3
6	PHCY206	Ultrasonics and Non-Destructive Testing	3	0	0	3
7	PHCY207	Optoelectronics and its devices	3	0	0	3
8	PHCY208	Biophotonics	3	0	0	3

Second year (Semester III)

S.No.	Course Code	Course Name	L	T	P	C
1	PHCY109	Mathematical methods for nonlinear science	3	0	0	3
2	PHCY110	Measurements and Instrumentation	3	0	0	3
3	PHCY111	Biomedical Instrumentation	3	0	0	3
4	PHCY112	Radiation Physics	3	0	0	3
5	PHCY113	Laser spectroscopy and its applications	3	0	0	3
6	PHCY114	Nanophotonics	3	0	0	3
7	PHCY115	Nanoscience and Technology	3	0	0	3
8	PHCY116	Thin film science and technology	3	0	0	3
9	PHCY117	Corrosion science and technology	3	0	0	3

10	PHCY118	Biomaterials	3	0	0	3
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List of Course work papers- Ph.D.

S.No.	Course Code	Course Name	L	T	P	C
1	PHCZ101	Chaos and Solitons	3	0	0	3
2	PHCZ102	Linear and nonlinear electronics	3	0	0	3
3	PHCZ103	Nano electronics	3	0	0	3
4	PHCZ104	Physicochemical methods for characterization of nanomaterials	3	0	0	3
5	PHCZ105	Imaging techniques for nanotechnology	3	0	0	3
6	PHCZ106	Digital signal processing	3	0	0	3
7	PHCZ107	Crystallography and crystal growth	3	0	0	3
8	PHCZ108	Advanced Statistical Mechanics	3	0	0	3

M. Sc. (Biotechnology)

Sl. No.	Course Code	Course Title	L	T	P	C
SEMESTER I						
1	LSC6101	Advanced Biochemistry	4	0	0	4
2	LSC6102	Cell and Molecular Biology	4	0	0	4
3	LSC6103	Genetic Engineering	4	0	0	4
4		Elective I	3	0	0	3
5		Elective II	3	0	0	3
6	LSC6104	Lab I (Biochemistry/Molecular Biology/ Genetic Engineering)	0	0	6	3
Total						21
SEMESTER II						
1	GEC6202	Research Methodology	3	0	0	3
2	LSC6201	Immunology	4	0	0	4
3	LSC6202	Bioinformatics	4	0	0	4
4		Elective III	3	0	0	3
5		Elective IV	3	0	0	3
6	LSC6203	Lab II (Immunology/ Bioinformatics)	0	0	6	3
7	LSC6204	Industrial Internship	0	0	0	1
8	LSC6205	Mini Project	0	0	0	1
Total						22
SEMESTER III						
1	LSC7101	Project Phase I	0	0	4	2
2	LSC7102	Plant and animal Biotechnology	4	0	0	4
3	LSC7103	Bioprocess Technology	4	0	0	4
4		Elective 5	3	0	0	3
5		Elective 6	3	0	0	3
6	LSC7104	Lab III (Plant and animal Biotechnology/ Bioprocess Technology)	0	0	6	3
Total						17
SEMESTER IV						
1	LSC7101	Project Phase II	0	0	36	18
Total						2+18=20
Total Credits						80

Note: Two credits earned in the Project Phase I will be added when the complete the Phase II project

LIST OF ELECTIVES

	Course Code	Course Title	L	T	P	C
SEMESTER I						
1	LSCY101	Bio safety and Bioethics, Bio entrepreneurship and IPR	3	0	0	3
2	LSCY102	Microbiology	3	0	0	3
3	LSCY103	Food Process Technology	3	0	0	3
4	LSCY104	Analytical Techniques	3	0	0	3
5	LSCY105	Environmental Biotechnology	3	0	0	3
6	LSCY106	siRNA/RNA Interference	3	0	0	3
SEMESTER II						
1	LSCY201	Recombinant DNA Technology	3	0	0	3
2	LSCY202	Advanced Instrumentation	3	0	0	3
3	LSCY203	Molecular Diagnostics	3	0	0	3
4	LSCY204	Omics	3	0	0	3
5	LSCY205	Biofuel and Bio energy	3	0	0	3
6	LSCY206	Molecular Farming	3	0	0	3
SEMESTER III						
1	LSCY111	Biopharmaceuticals	3	0	0	3
2	LSCY112	Molecular and Immune Diagnostics	3	0	0	3
3	LSCY113	Tissue and Antibody Engineering	3	0	0	3
4	LSCY114	Bio nanotechnology	3	0	0	3
5	LSCY115	Protein Engineering	3	0	0	3
6	LSCY116	Stem Cell Technology	3	0	0	3

M. Sc. (Microbiology)
(Four Semesters / Full Time)

Sl. No.	Course Code	Course Title	L	T	P	C
SEMESTER I						
1	LSC6102	Cell and Molecular Biology	4	0	0	4
2	LSC6121	Biochemistry	4	0	0	4
3	LSC6122	Microbial Genetics	4	0	0	4
4		Elective I	3	0	0	3
5		Elective II	3	0	0	3
6	LSC6123	Lab I (Biochemistry/Molecular Biology/ Microbial Genetics)	0	0	6	3
Total						21
SEMESTER II						
1	GEC6202	Research Methodology	3	0	0	3
2	LSC6201	Immunology	4	0	0	4
3	LSC6202	Bioinformatics	4	0	0	4
4		Elective III	3	0	0	3
5		Elective IV	3	0	0	3
6	LSC6203	Lab II (Immunology/ Bioinformatics)	0	0	6	3
7	LSC6221	Industrial Internship	0	0	0	1
	LSC6222	Mini Project	0	0	0	1
Total						22
SEMESTER III						
1	LSC7121	Project Phase 1	0	0	2	2
2	LSC7122	Industrial, Food and Agricultural Microbiology	4	0	0	4
3	LSC7123	Environmental and Medical Microbiology	4	0	0	4
4		Elective 5	3	0	0	3
5		Elective 6	3	0	0	3
6	LSC7124	Lab III (Industrial, Food and Medical Microbiology)	0	0	6	3
Total						17
SEMESTER IV						
1	LSC7121	Project Phase – II	0	0	36	18
Total						2+18=20
Total Credits: 80						

Note: Two credits earned in the Project Phase 1 will be added when the complete the Phase II project

LIST OF ELECTIVES

Sl. No.	Course Code	Course Title	L	T	P	C
SEMESTER I						
1	LSCY101	Biosafety, Bioethics, Bioentrepreneurship, Intellectual Property rights	3	0	0	3
2	LSCY104	Analytical Methods	3	0	0	3
3	LSCY121	Animal Biotechnology	3	0	0	3
4	LSCY122	Plant Biotechnology	3	0	0	3
5	LSCY123	Parasitology	3	0	0	3
6	LSCY124	Virology	3	0	0	3
SEMESTER II						
1	LSCY201	Recombinant DNA Technology	3	0	0	3
2	LSCY203	Molecular Diagnostics	3	0	0	3
3	LSCY205	Biofuel and Bioenergy	3	0	0	3
4	LSCY211	Diagnostics of Infectious Diseases	3	0	0	3
5	LSCY212	Metagenomics	3	0	0	3
	LSCY213	Bioprocess (Microbial)	3	0	0	3
SEMESTER III						
	LSCY111	Biopharmaceuticals	3	0	0	3
	LSCY112	Molecular and Immuno Diagnostics	3	0	0	3
	LSCY113	Tissue and Antibody Engineering	3	0	0	3
	LSCY114	Bio-nanotechnology	3	0	0	3
	LSCY115	Protein Engineering	3	0	0	3
	LSCY131	Bioremediations	3	0	0	3

M.A. ISLAMIC STUDIES (REVISED 2017)**Semester I**

S.No.	Course Code	Name of the Course	L	T	P	C
Core Courses						
1	ISC 6101	Thematic Tafseer : Commentary of Holy Qur'aan	3	0	0	3
2	ISC 6102	Analysis of Hadeeth Literature	3	0	0	3
Allied Courses						
3	ISC 6103	Advent of Islam in India	3	0	0	3
4	ISC 6104	Communicative Arabic	3	0	0	3
5	ISC 6105	Early Muslim Society & Biography of great companions	3	0	0	3

Semester II

S.No.	Course Code	Name of the Course	L	T	P	C
Core Courses						
1	ISC 6211	Uloom - Sciences of Qur'aan and Hadeeth	3	0	0	3
2	ISC 6212	Usool al Fiqh: Maslaha Mursalah (Welfare) & Maqaasid al Sharee'ah (Objectives of Shareeah)	3	0	0	3
3	ISC 6213	Evolution of Islamic Jurisprudence	3	0	0	3
Allied Courses						
4	ISC 6214	Islamic Economics	3	0	0	3
5	ISC 6215	Advanced Arabic Grammar	3	0	0	3

Semester III

S.No.	Course Code	Name of the Course	L	T	P	C
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Core Courses						
1	ISC 7101	Comparative Fiqh - Jurisprudence	3	0	0	3
2	ISC 7102	Law of Administration in Shari'ah: Al Siyasa al Sharaiyyah	3	0	0	3
Allied Courses						
3	ISC 7103	Islam in Spain - Andalusia	3	0	0	3
4	ISC 7104	History of Muslim Civilization: Contribution to Science and Technology	3	0	0	3
5	ISC 7105	Muslim Thinkers & Reform Movements in Modern Era	3	0	0	3

Semester IV

S.No.	Course Code	Name of the Course	L	T	P	C
Core Courses						
1	ISC 7211	A Study of Shari’ah Rulings in Qur’aan: Ahkaamul Qur’aan	3	0	0	3
2	ISC 7212	Ijthihad and Research Methodology of Imams (Or) Research Methodology	4	0	0	4
	ISC 7216					
Allied Courses						
3	ISC 7213	Interest Free Banking	3	0	0	3
4	ISC 7214	Business Arabic and Translation	3	0	0	3
5	ISC 7215	Research / Dissertation	4	0	0	4

M.TECH. (BIOTECHNOLOGY)

SEMESTER I

Sl. No	Course Code	Course Title	L	T	P	C
1	LSB6101	Advanced Biochemistry	4	0	0	4
2	LSB6102	Cell & Molecular Biology	3	0	0	3
3	LTB6101	Applied Biostatistics for Biotechnologists	3	0	0	3
4	LTB6102	Immunotechnology	4	0	0	4
5	LSB6105	Biomedical Instrumentation Technology	3	0	0	3
6	LTB6103	Microbial Biotechnology	3	0	0	3
7	LSB6107	Biochemistry Lab	0	0	3	1
8	LSB6108	Cell Biology Lab	0	0	3	1
9	LSB6109	Immunotechnology Lab		0	0	3

Credits

23

SEMESTER II

Sl. No	Course Code	Course Title	L	T	P	C
1	LTB6201	Genomics & Proteomics	4	0	0	4
2	LTB6202	Bioprocess Engineering & Downstream Processing	4	0	0	4
3	LSB6203	Genetic Engineering	3	0	0	3
4	LSB6204	Computational Biology	3	0	0	3
5	LTB6203	Environmental Biotechnology	3	0	0	3
6	LSB6205	Computational Biology Lab	0	0	3	1
7	LSB6206	Genetic Engineering Lab	0	0	3	1
8	LTB6204	Mini Project	0	0	3	1

Credits

20

SEMESTER III

Sl. No	Course Code	Course Title	L	T	P	C
1	LTB7101	Pharmacogenomics	3	0	0	3
2	LSB7102	Plant & Medical Biotechnology	4	0	0	4
3		Elective II	3	0	0	3
4		Elective III	3	0	0	3
5	LTB7102	Mini Project	3	0	0	6
Credits						19

SEMESTER IV

Sl. No	Course Code	Course Title	L	T	P	C
1	LTB7201	Project work	0	0	20	8
Credits						18
TOTAL CREDITS						80

ELECTIVES

Course Code	Electives I	L	T	P	C
LSBY021	Bioenterpreneurship	3	0	0	3
LSBY022	IPR and Patent Law	3	0	0	3
LSBY023	Biosafety & Bioethics	3	0	0	3
LSBY028	Bionanotechnology	3	0	0	3
Electives II					
LSBY024	Molecular Diagnostics	3	0	0	3
LSBY025	Food Process technology	3	0	0	3
LSBY026	Animal Biotechnology	3	0	0	3
LSBY029	Industrial Biotechnology	3	0	0	3

M. Sc. (Biochemistry and Molecular Biology)
(Four Semesters / Full Time)

Sl. No.	Course Code	Course Title	L	T	P	C
SEMESTER I						
1	LSC6101	Advanced Biochemistry	4	0	0	4
2	LSC6102	Cell and Molecular Biology	4	0	0	4
3	LSC6103	Genetic Engineering	4	0	0	4
4		Elective I	3	0	0	3
5		Elective II	3	0	0	3
6	LSC6104	Lab I (Biochemistry, Molecular Biology and Genetic Engineering)	0	0	6	3
Total						21
SEMESTER II						
1	GEC6202	Research Methodology	3	0	0	3
2	LSC6201	Immunology	4	0	0	4
3	LSC6202	Bioinformatics	4	0	0	4
4		Elective II	3	0	0	3
5		Elective III	3	0	0	3
6	LSC6203	Lab II (Immunology/ Bioinformatics)	0	0	6	3
7	LSC6241	Industrial Internship	0	0	0	1
8	LSC6242	Mini Project	0	0	0	1
Total						22
SEMESTER III						
1	LSC7141	Project Phase I	0	0	2	2
2	LSC7142	Clinical Biochemistry	4	0	0	4
3	LSC7143	Genomics, Proteomics and Metabolomics	4	0	0	4
4		Elective V	3	0	0	3
5		Elective VI	3	0	0	3
6	LSC7144	Lab III (Clinical Biochemistry /Omics)	0	0	6	3
Total						17

Sl. No.	Course Code	Course Title	L	T	P	C
SEMESTER IV						
1	LSC7141	Project Phase II	0	0	36	18
Total			2+18=20			
Total Credits: 80						

Note: Two credits earned in the Project Phase 1 will be added when the complete the Phase II project

LIST OF ELECTIVES

Sl. No.	Course Code	Course Title	L	T	P	C
SEMESTER I						
1	LSCY101	Biosafety, Bioethics, Bioentrepreneurship, Intellectual Property rights	3	0	0	3
2	LSCY102	Microbiology	3	0	0	3
3	LSCY103	Food Process Technology	3	0	0	3
4	LSCY104	Analytical Methods	3	0	0	3
5	LSCY105	Environmental Biotechnology	3	0	0	3
6	LSCY106	SiRNA/RNA Interference	3	0	0	3
SEMESTER II						
1	LSCY201	Recombinant DNA Technology	3	0	0	3
2	LSCY202	Advanced Instrumentation	3	0	0	3
3	LSCY203	Molecular Diagnostics	3	0	0	3
4	LSCY204	Omics	3	0	0	3
5	LSCY205	Biofuel and Bioenergy	3	0	0	3

Sl. No.	Course Code	Course Title	L	T	P	C
6	LSCY206	Molecular Farming	3	0	0	3

SEMESTER II

1	LSCY111	Biopharmaceuticals	3	0	0	3
2	LSCY112	Molecular and immuno Diagnostics	3	0	0	3
3	LSCY113	Tissue andAntibody Engineering	3	0	0	3
4	LSCY114	Bio-nanotechnology	3	0	0	3
5	LSCY115	Protein Engineering	3	0	0	3
6	LSCY116	Stem Cell Technology	3	0	0	3

M. ARCH (GENERAL)

Sl. No.	Course code	Course Title	L	T	C
1	AR A 6101	Advanced Building Services	3	0	3
2	AR A 6102	Research Methodologies	3	0	3
3	AR A 6103	Digital Tools For Architectural Representation / Digital Tools	2	8	6
4	AR A 6104	For Contemporary Practices Architectural Design Studio I	0	18	9

SEMESTER II

Sl. No.	Course code	Course Title	L	T	C
1	AR A 6201	Advanced Building Materials and Technology	3	0	3
2	AR A 6202	Performance & Energy Evaluation	2	5	6
3	AR A 6203	Architectural Design Studio II	0	18	9
4		Elective - 1			

SEMESTER III

Sl. No.	Course code	Course Title	L	T	C
1	AR A 7101	Environmental Services			
2	AR A 7102	Sustainability Practices in Design			
3	AR A 7103	Architectural Design Studio III			
4		Elective - 2			

SEMESTER IV

Sl. No.	Course code	Course Title	L	T	C
1	AR A 7201	Design Thesis			

ELECTIVES

Sl. No.	Course code	Course Title	L	T	C
1	AR A Y101	Laws, Acts & Governance	3	0	3
2	AR A Y102	Project Management	3	0	3
3	AR A Y103	GIS Modelling	3	0	3
4	AR A Y104	Codes And Standards	3	0	3
5	AR A Y105	Urban Design	3	0	3
6	AR A Y106	Architectural Conservation	3	0	3
7	AR A Y107	Landscape Architecture	3	0	3
8	AR A Y108	Disaster Resilient Design	3	0	3